Workload Monitoring and Observability

Use **VCF Operations** as a VI administrator to monitor operating systems and applications using managed and open-source Telegraf. Discover services and applications using the Service Discovery adapter.

In This Chapter

- Configuring Business Applications
- Monitoring Application Services using Product-Managed Telegraf
- Monitoring Application Services and Operating Systems using Open Source Telegraf
- Discovering Services and Applications using the Service Discovery Adapter

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Configuring Business Applications

A Business Application is a VCF Operations construct that consists of a set of interconnected applications, services and hosts, which are configured to offer a service to the organization. Business Applications can be internal, like organization email system or customer-facing, like an organization website. The Business Applications page in VCF Operations is where you see Business Applications and their health.

A Business Application is a container construct that represents a collection of interdependent hardware and software components that deliver a specific capability to support your business. You create a Business Application to determine how your environment is affected when one or more components in the Business Application experiences problems, and to monitor the overall health and performance of the Business Application. VCF Operations can also display applications discovered through Service Discovery, and by importing applications from VCF Operations for networks.

The Business Application Page

The Business Applications page is where you view all your Business Applications and their health in a sortable table. The health of the Business Application is determined by the aggregated health of the underlying objects. The **Health** column in the table displays the status.

Where You View the Business Applications Page

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Business Applications**. All the available Business Applications and applications are displayed in the table.

The table displays the following types of objects:

Table 1683: Business Applications Page

Business Applications Source	Business Applications Type
Auto discovered and monitored applications which have more than one node	• Imported through VCF Operations for networks
Manually created Business Applications.	• Can contain custom groups. • Can contain any object to tier relationships. • Can contain applications.

Columns in the Business Application Table

The Business Applications table displays the following columns:

Column Name	Description
Vertical Ellipses	Offers options for the user to perform actions on the Business Application. .
Health	Displays the health of the Business Application by showing different icons. On hovering the cursor over the icon, you can see the health percentage.
Name	Displays the name of the Business Application and an icon indicates the type of Business Application.
Description	Displays the description of the Business Application you entered when creating it.
Application Tag	Displays the application tags you entered when you created the Business Application.
Business Criticality	Displays one of the following criticality states for the Business Application: • Medium • Critical • Low
Environment	Displays the environment where the Business Application resides in.

Actionable Options on the Business Applications Page

Table 1684: Actionable Options on the Business Applications Page

Option	Description
Vertical Ellipses	Click the vertical ellipses beside a Business Application to perform one of the following actions: • Edit (Not available for VCF Operations created Business Applications) • Delete • Troubleshoot with Workbench Go to Details. This opens the Business Applications Summary tab in the Inventory page.
Add Business Application	Click to create a new Business Application. The following options are available: • Define Manually • Based on Flows
Show Columns	Click to hide/show columns in the table.
Preview	Click on a Business Application name in the table to see a visual preview of the objects and their relationships.

Add Business Application

A Business Application is a first class citizen with tiers. Business Applications are groups of related objects in your environment that mimic an application in your business. You can create application to application relationships. Once you add a Business Application to VCF Operations, you can use the Business Applications page to track the health of objects in the application. You can optimize capacity, cost, and run What-if Analysis on Business Applications.

How Business Applications Work

The tier is a convenient way to organize objects that perform a specific task in an application in your organization. For example, you can group all of your database servers together in a tier. Business Applications need not contain tier objects. Application or custom group objects can be connected to a Business Application without a tier object, and a tier object can be empty.

The objects in a tier are static if the Business Application is created manually and not by VCF Operations. If the set of objects in a tier changes, you must manually edit the Business Application. However, if the Business Application is created by VCF Operations, the changes in the actual application will be reflected in the Business Application object without manual edit.

Construct a Business Application to view a particular segment of your business. The Business Application shows how the performance of one object affects other objects in the same Application, and helps you to locate the source of a problem. For example, if you have an application that includes all the database, Web, and network servers that process sales data for your business, you see a yellow, orange, or red status if the application health is degrading. From the Business Applications page, you can investigate which server is causing or exhibiting the problem.

The Add Business Application Screen

When you click **Add**, VCF Operations displays a blank canvas on the left pane, and options to select objects on the right pane. The **Select Members** section in the right pane has two tabs. The **Inventory** tab displays a list of all objects in you environment in a tree structure. You can search for objects here. The **Custom Groups** tab displays all the custom groups in your environment. You can use the search bar to search for custom groups.

Drag and drop objects from the right pane to the tier objects on the left pane. You can also drag and drop objects directly to the Business Application. Click **Save** to create the Business Application.

The new Business Application canvas is what you use to build your Business Application.

Table 1685: New Business Application Canvas

Option	Description
Vertical icon	Change the layout of the canvas from horizontal to vertical.
Undo icon	Undo the last action.
Redo icon	Redo the last action.
Fit View icon	Fit the Business Application structure within the boundary of the canvas. Use this option after you zoom into the canvas.
Add Tier icon	Add a new tier to the Business Application
Edit Tier icon	Available when you click a tier. Click the edit icon in a tier to edit the name and description of the tier.
Edit Business Application Details icon	Available when you click a Business Application. Click the edit icon in the Business Application to open the Business Applications Detail dialog box. Alternatively, click the DETAILS link next to the name of the Business Application.
Preview	Available for custom group and application objects when you have added objects to a tier. Click the preview icon after you have added an Application to a tier to see how services and objects within the application are related. This interactive element is displayed on the right pane. You can view links, switch to spherical view or list view. If you have too many objects to can use the filter option to find objects.

The **Select Members** pane is where you select objects to add to the Business Application. Drag an object to add to a tier in the canvas. Applications and custom groups can be nested under a Business Application. Custom groups can be added to applications and vice versa.

To find an object, search by name. Each object listed includes identifier information to help distinguish between objects of similar names.

Table 1686: Select Members Pane

Option	Description
Search	Available in the **Inventory** and **Custom Groups** tabs. Search for objects to add to the Business Application. Filter your search by clicking the filter icon. In the **Inventory** tab, by using the filter option, you can choose to see only objects which are Applications.
Inventory	Browse the inventory to select objects to add to the Business Application using drag and drop.
Custom Groups	Browse custom groups to add to the Business Application using drag and drop. You can nest custom groups or add them to the Business Application directly. You can add custom groups to Applications.

The Business Applications Details dialog box is where you add information to build out a model of your Business Application resources through applications and other infrastructure objects in your environment.

Table 1687: Business Application Details

Option	Description
Description	Provide a description for your Business Application.
Application Tag	Provide a tag for your Business Application.
Business Criticality	This is your Business Application business criticality. Select between Medium, Critical and Low. Default is Medium.
Environment	This is where your Business Application is deployed. Select from one of the following options: • Disaster Recovery • Development • Production • Staging • Test

What Are The Different Types of Telegraf Agents in VCF Operations?

The three types of Telegraf agents are product-managed agent, open source agent, and user-managed agent in VCF Operations.

- **Product-managed agent**: Managed by VCF Operations. You can install and uninstall (only for VMs managed by the vCenter Server cloud account), activate application services, and manage the life cycle of the Telegraf agents such as upgrade the agent, start the agent, stop the agent, and so on. For more details, see Additional Operations from the Manage Telegraf Agents Page .
- **Open source agent**: Refers to the open source Telegraf agent that you use to monitor curated and non-curated application services completely by yourself. You can install the open source Telegraf agent using a script and also configure it using the helper script to send data to VCF Operations. For more information, see Monitoring Applications using Open Source Telegraf on a Linux Platform and Monitoring Applications using Open Source Telegraf on a Windows Platform. You cannot manage the life-cycle of open source telegraf agents in VCF Operations, such as activate an application service, and stop/start upgrade. You can only view configured application instances and their data collection. Agent management is carried out on the target machine by the user.
- **User-managed agent**: User-managed agents were used to monitor physical servers in prior releases. Please note the following:

  - Available only if you upgrade from previous versions of VCF Operations. If you want to continue with the existing user-managed agents you can only view the metrics collected and you do not have to make any changes in VCF Operations.
  - Deprecated for fresh install. You will not be able to install new user-managed agents.
  - As a first time user if you want to monitor physical servers, use the product-managed Telegraf agent and use the helper script for installation of the Telegraf agent. For more information, see Install/Uninstall an Agent Using a Script on a Linux Platform and Install/Uninstall an Agent Using a Script on a Windows Platform

Monitoring Application Services using Product-Managed Telegraf

You can monitor application services in VCF Operations. You can use the product-managed agent to monitor physical servers and VMs that are managed or not managed by the vCenter cloud account (target machine).

For example, as an administrator, you might need to ensure that the infrastructure provided for running the application services is sufficient and that there are no problems. If you receive a complaint that a particular application service is not working properly or is slow, you can troubleshoot by looking at the infrastructure on which the application is deployed. You can view important metrics related to the applications and share the information with the team managing the applications.

Product-Managed Telegraf on Different Types of Machines

The following (object types) are now supported for application monitoring based on the type of machine the product-managed Telegraf agent is running on.

- `Virtual Machine`: Refers to the vCenter. All the capabilities of product-managed Telegraf agents and monitoring of the vCenter VMs remain the same from previous releases. For more information, see Install and Uninstall a Telegraf Agentand Additional Operations from the Manage Telegraf Agents Page .
- `Endpoint`: VCF Operations additionally supports the following types of machines that are categorized under an object type called `Endpoint`:
  - Physical servers
  - Machines that are not directly controlled by VCF Operations VMs that are not managed by the vCenter cloud account.

    To install or uninstall a product-managed Telegraf agent on a remote machine or a physical server, use the helper script. For more information, see Install/Uninstall an Agent Using a Script on a Linux Platform and Install/Uninstall an Agent Using a Script on a Windows Platform. To manage the life cycle of agents on a remote machine or a physical server, use the VCF Operations UI or suite-api. For more information, see Additional Operations from the Manage Telegraf Agents Page .

  In the application monitoring documentation for product-managed Telegraf agents, the term **target machine/s** refers to physical servers and the VMs that are managed and not managed by the vCenter cloud account.

High Availability for Application Monitoring

You can now use application monitoring high availability activated collector groups in failover mode to monitor applications from the application that is monitored even if your cloud proxy is down or if any of the cloud proxy components fail. Here are some important points:

- Cloud proxies in an application monitoring high availability activated collector group work in a primary-secondary mode.
- You cannot convert an existing Telegraf agent on all target machines, to high availability. To convert an existing target machine to high availability, you must install or re-install the agent with an application monitoring high availability activated collector group.
- You will lose all application data when you add a cloud proxy used by Telegraf agents, to the application monitoring high availability activated collector group.
- Failover or failback has a downtime of three collection cycles.
- Addition or removal of a cloud proxy from the application monitoring high availability activated collector has a downtime of three collection cycles.

Steps to Activate Application Monitoring High Availability on Collector Groups

Here are the steps to activate application monitoring high availability on collector groups and use it for application monitoring and data collection:

1. Activate high availability for specific cloud proxies when you add a collector group. For more information, see Adding a Collector Group.
2. (Optional) You can view the cloud proxies that are configured for high availability from the **Cloud Proxies** page. For more information, see Monitoring the Health of Cloud Proxies.
3. While installing an agent, select a collector group that has application monitoring high availability activated. For more information, see Install an Agent from the UI.
4. (Optional) You can view the following agent details such as:
   a. If high availability is activated for the collector group (Collector Group column).
   b. The virtual IP configured on an application monitoring high availability activated collector group to which all target machines send data (Virtual IP details column).
   c. The IP address of the primary cloud proxy that is collecting data (Cloud Proxy column).

   You can view these details from the **Manage Telegraf Agents** page (in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, select **Manage Telegraf Agents**). For more information, see Additional Operations from the Manage Telegraf Agents Page .

Upgrading to VCF 9.1

- If you have upgraded to VCF 9.1, complete Telegraf agents update from the **Manage Telegraf Agents** tab. For more information, see Additional Operations from the Manage Telegraf Agents Page . You must complete updating Telegraf agents if you use:
  – An individual cloud proxy.
  – Cloud proxies that are a part of a non-HA collector group.
  – Cloud proxies that are a part of a HA-enabled collector group.
  – Cloud proxies that are a part of a load-balanced HA-enabled collector group.
- After you have updated all the Telegraf agents, you can perform other operations such as:
  – Edit/update of the collector group.
  – Reconfiguration of collector groups.
  – Agent management operations.

Load Balancing

In VCF 9.1 and later versions, load balancing for product managed agents is supported for all cloud proxies that are a part of HA-enabled collector groups.

Steps to Acivate Load Balancing for Existing Users

For users with an HA enabled collector group, complete the following steps:

1. Content Update
   – As an existing user, update Telegraf agents that are at a lower version. For more information, see Additional Operations from the Manage Telegraf Agents Page . This operation is applicable for all product managed agents.
2. Activate load balancing in the HA-enabled collector group settings by selecting the **Activate for Load Balancing** option. For more information, see Adding a Collector Group.

For users with load balanced enabled collector group, complete the following steps:

1. Content Update
   – As an existing user, update Telegraf agents that are at a lower version. For more information, see Additional Operations from the Manage Telegraf Agents Page This operation is applicable for all product managed agents.
2. Reconfigure the collector group
   – As an existing user, reconfigure the collector group by using the **Retry Collector Group Configuration** option. For more information, see Configuring Collector Groups.

Load Balancing for New Users

As a new user, you can activate load balancing in the HA enabled collector group settings by selecting the **Activate for Load Balancing** option. For more information, see Adding a Collector Group.

Note:

If a cloud proxy is removed or added from a load balanced HA collector group, you must rebalance the collector group for load balancing to take place. For more information about rebalancing the collector group, see Configuring Collector Groups.

Consolidation of Ports

In VCF 9.1, port 8443 is deprecated and will reach end of support in a future version. It is recommended that you use port 443 so that the end point VM/physical server has access to port 443 of an individual cloud proxy or cloud proxies that are a part of a collector group.

- As a new user, by default, the end point VM/physical server connects to port 443 of an individual cloud proxy or cloud proxies that are a part of a collector group.
- For existing users who upgrade to VCF 9.1, it is recommended that you reinstall your open source Telegraf workloads to use port 443 instead of ports 8443 within the next few releases.
- Existing users must reconfigure the collector group and update product managed Telegraf agents as described below:
  - Reconfigure the collector group
    - To use port 443, for cloud proxies that are part of the HA enabled collector group, reconfigure the collector group by using the **Retry Collector Group Configuration** option. For more information, see Configuring Collector Groups. The server certificate is generated and the changes to enable mutual authentication will be carried out by cloud proxy.
  - Content Update
    - Update agents that are at a lower version. For more information, see Additional Operations from the Manage Telegraf Agents Page . This operation is applicable for all product managed agents.
- For individual cloud proxy/non-HA collector group.
  - Content update is required to enable mutual authentication. Update agents that are at a lower version. For more information, see Additional Operations from the Manage Telegraf Agents Page .

Supported Application Services

OS and Application monitoring helps virtual infrastructure administrators and application administrators to discover operating systems and applications running in provisioned guest operating systems at a scale and to collect run-time metrics of the operating system and application for monitoring and troubleshooting respective entities.

The following 23 application services are supported.

Application Services
Active Directory
Active MQ
Apache HTTPD
Cassandra
HyperV
Java Application
JBoss Server
MongoDB
MS Exchange
Microsoft IIS
Microsoft SQL Server
MySQL

Application Services
Network Time Protocol
Nginx
Oracle DB
Pivotal TC Server
PostgreSQL
RabbitMQ
Riak KV
SharePoint Server
Tomcat Server
Oracle WebLogic Server
WebSphere

VeloCloud Application Services

Eight VeloCloud application services are supported in VCF Operations. The supported application services are listed here. Some of the application services have mandatory properties which you must configure. Some of the application services have pre-requirements that you must configure first. After you configure the properties, data is collected. The following services are supported in VCF Operations:

Application Services
VeloCloud Orchestrator
Nginx
Clickhouse
Network Time Protocol
MySQL
Redis
Java Application
VeloCloud Gateway

Supported Platforms

VCF Operations supports monitoring for the following platforms and app combinations with API support.

Platforms Supported by VCF Operations for OS and Application Monitoring

Platform	Version	Architecture	Application
Red Hat Enterprise Linux	8.x 9.x	64-bit	OS Metrics and all supported applications.
CentOS	8.x	64-bit	OS Metrics and all supported applications.
Oracle Linux	8.x 9.x	64-bit	OS Metrics and all supported applications.
Rocky Linux	8.x 9.x	64-bit	OS Metrics and all supported applications.

Platform	Version	Architecture	Application
Ubuntu	16.04 LTS 18.04 LTS 20.x	64-bit	OS Metrics and all supported applications.
VMware Photon Linux	1.0 2.0 3.0 4.0 5.0	64-bit	Only OS metrics and custom monitoring actions are supported for photon OS. VCF Application Remote Collector 8.3, 8.2, 8.1, and 7.5 runs on Photon 1.0. Site Recovery Manager 8.2 runs on Photon 2.0 vSphere- vSphere 6.7 & 6.5 runs on Photon OS 1.0 vSAN 6.7 & vSAN 6.5 runs on Photon OS 1.0 Unified Access Gateway 3.7 runs on Photon 3.0 & 3.6 runs on Photon 2.0.
SUSE Linux Enterprise Server	12.x 15.x	64-bit	OS Metrics and all supported applications.
Windows	Windows 2012 Windows Server 2012 R2 Windows Server 2016 Windows Server 2019 Windows Server 2022	64-bit	OS Metrics and all supported applications.

Prerequisites

To monitor your application services and operating systems, complete all the prerequisites so that cloud proxy can communicate successfully with the target machines.

Port Information and Communication with the Target Machines

Port information and communication details for installing the Telegraf agent from the UI or via the helper script are detailed on this page.

Ensure that the cloud proxy's ports are reachable from target machines both via the IP and FQDN.

For the latest port information, see VMware Ports and Protocols.

Figure 486: Port Information and Communication with the vCenter, VMs, VCF Operations, and Cloud Proxy (Agent Install from the UI)

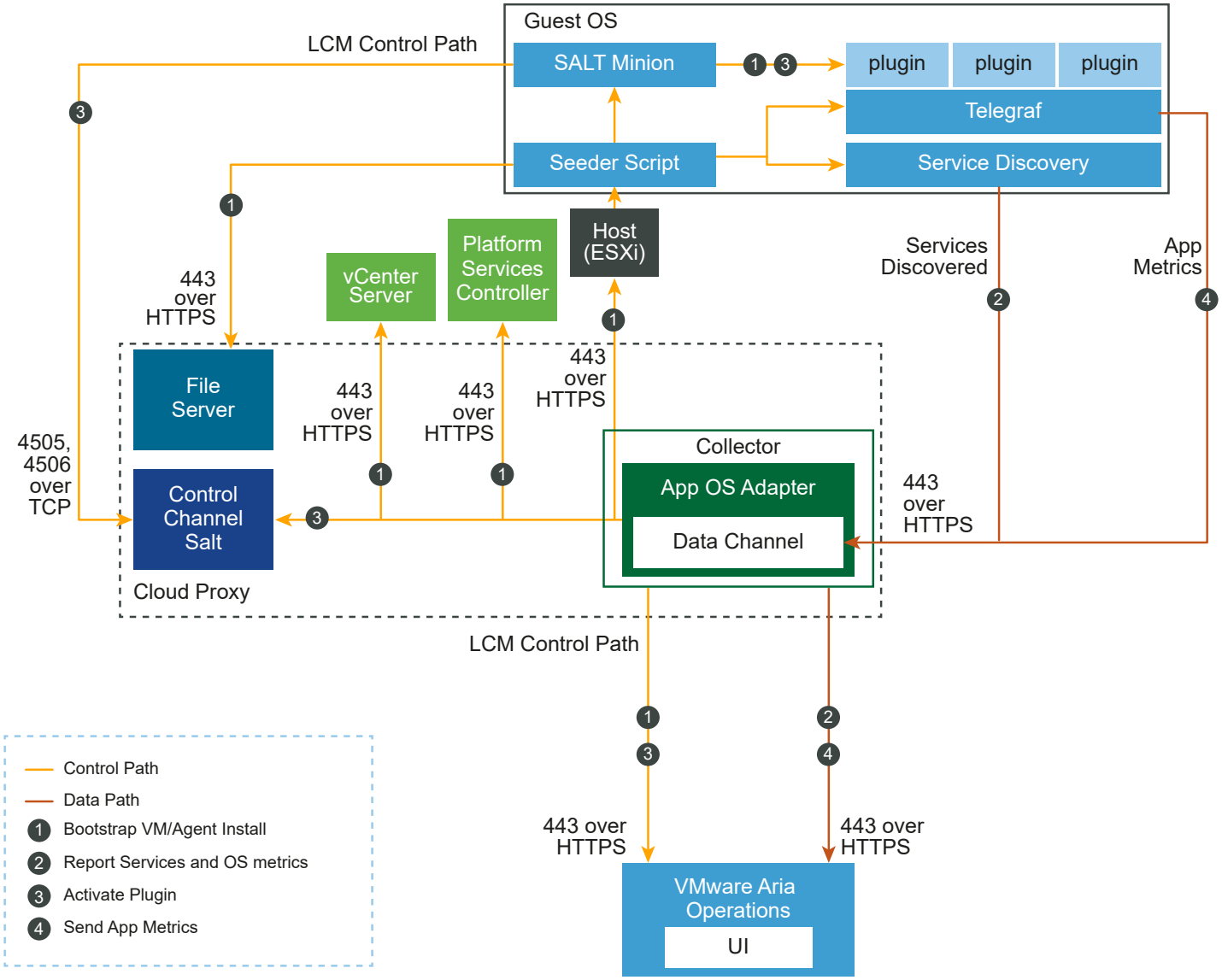
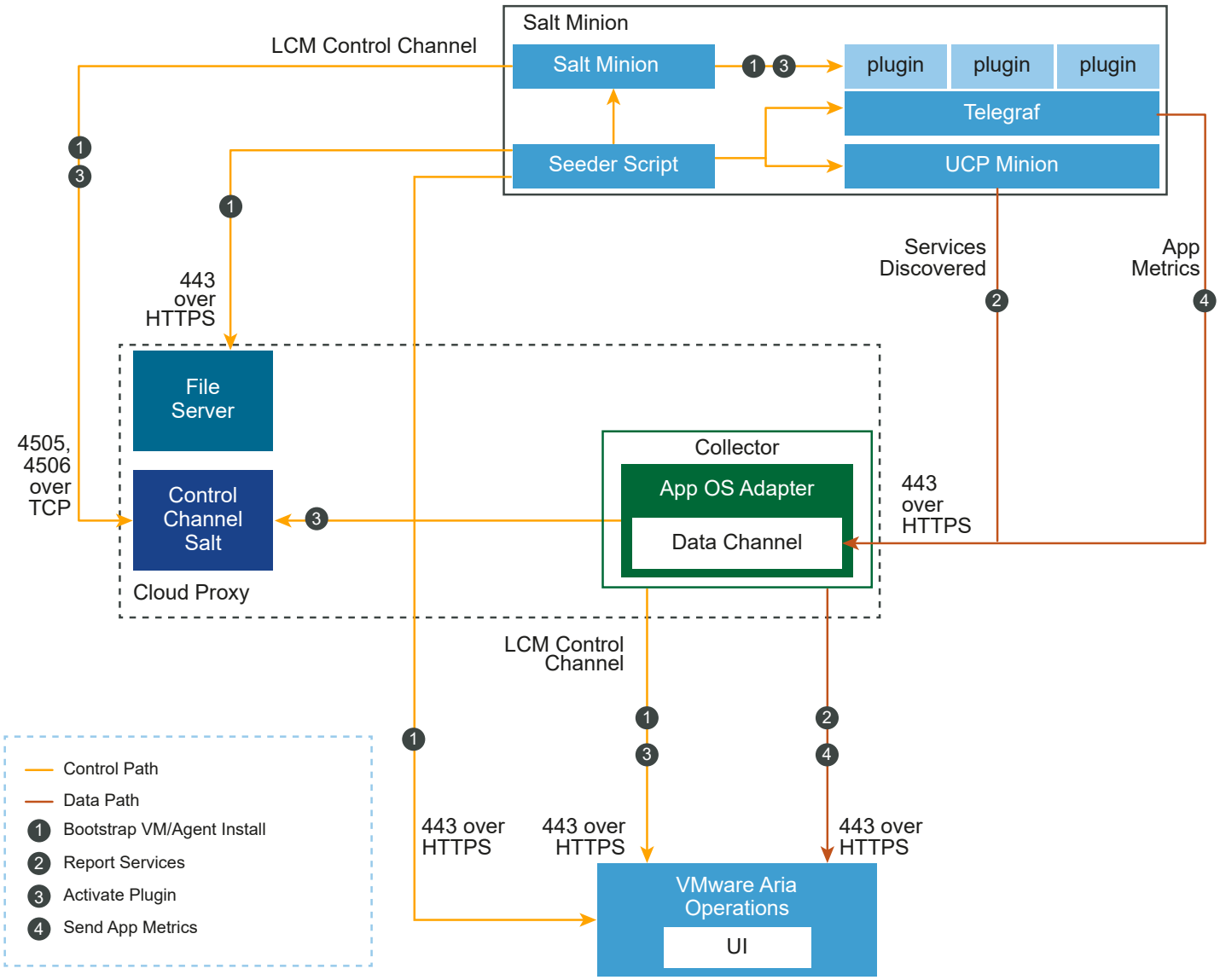


Figure 487: Port Information and Communication with the Target Machines and VMs for Script-Based Agent Install



Communication with Cloud Proxy and vCenter

Complete all the prerequisites required so that cloud proxy can communicate with vCenter. In the case of application monitoring high availability activated collector groups, these prerequisites are applicable for cloud proxies that are a part of the collector group.

> **Note:**
> Applicable to VMs managed by the vCenter cloud account.

- Port 443 in the ESX where the workload VMs are deployed must be accessible to cloud proxy.
- Port 443 in vCenter is accessible to cloud proxy.
- Configure a vCenter adapter. The vCenter user should have read access at the vCenter level and should also have the following permissions: `Guest operation modifications`, `Guest operation program execution`, and `Guest operation queries`.

> **Note:**
> For script-based agent install and uninstall, the `Guest operation modifications`, `Guest operation program execution`, and `Guest operation queries` permissions are not required.

Communication with Cloud Proxy and Target Machines

Complete the prerequisites required for the cloud proxy handshake with target machines.

For the handshake of cloud proxy with the target machines, the following prerequisites are required:

- The target machine where Telegraf should run, cloud proxy, and VCF Operations should be time synchronized. In case of VMs managed by vCenter cloud account, the ESX instance where the VM is deployed, vCenter, VMs, VCF Operations and cloud proxy should be time synchronized.
- The target machines must have access to ports 443, 4505, and 4506 on cloud proxy and on the virtual IP of the application monitoring high availability activated collector group. You can verify access by running the following commands on the target machine:
  - For Linux Machines (the "echo $?" command should return "0"):

    ```
    timeout 10 bash -c "</dev/tcp/{cloudproxy_fqdn_or_virtual_IP}/4505"
    echo $?
    timeout 10 bash -c "</dev/tcp/{cloudproxy_fqdn_or_virtual_IP}/4506"
    echo $?
    timeout 10 bash -c "</dev/tcp/{cloudproxy_fqdn_or_virtual_IP}/443"
    echo $?
    ```

  - For Windows Machines:

    ```
    wget.exe --spider -t 1 -T 10 {cloudproxy_fqdn_or_virtual_IP}:4505
    wget.exe --spider -t 1 -T 10 {cloudproxy_fqdn_or_virtual_IP}:4506
    wget.exe --spider -t 1 -T 10 {cloudproxy_fqdn_or_virtual_IP}:443
    ```

    Note:
    If you do not have `wget.exe` on the Windows machine, navigate to the `%temp%` folder or it's parent folder in File Explorer and search for `wget.exe` after you attempt installation.

    Note:
    In the above commands, use virtual IP in the case of application monitoring high availability activated collector groups. For individual cloud proxy or cloud proxy which belongs to application monitoring high availability deactivated collector groups, use the cloud proxy FQDN.
- The necessary privileges for a user which are required for agent installation are mentioned in the User Account Prerequisites page.
- Target machine configuration requirements.
  - Linux requirements
    Commands: `/bin/bash, sudo, tar, awk, curl`
    Packages: `coreutils (chmod, chown, cat), shadow-utils (useradd, groupadd, userdel, groupdel), net-tools`
    Configure mount point on `/tmp` directory to allow script execution.
  - Windows requirement
    - The Visual C++ version must be higher than 14.
    - Performance monitors on a Windows OS VM must be activated.
  - Windows 2012 R2 requirement
    The target machine must be updated with the Universal C Runtime. Refer to the following link for more information.
- VMware Tools must be installed and running on the vCenter VM on which you want to install the agent. For information about supported VMware Tools versions, click this link.

  Note:
  Applicable only to VMs managed by the vCenter cloud accounts.
- Add a `tmp` folder with "exec" permission to install agents on the latest UAG Photon OS VM. To configure a mount point on the `/tmp` directory to allow script execution, run the following command: `<mount -o remount,exec /tmp>`

User Account Prerequisites

There are certain user account prerequisites required when installing agents.

Windows Target Machine User Account Requirements

- To install agents,
  - The user must be either an administrator, or
  - A non-administrator who belongs to the administrator group.

Linux Target Machine User Account Requirements

For Linux target machines, there are two user accounts for the Telegraf agent, such as the install user and the run-time user. User credentials which are provided during agent installation, are for the install user. The *arcuser* is a run-time user and needs a set of privileges which are necessary for the agent's components to run.

- `/tmp` mount point should be mounted with exec mount option.
- The following are minimal necessary permissions of the user to install agents and should be mentioned in `sudoers` file:
  For example, for a user called **telegrafinstall**, you can find the `sudoers` file in the `/etc/sudoers` file or in the folder `/etc/sudoers.d/`:

```
Defaults:telegrafinstall !requiretty

Cmnd_Alias ARC_INSTALL_USER_COMMANDS=/usr/bin/cp*,/bin/cp*,/usr/bin/mkdir*,/bin/mkdir*,/usr/bin/chmod*,/
bin/chmod*,/opt/vmware/ucp/bootstrap/uaf-bootstrap.sh,/opt/vmware/ucp/ucp-minion/bin/ucp-minion.sh

telegrafinstall ALL=(ALL)NOPASSWD: ARC_INSTALL_USER_COMMANDS
```

Run-Time User Prerequisites

There are two ways in which a run-time user is created in Linux target machines: automatically and manually. A run-time user has a standard name and group, which is the *arcuser* and *arcgroup* respectively. If the **Create run time user on linux virtual machines, with required permissions as part of agent installation** check box is selected, the *arcuser* and *arcgroup* are created automatically. The check box is selected by default. A run-time user is also created automatically during a script-based install. If you choose to manually create the *arcuser* and *arcgroup*, here are the steps to do it manually:

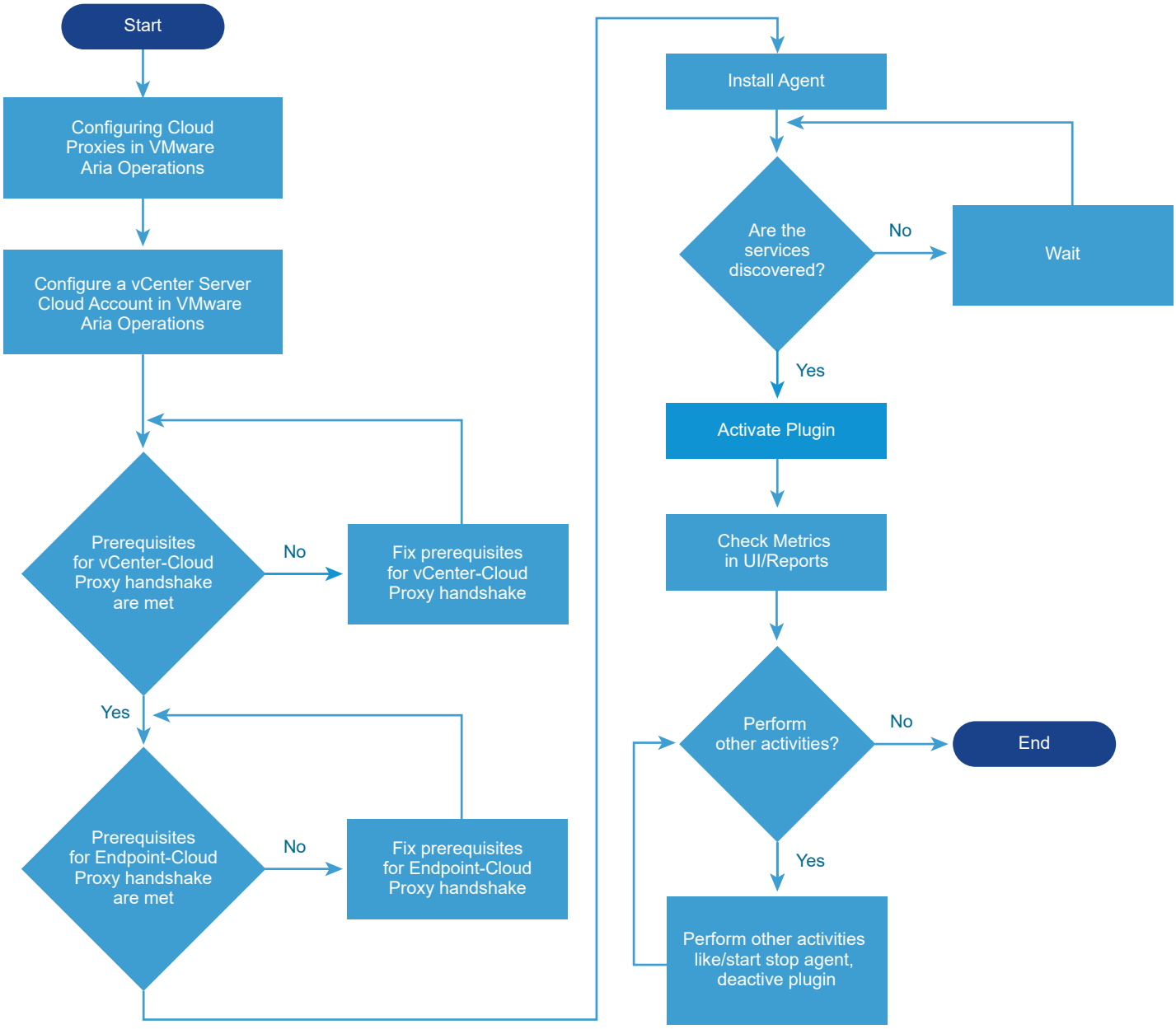
Create the *arcgroup* and *arcuser* and associate the *arcgroup* as the primary group of the *arcuser*.

1. The *arcgroup* must be the primary group of the *arcuser*.
   The following commands can be used to create the *arcgroup* and *arcuser*:
   ```
   groupadd arcgroup
   useradd arcuser -g arcgroup -M -s /bin/false
   ```
2. The *arcuser* must be created with no home directory and no access to the login shell.
   For example, the `/etc/passwd` entry for the *arcuser* is as follows after adding *arcuser* and *arcgroup*.
   ```
   arcuser:x:1001:1001::/home/arcuser:/bin/false
   ```
3. The **arcuser** must have password-less specific set of privileges as mentioned below, which must be written in **/etc/sudoers** file or in the folder **/etc/sudoers.d/**:
   ```
   Defaults:arcuser !requiretty
   Cmnd_Alias VAPCOMMANDS=/usr/bin/systemctl * ucp-telegraf.service, !/usr/bin/systemctl * * ucp-
   telegraf.service, /bin/systemctl * ucp-telegraf.service, !/bin/systemctl * * ucp-telegraf.service, /usr/
   bin/systemctl * ucp-minion.service, !/usr/bin/systemctl * * ucp-minion.service, /bin/systemctl * ucp-
   minion.service, !/bin/systemctl * * ucp-minion.service, /usr/bin/systemctl * salt-minion.service, !/
   usr/bin/systemctl * * salt-minion.service, /bin/systemctl * salt-minion.service, !/bin/systemctl * *
    salt-minion.service, /usr/bin/systemctl * ucp-salt-minion.service, !/usr/bin/systemctl * * ucp-salt-
   minion.service, /bin/systemctl * ucp-salt-minion.service, !/bin/systemctl * * ucp-salt-minion.service, /
   usr/bin/netstat, /bin/netstat, /opt/vmware/ucp/tmp/telegraf_post_install_linux.sh, /opt/vmware/ucp/
   bootstrap/uaf-bootstrap.sh, /opt/vmware/ucp/content/runscript.sh, /opt/vmware/ucp/ucp-minion/bin/ucp-
   minion.sh, /usr/bin/systemd-run, /bin/systemd-run
   arcuser ALL=(ALL) NOPASSWD: VAPCOMMANDS
   ```

Steps to Monitor Applications

You can monitor and collect metrics for your application services and operating systems.

The following flowchart describes how you can set up VCF Operations for application monitoring in the case of VMs managed by the vCenter cloud account.



Follow these steps to monitor applications.

1. Configure cloud proxy on which the OS and Application adapter instance is created.
   For more information, see Add a Cloud Proxy.
2. Configure a vCenter cloud account.

For more information, see Configuring a vCenter Account in VCF Operations.

Note:
This step is not applicable for target machines.

3. Complete all the prerequisites.
   For more information, see Prerequisites.
4. Install agents on selected target machines.
   For more information, see Install an Agent from the UI or Install/Uninstall an Agent Using a Script on a Linux Platform and Install/Uninstall an Agent Using a Script on a Windows Platform.
5. Activate an application service.
   For more information, see Activate an Application Service.
6. View the summary of application services and operating systems that are discovered in VCF Operations.
   For more information about monitoring your applications in VCF Operations, see Summary of Discovered and Supported Operating Systems and Application Services .

Install and Uninstall a Telegraf Agent

You can install and uninstall Telegraf agents on VMs managed by the vCenter cloud account from the user interface of VCF Operations or by running a script on physical servers and on machines not managed by VCF Operations.

Install an Agent from the UI

You must select the vCenter VMs on which you want to install the agent. All the vCenter VMs of the vCenter cloud account are listed in the **Manage Telegraf Agents** page.

Note:
You cannot install an agent from the VCF Operations user interface, on physical servers and target machines (Discovered in VCF Operations as the `Endpoint` object type).

Ensure that you have completed all the prerequisites. For more information, see Prerequisites.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, select **Manage Telegraf Agents**. You see the **Manage Telegraf Agents** page.
2. From the **Manage Telegraf Agents** page, select the VMs managed by the vCenter cloud account on which you want to install the agent.
3. From the **Manage Telegraf Agents** page, click **Agent Actions**, and then click **Install**.
4. If the VCF Operations Application Monitoring Adapter instance has not been created for the specific vCenter, you see the **Installing Telegraf Agent** dialog box. You can configure high availability for application monitoring via collector groups:
   - Select **Cloud Proxy Groups, with High-Availability** if you want to associate an application monitoring high availability activated collector group to the vCenter. You can choose the collector group from the **HA Enabled Group** drop-down list, or
   - Select **Cloud Proxies, without High-Availability** if you want to associate a cloud proxy to the vCenter. You can choose a cloud proxy from the **Cloud Proxy** drop-down list.
   - Click **Done**.

   Note:
   If the vCenter is on the default collector, the cloud proxy options are blank. If the vCenter is on a specific cloud proxy, the cloud proxy option is automatically populated. To deploy the VCF Operations Application Monitoring Adapter instance on a different cloud proxy, select another cloud proxy or application monitoring high availability activated collector group.

5. You see the **Manage Agent** dialog box.
6. From the **How do you want to provide VM Credentials** page, complete the following steps:
   a) If you have a common user name and password for all selected VMs, select the **Common username and password** option.
   b) If you have different user names and passwords for all selected VMs, select the **Enter virtual machine credentials** option.
   c) Under the section called **Telegraf Configurations**, select the **Telegraf agent configurations - Use backup of configurations if available** option if you want to restore the configuration details that were backed up for available agents on the VM, during a previous uninstall of the agents. For more information see, Uninstall an Agent.

      Note:
      Some agents such as Active MQ, Java Application, JBoss Server, Pivotal TC Server, Tomcat Server, Oracle WebLogic Server, and Riak KV might not work after you upgrade VCF Operations from a previous version to 9.x and choose to restore the agent configurations. As a workaround, after you restore agent configurations during the install, select the agents that were restored after you upgraded VCF Operations to 9.x and update the agents by selecting the **vertical ellipsis** > **Update** against the agent.
   d) Click **Next**.
7. From the **Provide Credentials** page, depending on whether you have a common credential for all VMs managed by the vCenter cloud account or different credentials for all VMs managed by the vCenter cloud account, enter the following details:
   a) If the selected VMs have a common user name and password, enter the common user name and password.
   b) For different user names and passwords for each VM, download the CSV template and add the required details such as the user name, password for each VM managed by the vCenter cloud account. Use the **Browse** button to select the template.
   c) The **Create run time user on Linux virtual machines, with required permissions as part of agent installation** check box is selected by default. For more information, see User Account Prerequisites.
   d) Click **Next**.
8. From the **Summary** page, you can view the list of VMs managed by the vCenter cloud account on which the agent is to be deployed.
9. Click **Install Agent**. Refresh the UI to view the agents that are installed.

   On User Account Control (UAC) deactivated Windows VMs managed by the vCenter cloud account, the Telegraf agent installation completes with the **Agent running** agent status and **Install Success** Last operation status, if all prerequisites are performed.

   UAC Activated on Windows VMs Managed by the vCenter Server Cloud Account

   In the case of UAC activated VMs, for a non-administrator user who belongs to the administrator group, Telegraf agent installation completes with the **Not Started** agent status and **Download Success** last operation status, if all the prerequisites are performed. The bits are downloaded to the VM. You must manually install the bits. From `C:\VMware\UCP\downloads`, run a bootstrap launcher, and then the agent status becomes **Agent Running** and the Last operation status becomes **Install Success**.

   - Open PowerShell with administrator privileges.
   - Go to `$SYSTEMDRIVE\VMware\UCP\downloads` folder: `cd $SYSTEMDRIVE\VMware\UCP\downloads`
   - Run the `cmd /c uaf-bootstrap-launcher.bat > uaf_bootstrap.log 2>&1` command.
   - View the results from the command in `uaf_bootstrap.log`.
   - Verify the status of agent installation from the **Agent Status** and **Last Operation Status** columns in the **Manage Telegraf Agents** page.
10. The agent discovers the application services that are installed on the target machines. You can view the application services by clicking the drop-down arrow against the VMs managed by the vCenter cloud account on which the agent

is installed in the **Manage Telegraf Agents** page. You can view the status of agent installation from the **Agent Status** column in the **Manage Telegraf Agents** page.

11. After installation, an OS object will be created under VMs managed by the vCenter cloud account topology. To view the object, select a VM from **Applications** > **Manage Telegraf Agents** page. Click the vertical ellipsis and select **Go to Details**, and click the **Metrics** tab. Expand the **Object relationship** widget, and select the OS object.
12. If agent installation fails, a detailed error message is displayed when you hover over the red exclamation icon before the Install Failed text in the **Last operation status** column. You can take appropriate actions based on the error message to resolve the agent installation issues. For troubleshooting information, see Troubleshooting Agent Installation.

You can manage the services on each agent.

For information about uninstalling an agent, see Uninstall an Agent.

Install/Uninstall an Agent Using a Script on a Linux Platform

You can install or uninstall an agent on a target machine using a script.

- Complete all the prerequisites. For more information, see Prerequisites.
- The unzip package must be available on the target machine.
- The user must have access permissions to the download folder.
- The guest IPs must be properly configured and match only one VM monitored by VCF Operations. If there are multiple VMs with the same guest IPs monitored by VCF Operations, the script may not generate a proper mapping.

  Note:

  This prerequisite is applicable only to the VMs managed by the vCenter cloud account. For more information about object types, see Monitoring Application Services using Product-Managed Telegraf.
- The VM managed by the vCenter cloud account must be available in VCF Operations.

  Note:

  This prerequisite is applicable only to the VMs managed by the vCenter cloud account.
- The cloud account must be configured for the vCenter to which the VM belongs.

  Note:
  This prerequisite is applicable only to the VMs managed by the vCenter cloud account. For more information about object types, see Monitoring Application Services using Product-Managed Telegraf.
- Port 443 in VCF Operations must be accessible to the target machine.
- The VCF Operations user must have the following permissions:
  - Administration > Control Panel > REST-APIs > All other Read, Write APIs
  - Administration > Control Panel > REST-APIs > Read access to APIs
  - Workload Operations > Applications > Manage Telegraf Agent
  - Inventory > Actions > Bootstrap virtual machines
  - Inventory > Actions > Download bootstrap

  Note:

  To check permissions, navigate to **Operate** > **Administration** > **Control Panel** > **Access Control** > **Roles**, select the role to which the user is assigned. You can view the permissions on the right side at the end of the page.
- Only IPv4 is supported at present for cloud proxy.
- VMTools version >=10.2.

  Note:

  Applicable only to VMs managed by the vCenter cloud account.

1. Log in to the target machine on which you want to install/uninstall the agent and download the sample script from cloud proxy from the following location: `https://<CloudProxy>/downloads/salt/telegraf-utils.sh`.

   Run one of the following commands:

   ```
   wget --no-check-certificate https://<CloudProxy>/downloads/salt/telegraf-utils.sh
   ```

```
curl -k "https://<CloudProxy>/downloads/salt/telegraf-utils.sh" --output telegraf-utils.sh
```

Note:
Use the relevant cloud proxy **IP address** for <CloudProxy> in the preceding commands and location specified. For application monitoring high availability activated collector groups, provide the virtual IP.

2. Make the script executable by running the following command:

```
chmod +x telegraf-utils.sh
```

3. To execute the script and install/uninstall the agent, run the following command:

```
telegraf-utils.sh  product-managed -c <cloud_proxy_ip_or_collector_group_name> -t <token> (-v
 vmwarearіaoperations_ip_or_fqdn | [-g gateway_url -a csp_auth_url]) [-d download_tmp_dir -s sleep_seconds
 -i list_of_IPs -o operation] -k skip_certificate -h certificate_path -b uuid -r  retain_telegraf_config

Description of arguments:
      -c : [REQUIRED] cloud_proxy_ip_or_collector_group_name - Cloud Proxy IP or Collector Group Name
      -t : [REQUIRED] token - This can be on-prem vmwarearіaoperations_auth_token.
            on-prem: Auth Token of the user/account. For getting new token
                     ( https://<VMwareAriaOperations_IP>/suite-api/ or curl -ks -X POST https://
<VMwareAriaOperations_IP>/suite-api/api/auth/token/acquire -H \"Content-Type: application/json\"
 -H \"Accept: application/json\" -d \"{\"username\":\"<VMwareAriaOperations_USER>\",\"password\":
\"<VMwareAriaOperations_USER_PASSWORD>\"}\")
      -d : [OPTIONAL] download_tmp_dir - Temporary directory for agent installation. Default: current
 directory
      -v : [CONDITIONAL] [ON-PREM-SPECIFIC] vmwarearіaoperations_ip_or_fqdn - IP/FQDN of VMware Aria
 Operations master node and required for on-prem.
      -o : [OPTIONAL] operation - The type of operation to be performed {install,uninstall} Default:
 install
      -s : [OPTIONAL] sleep_seconds - Time (in seconds) to delay the script execution. This optional
 parameter will be helpful when this script is used in vRA to have agent installed on deploying VM.
 Recommended time 600 seconds.
      -i : [OPTIONAL] list_of_IPs - This is an optional parameter. Comma separated IPs that are set as
 properties of a VM in VMware Aria Operations (VM -> Properties -> Network -> <Integer> -> IP Address). If
 single adapter has multiple IP Addresses, then delimit them using '_'
                    ex:- If your VM has 3 adapters with IP Address as follows: Adapter1: 10.0.0.1
 Adapter2: 10.0.0.2 & 10.0.0.3 Adapter3: 10.0.0.4
                    Then this parameter should be given as "10.0.0.1,10.0.0.2_10.0.0.3,10.0.0.4"
      -k : [MANDATORY]  skip_certificate - To skip certificate verification when downloading certificate
 bundle from VCF Operations. Accepted values : {1,0}.  1 indicates skip and 0 indicates provide CA
 certificate part of -h argument.
      -h : [CONDITIONAL]  certificate_path - Path of the certificate used for SSL verification. This is
 mandatory field if skip_certificate is set to 0.
      -b : [OPTIONAL] uuid - Overrides VM/Endpoints identifier. If not provided, will use the system
 generated identifier.
      -r : [OPTIONAL] retain_telegraf_config - To retain the available telegraf plugins configuration    -
 Accepted values : {1,0}. Default : 1. 1 indicates true and 0 indicates false

Example: /bin/bash telegraf-utils.sh product-managed -t 8dab02cc-277c-4392-b910-
bd2e98c7e741::8bcde100-6318-44d7-a8dc-11f4ff84b3b -k 0 -h /root/cert.pem -v 10.10.10.100 -c 10.10.10.101
```

Note:
The `-c` argument is mandatory to run the helper script.

To verify the bootstrap status, view the `uaf-bootstrap-results` file. If the installation fails, look for error messages in `uaf_bootstrap.log`.

If the script is successful, the agent status will be updated in the **Manage Telegraf Agents** tab after one collection cycle that takes 5–10 minutes.

Note:
When you use an automation script, concurrent agent installation with a batch size of 20 is supported.

Note:

It can take up to two collection cycles for endpoints monitored by the installed Telegraf agents to appear in the **Inventory** section of VCF Operations. It can take up to four collection cycles for installed Telegraf agents to appear in the **Manage Telegraf Agents** section of VCF Operations.

Install/Uninstall an Agent Using a Script on a Windows Platform

You can install an agent on a target machine using a script.

- Complete all the prerequisites. For more information, see Prerequisites.
- The unzip command must be available on the target machine.
- The user must have access permissions to the download folder.
- The cloud account must be configured for the vCenter to which the VM belongs.

  Note:
  This prerequisite is applicable only to the VMs managed by the vCenter cloud account.
- The VM managed by the vCenter cloud account must be available in VCF Operations.

  Note:

  This prerequisite is applicable only to the VMs managed by the vCenter cloud account.
- The guest IPs must be properly configured and match only one VM monitored by VCF Operations. If there are multiple VMs with the same guest IPs monitored by VCF Operations, the script may not generate a proper mapping.

  Note:

  This prerequisite is applicable only to the VMs managed by the vCenter cloud account. For more information about object types, see Monitoring Application Services using Product-Managed Telegraf.
- Port 443 in VCF Operations must be accessible to the target machine.
- The VCF Operations user must have the following permissions:
  - Administration > Control Panel > REST-APIs > All other Read, Write APIs
  - Administration > Control Panel > REST-APIs > Read access to APIs
  - Operations > Applications > Manage Telegraf Agent
  - Inventory > Actions > Bootstrap virtual machines
  - Inventory > Actions > Download bootstrap

  Note:
  To check permissions navigate to **Operate** > **Administration** > **Control Panel** > **Access Control** > **Roles**, select the role to which the user is assigned. You can view the permissions on the right side at the end of the page.
- Only IPv4 is supported at present for cloud proxy.
- Windows PowerShell must be >= 5.0.
- VMTools version must be >=10.2.

  Note:
  Applicable only to VMs managed by the vCenter cloud account.

1. Log in to the target machine on which you want to install/uninstall the agent and download the sample script from cloud proxy from the following location: `https://<CloudProxy>/downloads/salt/telegraf-utils.ps1`

   Note:

   Do not use a space in the configuration path where the `telegraf-utils.ps1` script is placed. Paths with spaces can be passed as a short name notation, such as `c:\PROGRA~1` for `c:\Program Files`.

   Run one of the following commands.

   ```
   PowerShell command:
   Invoke-WebRequest "https://<CloudProxy>/downloads/salt/telegraf-utils.ps1" -OutFile telegraf-utils.ps1
   ```

```
Or if you have the wget tool:
wget --no-check-certificate https://<CloudProxy>/downloads/salt/telegraf-utils.ps1
```

Note:

- Use the relevant cloud proxy **IP address** for <CloudProxy> in the preceding commands and location specified. For application monitoring high availability activated collector groups, provide the virtual IP.
- For troubleshooting information, see Script Download Fails on a Windows Platform.

If the script download fails with the following message: `The request was aborted: Could not create SSL/TLS secure channel`, follow the steps mentioned in Script Download Fails on a Windows Platform.

2. To execute the script and install/uninstall the agent, run the following command:

```
telegraf-utils.ps1  product-managed -c cloud_proxy_ip_or_collector_group_name -t token (-v
 vmwarearіaoperations_ip_or_fqdn | [-g gateway_url -a csp_auth_url]) [-d download_tmp_dir -s sleep_seconds
 -i list_of_IPs -o operation] -k skip_certificate -h certificate_path -b uuid -r  retain_telegraf_config

Description of arguments:
      -c : [REQUIRED] cloud_proxy_ip_or_collector_group_name - Cloud Proxy IP or Collector Group Name
      -t : [REQUIRED] token - This can be on-prem vmwarearіaoperations_auth_token.
            on-prem: Auth Token of the user/account. For getting new token
                     ( https://<VMwareAriaOperations_IP>/suite-api/ or curl -ks -X POST https://
<VMwareAriaOperations_IP>/suite-api/api/auth/token/acquire -H \"Content-Type: application/json\"
 -H \"Accept: application/json\" -d \"{\"username\":\"<VMwareAriaOperations_USER>\",\"password\":
\"<VMwareAriaOperations_USER_PASSWORD>\"}\")
      -d : [OPTIONAL] download_tmp_dir - Temporary directory for agent installation. Default: current
 directory
      -v : [CONDITIONAL] [ON-PREM-SPECIFIC] vmwarearіaoperations_ip_or_fqdn - IP/FQDN of VMware Aria
 Operations master node and required for on-prem.
      -o : [OPTIONAL] operation - The type of operation to be performed {install,uninstall} Default:
 install
      -s : [OPTIONAL] sleep_seconds - Time (in seconds) to delay the script execution. This optional
 parameter will be helpful when this script is used in vRA to have agent installed on deploying VM.
 Recommended time 600 seconds.
      -i : [OPTIONAL] list_of_IPs - This is an optional parameter. Comma separated IPs that are set as
 properties of a VM in VMware Aria Operations (VM -> Properties -> Network -> <Integer> -> IP Address). If
 single adapter has multiple IP Addresses, then delimit them using '_'
                    ex:- If your VM has 3 adapters with IP Address as follows: Adapter1: 10.0.0.1
 Adapter2: 10.0.0.2 & 10.0.0.3 Adapter3: 10.0.0.4
                    Then this parameter should be given as "10.0.0.1,10.0.0.2_10.0.0.3,10.0.0.4"
      -k : [MANDATORY]  skip_certificate - To skip certificate verification when downloading certificate
 bundle from VCF Operations. - Accepted values : {1,0}. 1 indicates skip and 0 indicates provide CA
 certificate part of -h argument.
      -h : [CONDITIONAL]  certificate_path - Path of the certificate used for SSL Verification. This is a
 mandatory field if skip_certificate (-k) is set to 0 or '-k' argument is not passed.
      -b : [OPTIONAL] uuid - Overrides VM/Endpoints identifier. If not provided, will use the system
 generated identifier.
      -r : [OPTIONAL] retain_telegraf_config - To retain the available telegraf plugins configuration -
 Accepted values : {1,0}. Default : 1

Example: .\telegraf-utils.ps1 product-managed -t 8dab02cc-277c-4392-b910-bd2e98c7e741::8bcde100-6318-44d7-
a8dc-11f4ff84b3b -k 0 -h /root/cert.pem -v 10.10.10.100 -c 10.10.10.101
```

Note:
The `-c` argument is mandatory to run the helper script.

To verify the bootstrap status, view the `uaf-bootstrap-results` file. If the installation fails, look for error messages in `uaf_bootstrap.log`.

If the installation is successful, the agent status will be updated in the **Manage Telegraf Agents** tab after one collection cycle that takes 5–10 minutes.

Note:
When you use an automation script, concurrent agent installation with a batch size of 20 is supported.

Note:

It can take up to two collection cycles for endpoints monitored by the installed Telegraf agents to appear in the **Inventory** section of VCF Operations. It can take up to four collection cycles for installed Telegraf agents to appear in the **Manage Telegraf Agents** section of VCF Operations.

Uninstall an Agent

You must select the VMs managed by the vCenter cloud account on which you want to uninstall the agent.

Note:
You cannot uninstall an agent from the UI for physical servers and VMs unmanaged by the vCenter cloud account (`Endpoint` object type in VCF Operations).

Ensure that you have completed all the prerequisites. For more information, see Prerequisites.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, select **Manage Telegraf Agents**. You see the **Manage Telegraf Agents** page.
2. From the **Manage Telegraf Agents** page, select the VMs that are managed by the vCenter cloud account on which you want to uninstall the agent.
3. From the **Manage Telegraf Agents** page, click the horizontal ellipsis, and then click **Uninstall**. You see the **Manage Agent** dialog box.
4. From the **How do you want to provide VM Credentials** page, complete the following steps:
   a) If you have a common user name and password for the selected VMs, select the **Common username and password** option.
   b) If you have different user names and passwords for all the VMs, select the **Enter virtual machine credentials** option.
   c) Under the section called **Telegraf Configurations**, select the **Telegraf agent configurations - Save a backup of configurations** option if you want to backup active agent configurations available on the VM. You can restore these agent configurations during install of an agent. For more information, see Install an Agent from the UI.
   d) Click **Next**.
5. From the **Provide Credentials** page, depending on whether you have a common credential for all the selected VMs or different credentials for all VMs, enter the following details:
   a) If your VM has a common user name and password, enter the common user name and password.
   b) For multiple user names and passwords for each VM, download the CSV template and add the details. Use the **Browse** button to select the template.
   c) Click **Next**.
6. From the **Summary** page, you can view the list of VMs on which the agent is uninstalled.
7. Click **Uninstall Agent**. Refresh the UI to view the progress of agent uninstallation.

   The **Agent Status** column and the missing drop-down arrow against the VM in the workspace, indicate that uninstallation is complete and that there are no application services discovered on each agent.

   UAC Activated on Windows Machine

   The bits are downloaded to the VM. You have to manually uninstall the bits.

   - Open PowerShell with administrator privileges.
   - Go to `$SYSTEMDRIVE\VMware\UCP\downloads` folder: `cd $SYSTEMDRIVE\VMware\UCP\downloads`
   - Run the `cmd /c uaf-bootstrap-launcher.bat > uaf_bootstrap.log 2>&1` command.

- View the results from `uaf_bootstrap.log`.
- Verify the status of agent installation from the **Agent Status** and **Last Operation Status** columns in the **Manage Telegraf Agents** page.

Activate an Application Service

To monitor application services running on the target machines, you must configure Telegraf agents in VCF Operations.

> **Note:**
> The maximum allowed application services count per target machine is 200.

After you have installed the agent, you can activate plugins to monitor application services. Additionally, you can monitor a discovered service on VMs managed or unmanaged by the vCenter cloud account and physical servers.

You can configure discovered services and carry out custom monitoring actions. For custom monitoring actions, see Custom Script, Activate Remote Checks, Monitor Windows Services, and Monitor Linux Processes.

Prerequisite

- If plugin activation requires the location of a file (for example, client certificates for SSL Trust) on the target machine, the location and the files should have appropriate read permissions for the *arcuser* to access those files. Configuration of application services and custom monitoring using Telegraf is referred to as plugins.

  > **Note:**
  > If the plugin displays a permission denied status, provide the *arcuser* with permissions to the file locations that you have specified during plugin activation.
- Linux process activation for PID files works only if the PID file and its parent directories have read permission for **Others**.

Activate an Application Service on a Single Target Machine

To monitor an application service on a single target machine, complete the following steps:

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, click **Manage Telegraf Agents**.
2. Select the target machine on which the agent is running. You can use the filter functionality too. For example, filter by **Agent Status** > **Agent Running**.
3. Expand the drop-down arrow against the target machine on which the agent is installed. You see the **Services Discovered** section.
4. From the **Services Discovered** section, select a discovered service or custom monitored service from the **Custom Monitoring** section, click the vertical ellipsis and then click **Add**.
5. Activate the application service from dialog box that is displayed on the right side.
6. Enter the details for each instance that you add and click **Save**. For configuration details of each application, see Configuring Supported Application Services.
   Fields with a star are mandatory.
   For more information about the status details that appear against the application services, see the table called Status Details in Additional Operations from the Manage Telegraf Agents Page .
   To edit or delete instances of application services, click the **Edit** or **Delete** options from the vertical ellipsis against application service you added. After the services have been added and saved, click the drop down arrow against the application service to view the list of services and their status.

Activate an Application Service on Multiple Target Machines

To monitor an application service on multiple target machines, complete the following steps:

1. In the VCF Operations navigation bar at the top, click **Operation** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, click **Manage Telegraf Agents**.
2. Select the target machines on which the agent is running where the agent type is `product-managed`. You can use the filter functionality too. For example, filter by **Agent Status** > **Agent Running** and **Agent Type** > **Product-managed agent**.

Note:
You can only activate an application service for all object types (Endpoint and Virtual Machine) with a product-managed Telegraf agent installed and running.

3. Click the **Horizontal Ellipsis** and then click **Activate Service**. You see the **Activate Service** dialog box.
4. From the **Activate Service** dialog box, select a discovered service or custom monitoring service by clicking the drop-down arrow in the **Select a Service** field and click **Confirm**.
5. Activate the application service from dialog box that is displayed on the right side.
6. Enter the details for each instance that you add and click **Save**. For configuration details of each application, see Configuring Supported Application Services.
   Fields with a star are mandatory.
   For more information about the status details that appear against the application services, see the table called Status Details in Additional Operations from the Manage Telegraf Agents Page .
   To edit or delete instances of application services, click the **Edit** or **Delete** options from the vertical ellipsis against application service you added. After the services have been added and saved, click the drop down arrow against the application service to view the list of services and their status.
   For custom monitoring actions, see Custom Script, Activate Remote Checks, Monitor Windows Services, and Monitor Linux Processes.

The following special characters are permitted in the DB user field: `'[]{} (),.<> ?:!|/~@#$%^&*- _ +=`

You can provide DB name lists in the following format `['DBNAME_1', 'DBNAME_2', 'DBNAME_3']` where DBNAME_1, DBNAME_2, DBNAME_3 must not contain quotes such as `'` and `"`.

Application Availability

When an application service is activated, the **Application Availability** metric is collected and displays if the application service is running on the target machine or if it is down. `1` indicates that the application service is running on the target machine and `0` indicates that the application service is down. This metric is available for all supported application services except JAVA application service.

For information about deactivating a service, see Deactivate an Application Service.

Configuring Supported Application Services

Twenty-three application services are supported in VCF Operations. The supported application services are listed here. Some of the application services have mandatory properties which you must configure. Some of the application services have pre-requirements that you must configure first. After you configure the properties, data is collected.

Active Directory

Active Directory is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

Active MQ

ActiveMQ is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Server URL	Yes	http://localhost:8161
User name	Yes	User name for Active MQ. Example: admin
Password	Yes	Password

Name	Mandatory?	Comment
Installed Path	Yes	The path on the target machine where Active MQ is installed. Example: For Linux target machines: `/opt/apache-activemq` For Windows target machines: `C:\apache-activemq-5.15.2`

Apache HTTPD

Apache HTTPD is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Status Page URL	Yes	http://localhost/server-status?auto
User name	No	User name for Apache HTTPD service. `Example:root`
Password	No	Password
Response String Match	No	Optional substring or regex match in the body of the response (case sensitive). For example, `ok`. You must escape regex special characters `?, (, ), .` with a backslash. For example, for matching against `?doodles`, escape the `?` with `\?`, such as, `\?doodles`.
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

Cassandra Database

Cassandra database is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the application instance.
URL	Yes	http://localhost:8778 For Cassandra Database prerequirements, see Cassandra.
Installed Path	Yes	Valid file path.

Hyper-V

Hyper-V is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the application service.

Java

Java is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Base URL	Yes	http://localhost:8080
Installed Path	Yes	The path on the target machine where Java is installed. Example: For Linux target machines: /opt/vmware/ucp ; For Windows target machines: C:\VMware\UCP
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

JBoss

JBoss is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Base URL	Yes	http://localhost:8080
Installed Path	Yes	The path on the target machine where JBoss is installed.
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

MongoDB

MongoDB is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Port	Yes	The port where MongoDB is running. Example: 27017
Hostname	No	Optional hostname for the MongoDB Service.
Username	No	User name for MongoDB. Example: Root
Password	No	Password
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

MS Exchange

MS Exchange is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

MS IIS

MS IIS is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

MS SQL

MS SQL is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Instance	Yes	Instance name of the MS SQL server.
Port	No	The port where MS SQL is running. Example: 1433
Hostname	No	Optional hostname for the MS SQL Service.
Username	Yes	User name for MS SQL. Example: Root
Password	Yes	Password

MySQL

MySQL is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Port	Yes	The port where MySQL is running. Example: 3306
User name	Yes	User name for MySQL service. Example: Root
password	Yes	Password
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Hostname	No	Optional hostname for the MySQL Service
Databases	No	Comma-separated list of databases to monitor. Each of the database names to be monitored must be enclosed in single quotes and the databases themselves should be comma separated. For example, 'database1','database2','database3'.
TLS Connection	No	Allowed values are true, false, and skip-verify.

Nginx

Nginx is supported in VCF Operations.

Table 1688:

Name	Mandatory?	Comment
Display Name	Yes	Display name of the application instance.
Status Page URL	Yes	http://127.0.0.1:8081/nginx_status
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain and host verification. Expected: True/False.

NTPD

NTPD is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

Oracle Database

Oracle database is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the application instance.
OracleDB Username	Yes	User name for the Oracle database instance.
OracleDB Password	Yes	Password for the Oracle database instance.
OracleDB SID	Yes	SID of the Oracle database instance.

Pivotal Server

Pivotal Server is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Base URL	Yes	http://localhost:8080
Installed Path	Yes	The path on the target machine where Pivotal server is installed.
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

Postgres

Postgres is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Port	Yes	The port where PostgreSQL is running. Example: 5432
User name	Yes	User name for PostgreSQL service. Example: Root
Password	Yes	Password
SSL Connection	No	Allowed values are deactivate, verify-ca, verify-full.

Name	Mandatory?	Comment
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: true/false.
Hostname	No	Optional hostname for the PostgreSQL Service.
Default Database	No	The database for initiating connection with the server
Databases	No	Comma-separated list of databases to monitor. Each of the database names to be monitored must be enclosed in single quotes and the databases themselves should be comma-separated, for example , 'database1','database2','database3'.
Ignored Databases	No	Comma-separated list of databases that need not be monitored. Each of the database names to be excluded from monitoring must be enclosed in single quotes and the databases themselves should be comma-separated for example, 'database1','database2','database3'.

RabbitMQ

RabbitMQ is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Management Plugin URL	Yes	http://localhost:15672
User name	No	User name for RabbitMQ. Example: Guest
Password	No	Password
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.
Nodes	No	Each of the RabbitMQ data collection nodes should be in single quotes and the nodes themselves should be comma-separated. The list of nodes must be enclosed in square brackets. For example ['rabbit@node1','rabbit@node2',.....]

Riak

Riak is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Server URL	Yes	http://localhost:8098

Sharepoint

Sharepoint is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

Tomcat

Tomcat is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Base URL	Yes	http://localhost:8080
Installed Path	Yes	The path on the target machine where Tomcat is installed.
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.
SSL Key	No	Path to the SSL Key file on thetarget machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

Weblogic

Weblogic is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Base URL	Yes	http://localhost:7001
Installed Path	Yes	The path on the target machine where WebLogic is installed.
User name	Yes	User name for WebLogic. Example: admin
Password	Yes	Password
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificate	No	Path to the SSL Certificate file on the target machine.

Name	Mandatory?	Comment
SSL Key	No	Path to the SSL Key file on the target machine.
Skip SSL Verification	No	Use SSL but skip chain & host verification. Expected: True/False.

Websphere

Websphere is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
IBM Websphere Server URL	Yes	Example : http://localhost:9081
Websphere Authorization Token	Yes	To generate the token, follow the below steps: • Go to https://www.base64encode.org. • Type in the user and password created in the format: user:password • Click the Encode button. • Copy the resulting Base64 encoded string. Example: d2F2ZWZyb250OndhdmVmcm9udA==

Custom Monitoring

HTTP Check

HTTP is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the remote check instance.
URL	Yes	http://localhost
Method	Yes	GET/POST/PUT
Proxy	No	Proxy URL: http://localhost
Response Timeout	No	Timeout for the connection in seconds. For example, 10.
Follow Redirects	No	True/False if redirects from the server. For example, true/false (all small values).
Body	No	HTTP request body.
Response String Match	No	Substring or regex match in the response body.
SSL CA	No	Path to the SSL CA file on the target machine.
SSL Certificates	No	Path to the SSL certificate file on the target machine.
SSL Key	No	Path to the SSL key file on the target machine.
Skip Host & chain verification	No	Use SSL but skip chain and host verification. Expected: True/False.

Name	Mandatory?	Comment
Headers	No	HTTP Request Headers, which can look like the following. For example: `accept = "application/json" Authorization = "5609fe9d-cddd-4654-898b-26d9b67f137a::4ca86c16-ac01-4e17-90da-fb55eb36571a"`

ICMP Remote Check

ICMP is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the remote check instance.
FQDN/IP	Yes	Host name to send the packets. Example: *example.org*
Count	No	Number of ping packets to send per interval. For example, 1.
Ping Interval	No	Time to wait between ping packets in seconds. For example, 10.0. **Note:** Follow the decimals as mentioned in the example.
Timeout	No	Timeout to wait for ping response in seconds. For example, 10.0. **Note:** Follow the decimals as mentioned in the example.
Deadline	No	The total ping deadline in seconds. For example, 30.
Interface	No	Interface or source from which to send a ping.

TCP Check

TCP Check is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the remote check instance.
Address	Yes	<hostname>:port
Send	No	The given string is sent to the TCP. It can be any string of your choice.
Expect	No	The given string is expected from the TCP. It can be any string of your choice.
Timeout	No	Timeout for the connection to the TCP server. For example, 10.
Read Timeout	No	Timeout for the response from the TCP server. For example, 10.

UDP Check

UDP Check is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display name of the remote check instance.
Address	Yes	<hostname>:port
Send	Yes	The given string is sent to the UDP.
Expect	Yes	The given string is expected from the UDP.
Timeout	No	Timeout for the connection to the UDP server. For example, 10.
Read Timeout	No	Timeout for the response from the UDP server. For example, 10.

Configuring Supported VeloCloud Services

Eight VeloCloud application services are supported in VCF Operations. The supported application services are listed here. Some of the application services have mandatory properties which you must configure. Some of the application services have pre-requirements that you must configure first. After you configure the properties, data is collected.

VeloCloud Orchestrator

VeloCloud Orchestrator and the following services are supported in VCF Operations.

- VeloCloud Orchestrator
- Nginx

  Note:
  To activate the plugin for ngnix service you must use the loopback address in the url `http://127.0.0.1/nginx_status`.
- Clickhouse
- Network Time Protocol
- MySQL
- Redis
- Java Application

  Note:
  Java application gets discovered after bootstrapping a VeloCloud Orchestrator virtual machine, but you must ignore it, as we do not monitor the Java application.

In VeloCloud Orchestrator, we monitor the following services. For each of these services we display a metric which indicates the service status:

- Backend
- Portal
- Upload

VeloCloud Orchestrator details.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the VeloCloud Orchestrator instance.

Nginx

Nginx is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Status Page URL	Yes	http://127.0.0.1/nginx_status
SSL CA	No	Path to the SSL CA file on the end point VM.
SSL Certificate	No	Path to the SSL Certificate file on the end point VM.
SSL Key	No	Path to the SSL Key file on the end point VM.
Skip SSL Verification.	No	Use SSL but skip chain & host verification. Expected: True/False.

ClickHouse

ClickHouse is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Servers URL	Yes	http://127.0.0.1:8123
User name	No	User name for the ClickHouse service.
Password	No	Password

NTPD

NTPD is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

MySQL

MySQL is supported in VCF Operations.

To activate the MySQL plug-in and fetch the credentials, refer to the KB article 81153.

Use the port number 3306 to run MySQL and the telegraf credentials and activate the plug-in.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Port	Yes	The port where MySQL is running. Example: 3306
User name	Yes	User name for the MySQL service. Example: Root
password	Yes	Password
SSL CA	No	Path to the SSL CA file on the end point VM.

Name	Mandatory?	Comment
SSL Certificate	No	Path to the SSL Certificate file on the end point VM.
SSL Key	No	Path to the SSL Key file on the end point VM.
Hostname	No	Optional hostname for the MySQL Service
Databases	No	Comma-separated list of databases to monitor. Each of the database names to be monitored must be enclosed in single quotes and the databases themselves should be comma-separated. For example, 'database1','database2','database3'.
TLS Connection	No	Accepted values are true, false, and skip-verify.

Redis

Redis is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.
Redis URL	Yes	servers = ["tcp://localhost:6379"]
SSL CA	No	Secure Socket Layer Certification Authority.
SSL Certificate	No	Secure Socket Layer Certificate.
SSL Key	No	Secure Socket Layer Key
Skip SSL Verification.	No	Skips verification for SSL.

VeloCloud Gateway

VeloCloud Gateway and the following services are supported in VCF Operations

- Network Time Protocol
- VeloCloud Gateway

In VeloCloud Gateway, we monitor the following processes. For each of these processes, we display a metric which indicates the process status.

- bgpd
- watchquagga
- gwd
- mgd
- natd
- ssh
- vc procmon
- vcsyscmd

VeloCloud Gateway details.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the VeloCloud Gateway instance.

NTPD

NTPD is supported in VCF Operations.

Name	Mandatory?	Comment
Display Name	Yes	Display Name of the application instance.

Pre-Requirements for Application Services

For the Telegraf agent to collect metrics for some of the application services, you must make modifications in the target machines. After you make these modifications, the agent starts collecting metrics. You must SSH to the virtual machine where you have deployed the agent and modify the configuration files.

Apache HTTPD

For the agent to collect metrics, modify the conf file available in `/etc/httpd/conf.modules.d/status.conf` and activate the mod_status for the HTTPD plugin.

```
<IfModule mod_status.c>

<Location /server-status>

    SetHandler server-status

</Location>

ExtendedStatus On

</IfModule>
```

If the conf file is not available, you must create one. Restart the HTTPD service after modifying the conf file with the following command:

```
systemctl restart httpd
```

Java Application

Support for Java application monitoring is currently limited to applications that support jolokia based JMX monitoring. Currently, as part of Java plugin activation, both the jolokia.jar and jolokia.war files are placed in the "Installed Path". If the Java application to be monitored is a servlet container, then the jolokia.war file is automatically deployed to the "Installed Path" that you provided when you configured the Java application in VCF Operations. If auto deployment is not supported or not activated, restart the application after activating the plugin or refer to the application's product documentation for the deployment of manual .war.

For all other java applications, first activate the plugin then include the path to the jolokia.jar file in the `"-javaagent:"` `JVM flag as a command line parameter`. For example, `"-javaagent:/<path_to_jolokia>/jolokia.jar"`, `<path_to_jolokia>` is the "Installed Path" which was provided during application configuration. Restart the application.

> **Note:**
> For monitoring a customized third-party java application, ensure that:
>
> - The jolokia based JMX monitoring is supported.
> - In case auto deploy is turned off, manual .war deployment is supported for servlet container application.
> - `"-javaagent:" JVM flag` is added as part of the command line parameter (JVM_OPTS)

Nginx

Add the following lines to the conf file available in `/etc/nginx/nginx.conf`:

```
http {
    server {
        location /status {
            stub_status on;
   access_log off;
   allow all;
  }
    }
}
```

Restart the Nginx service with the following command:

```
systemctl restart nginx
```

Postgres

In the configuration file available in the `/var/lib/pgsql/data/pg_hba.conf`, change the value of `local all postgres peer` to `local all postgres md5` and restart the service with the following command:

```
sudo service postgresql restart
```

Cassandra

To monitor the Cassandra database application, the Jolokia jar must be included as a JVM input to the Cassandra database application. Complete the following steps:

1. Modify `/etc/default/cassandra`.
   ```
   echo "export JVM_EXTRA_OPTS=\"-javaagent:/usr/share/java/jolokia-jvm-1.6.0-
   agent.jar=port=8778,host=localhost\"" | sudo tee -a /etc/default/cassandra
   ```

   Note:

   You can download the latest version of the Jolokia agent from https://jolokia.org/download.html. For a .JAR file deployment, you have to restart the application service after including the full file path of the JAR in the JMX argument of the JAVA process which you are monitoring.
2. Alternatively, you can activate the agent by modifying `cassandra-env.sh`. Include the following line at the end of the `cassandra-env.sh`:
   ```
   JVM_OPTS="$JVM_OPTS -javaagent:/usr/share/java/jolokia-jvm-1.6.0-agent.jar=port=8778,host=localhost"
   ```
   After you see the JVM inputs, restart the Cassandra service.

Oracle Database

To monitor the Oracle database, complete these steps:

1. Download the instant client library from: `https://www.oracle.com/database/technologies/instant-client/downloads.html`.
   You must download the Oracle instant library and included it in the PATH.
2. Install Python 3.6 or later. Install the cx_Oracle module.
   ```
   python3 -m pip install cx_Oracle --upgrade
   ```
3. Create a User.
   ```
   CREATE USER <UserName> IDENTIFIED BY <yourpassword>;
    GRANT select_catalog_role TO <UserName>;
    GRANT CREATE SESSION TO <UserName>;
   ```
4. **For a Windows VM:**
   a. Append the Oracle instant client library path and the Python path to the PATH environment variable.
   b. Install the ARC agent.
5. **For a Linux VM:**
   a. Install the ARC agent.

b. Get the file `exec_oracle_python.sh(wget --no-check-certificate https://<CP_IP>/downloads/salt/content-accessories/exec_oracle_python.sh)` from `/opt/vmware/ucp/`.
c. Fill in `/opt/vmware/ucp/exec_oracle_python.sh` with absolute values for `LD_LIBRARY_PATH`, `ORACLE_HOME`, `PYTHON_BIN`, and `TNS_ADMIN`.
   `LD_LIBRARY_PATH`: Path to the Oracle instant client library. For example, `/opt/vmware/ucp/instantclient_21_4`.
   `ORACLE_HOME`: Directory in the file system where the Oracle software is installed. For example, `/u01/app/oracle/product/19.3.0/dbhome_1/`.
   `PYTHON_BIN`: Path to the Python executable. For example, `/usr/bin/python`.
   `TNS_ADMIN`: Environment variable that points to the directory where the SQL*Net configuration files are located. For example, `/u01/app/oracle/product/19.3.0/dbhome_1/network/admin`.
d. Activate the Oracle DB plugin.

Active MQ 5.16 and Later Versions

To activate Active MQ 5.16 and later versions, complete the following steps:

- Navigate to `/opt/activemq/apache-activemq-5.16.0/webapps/api/WEB-INF/classes/jolokia-access.xml`
- Remove or comment out the following lines:

```
<cors>
  <strict-checking/>
</cors>
```

- Restart the Active MQ service.

Microsoft SQL Server

The user account must have the following permissions to monitor the Microsoft SQL Server application service with Telegraf.

```
USE master;
GO
CREATE LOGIN [telegraf] WITH PASSWORD = N'mystrongpassword';
GO
GRANT VIEW SERVER STATE TO [telegraf];
GO
GRANT VIEW ANY DEFINITION TO [telegraf];
GO
```

Note:
If you want to monitor named instances of the Microsoft SQL Server application service, then the plugin activation input should not have the Port field configured. Use Port only when there is no instance in the server (default 1433).

Additional Operations from the Manage Telegraf Agents Page

After you have configured cloud proxy and installed a product-managed Telegraf agent, configure a vCenter Cloud account (relevant only to VMs managed by the vCenter cloud account). You can manage the agents on all the target machines from the **Manage Telegraf Agents** page.

From the **Manage Telegraf Agents** page, you can:

- Use the filter to easily find physical servers or VMs managed or unmanaged by the vCenter cloud accounts.
- View data collected on all target machines.
- Start, stop, and update agents on all target machines where the product-managed agent is installed. You can also discover and manage the services on each product-managed Telegraf agent that you install.

Note:
The maximum allowed application services count per target machine is 200.

Additionally, all the types of supported monitored object types such as `Virtual Machine` and `Endpoint` are consolidated in the **Manage Telegraf Agents** page so that you have visibility of the entire environment you are monitoring and can manage agents and monitor the content from one single page, depending on the agent you are using and type of the objects available. For more information about object types, see Monitoring Application Services using Product-Managed Telegraf.

Telegraf Agents in VCF Operations

For more information about Telegraf agents, see What Are The Different Types of Telegraf Agents in VCF Operations?.

> **Note:**
> Data collected from product-managed agents, open-source agents, and user-managed agents can be viewed from the **Manage Telegraf Agents** page. For objects that have open-source agents and user-managed Telegraf agents installed and running, you can only view data that is collected, you cannot install, uninstall, or manage the lifecycle of the agents.

Ensuring FIPS Compliance

In VCF 9.1, by default, product managed Telegraf agents are FIPs compliant.

FIPS Compliance for Product Managed Telegraf Agents with Versions Lower than 8.18.6 and 9.0.2

If there are product managed Telegraf agents lower than versions 8.18.6 and 9.0.2, you will not be able to carry out agent management actions because of FIPS non-compliance. In such cases, you will see the following banner in the **Manage Telegraf Agents** page: **Agents on some objects can't be upgraded. Use the filter option 'Agent Reinstallation Required' and resintall agents on all those objects**.

To ensure FIPS compliance, you must filter the objects by the **Agent Reinstallation Required** option from the **Manage Telegraf Agents** page. Only those objects where the Telegraf agent version are lower than 8.18.6 and 9.0.2 are displayed. You must reinstall these product managed Telegraf agents from **Agent Actions/Vertical Ellipsis** > **Install** to ensure FIPS compliance.

FIPS Compliance for Product Managed Telegraf Agents with Versions at 8.18.6 and 9.0.2

If there are product managed Telegraf agents at versions 8.18.6 and 9.0.2, you will see the following banner in the **Manage Telegraf Agents** page: **Agent update is available for some objects. Update the agents to the latest versions**.

To ensure FIPS compliance, you must update the product managed Telegraf agents from **Agent Actions Ellipsis/Vertical Ellipsis** > **Update**.

Where You Manage the Agents

To manage the agents and application services, in the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, click the **Manage Telegraf Agents** tab.

Table 1689: Options

Options	Description
Agent Actions > **Install**	Installs the agents on the selected VM/s managed by the vCenter cloud account. Select the VM/s managed by the vCenter cloud account on which you want to install the agent, click Agent Actions/vertical ellipsis and then click **Install**. For more information, see Install an Agent from the UI. **Note:** • **Install** from the UI is available only for VMs managed by the vCenter cloud account using the product-managed agent. • For `Endpoint` object types, product-managed agent install is not supported. • **Install** from the UI is not available for open-source agents and user-managed agents.
Agent Actions > **Uninstall**	Uninstalls the agent. Select the VM/s managed by the vCenter cloud account on which you want to uninstall the agent, click Agent Actions/vertical ellipsis and then click **Uninstall**. For more information, see Uninstall an Agent. **Note:** • Uninstall from the UI is available only for VMs managed by the vCenter cloud account using the product-managed agent. • For `Endpoint` objects project-managed agent uninstall is not supported. • **Uninstall** from the UI is not available for open-source agents and user-managed agents.
Agent Actions > **Update**	Updates agents that are at a lower version. Select the VMs managed and unmanaged by the vCenter cloud account and physical servers on which you want to update the agent, click Agent Actions/vertical ellipsis and then click **Update**. After the agents are updated, the last operation status changes to **Content Upgrade Success**. **Note:** You can use the **Update** option only for objects with a product-managed Telegraf agent installed and running. **Note:** After you upgrade VCF Operations and then update product-managed Telegraf agents that are running at an earlier version, there could be a data collection gap of a maximum of 2-3 collection cycles.
Agent Actions > **Start**	If you have temporarily stopped sending metrics to VCF Operations, you can use the **Start** option to start data collection for the application service. Select the VMs managed and unmanaged by the vCenter cloud account and physical servers on which you want to start the agent and click Agent Actions/vertical ellipsis and then click **Start**. **Note:** You can use the **Start** option only for objects with a product-managed Telegraf agent installed and running.
Agent Actions > **Stop**	During a maintenance period, you can temporarily stop sending application service metrics to VCF Operations. Select the VMs managed and unmanaged by the vCenter cloud account and physical servers on which you want to stop the agent and click Agent Actions/vertical ellipsis and then click **Stop**. **Note:** You can use the **Stop** option only for objects with a product-managed Telegraf agent installed and running.

Options	Description
Agent Actions > Activate Service	Allows you to activate a service. **Note:** You can only activate a service for objects with a product- managed Telegraf agent installed and running. **Note:** Applicable for bulk actions.
Vertical Ellipsis > **Go To Details**	Displays the **Summary** tab of the selected VMs managed and unmanaged by the vCenter cloud account and physical servers. **Note:** You can view the summary and details for objects with product-managed Telegraf agents, open-source Telegraf agents, and user-managed Telegraf agents installed and running.
Filter	Filters the VMs managed and unmanaged by the vCenter cloud account and physical servers based on the name of the VM or the host name in the case of `Endpoint` type objects, the operating system it runs on, object type, agent type, agent status, the application service discovered, the last operation status, the power status of the VMs managed by the vCenter cloud account, agent reinstallation required, and the agent version.

You can also view specific details from the options in the data grid.

Table 1690: Data Grid Options

Option	Description
Name	Name of the virtual machine or host name in case of physical servers and VMs unmanaged by the vCenter cloud account.
Object Type	There are four types of object types: • Virtual Machine: Includes only VMs managed by the vCenter cloud account. • Endpoint: Includes physical servers and VMs unmanaged by the vCenter cloud account. **Note:** The agent types supported by the `Endpoint` object type are: `product-managed agent`, `open-source agent`, and `user-managed agent`.
Operating System	Operating system running on the target machine.
Agent Status	Displays the status of the agent at the target machine. • Agent Running. Indicates that the agent is running. **Note:** Applicable only for product managed Telegraf agents. • Agent Stopped. Indicates that the agent has stopped. • Agent Unhealthy: Indicates that a component of the product-managed agent is not healthy. • Not Installed. Indicates that the agent is not installed.

Option	Description
Agent Type	Types of agents: • Product-managed agent: Telegraf agent managed by VCF Operations. • Open-source agent: Open source Telegraf agent. • User-Managed agent: Telegraf agent used to monitor physical servers. **Note:** The `User-managed agent` option is available only if you have upgraded from a previous version of VCF Operations.
Last Operation Status	Status of the last operation. **Note:** Applicable only for product-managed agents. The possible values are: • No Operation • Install Success • Install Failed • Install In Progress • Start Success • Start Failed • Start In Progress • Stop Success • Stop Failed • Stop In Progress • Update Success • Update Failed • Update In Progress • Uninstall Success • Uninstall Failed • Uninstall In Progress • Download Success • Client Certificate Renewal Pending • Client Certificate Renewal in Progress • Client Certificate Copied • Client Certificate Renewal Success • Client Certificate Renewal Failed
State	Power status of the VMs managed by the vCenter cloud account. The possible values are: • Powered On • Powered Off **Note:** Powered on and Powered Off status is applicable only for the `Virtual Machine` object type. • Unknown
Virtual IP Details	Virtual IP configured on an application monitoring high availability activated collector group to which target machines will send data. For collector groups that are not activated for application monitoring high availability, the column will be empty.
Collector Group	Displays the name of the collector group to which the cloud proxy mentioned in the **Cloud Proxy** column belongs.

Option	Description
Cloud Proxy	You can see either: • The cloud proxy IP through which Telegraf agent management is performed and data is collected, or • For high availability activated collector groups, the IP address of the primary cloud proxy that is collecting data is displayed.
Agent Version	Version of the agent. A gray dot is displayed if the target machine requires an update for product managed agents. • For product-managed agents: The version displayed is the same as the VCF Operations version. • For open-source agents: The version displayed is the open source Telegraf agent version that you use.
vCenter Name	Name of the vCenter Adapter cloud account to which that vCenter VMs resource belongs. **Note:** The field is empty for `Endpoint` object types.
Collection State	VCF Operations Application Management cloud account collection state which monitors the VM.
Collection Status	VCF Operations Application Management cloud account collection status which monitors the VM.

Table 1691: Status of Application Services

Icon	Description
Green tick icon against the application service	Indicates that the application service is activated and the application service instances are receiving data.
Red exclamation icon against the application service	Indicates that the application service has been activated but there is a problem with data collection. When there is more than one agent plugin of the same kind, and one of them is activated, but the other is not collecting data, a red exclamation icon is still displayed against the application service.
Red icon against the application service instance	Indicates that there is an error in receiving data with an error message displayed after the name of the application service instance.
Gray question icon against the application service	Indicates that the application services requires reactivation. The application service must be reactivated. For reactivation, see Activate an Application Service for more information
Gray pause icon against the application service instance	Indicates that the application service instance has been stopped.
Blue icon with three horizontal dots against the agent plugin	Indicates data is being received.
Progress status icon	After you have added the parameters and activated the application service, the progress status is displayed until data collection starts.

To manage the agent, follow these steps:

1. Install the agent.
   For more information, see Install an Agent from the UI.

   Note:

   - You can install the agent from the UI only for VMs managed by the vCenter cloud account.
   - To install agents on physical servers and VMs unmanaged by the vCenter cloud account, use the script. For information to install/uninstall product-managed Telegraf agents, see Install/Uninstall an Agent Using a Script

on a Linux Platformand Install/Uninstall an Agent Using a Script on a Windows Platform. For information to install open-source agents, see Monitoring Applications using Open Source Telegraf on a Linux Platform and Monitoring Applications using Open Source Telegraf on a Windows Platform.

2. Manage the application services on each agent.
   For more information, see Activate an Application Service.

   Note:
   You can manage application services only for objects with a product-managed Telegraf agent installed and running.

   For custom monitoring actions, see Custom Script, Activate Remote Checks, Monitor Windows Services, and Monitor Linux Processes.
3. Stop and start the agents on the target machines.
   When you stop an agent, you cannot activate or deactivate a plugin. If the target machine is powered off or if you lose connection with cloud proxy, you cannot configure or activate a plugin.

   Note:
   You can stop and start agents only for objects with a product-managed Telegraf agent installed and running.
4. Uninstall the agent.
   For more information, see Uninstall an Agent.

   Note:
   You can uninstall the agent from the UI only for managed vCenter VMs.
5. Update agents that are at a lower version.

Custom Script

You can run custom scripts in the target machine and collect custom data which can then be consumed as a metric.

Prerequisites

- All the scripts that you run using the custom script, must output a single integer value. If the output is not a single integer value, an error is displayed in the user interface.
- The custom script uses Telegraf's `exec` plugin to run scripts on a target machine's operating system. In Linux operating systems, a special user called *arcuser* with specific privileges, is created for installing the Telegraf agent. As a result, the `exec` plugin runs the scripts using that *arcuser* user. Ensure that the *arcuser* can run the scripts that use the custom script (the *arcuser* must have permissions to run the script). For example, the *arcuser* created automatically by cloud proxy, does not have privileges to run scripts which are stored under the `/root` directory.
- In Windows operating systems, a system user is used for installing the Telegraf agent. As a result, the `exec` plugin runs the scripts using that system user. Ensure that the system user has privileges and can run the custom script.
- The script must be placed in the `/opt/vmware` folder.
- The environment variable PATH must be set for PowerShell scripts to be executed on the Windows operating system if the user used the *`powershell -File`* prefix. For example, *`"%SYSTEMROOT%\System32\WindowsPowerShell\v1.0\"`* in the environment variable PATH variable.

How to Run Custom Scripts

- From the **Manage Telegraf Agents** tab, filter by **Agent Status** > **Agent Running**.
- Expand the drop-down arrow against the target machine on which the agent is installed. You see the **Custom Monitoring** section.
- Against the **Custom Script** option, click the vertical ellipsis and then click **Add**.
- From the **Manage Custom Service** dialog box, you can add and configure the Windows services to be monitored.

Instance Settings

Option	Description
Status	Activate the custom script execution.
Display Name	Add a suitable name for the script. The * is an invalid character and must not be used in the name.

Filepath	Enter the path to the script file on the target machine.
	Example:
	For Linux machines:
	`/opt/vmware/scripts/customscript.py`
	For Windows machines:
	`C:\scripts\customscript.ps1`

Prefix	Enter a prefix if necessary. Example: For Linux machines: python2.7, /bin/bash, or perl, etc For Windows machines: *`powershell -File`*
Args	List the arguments in the script.
Timeout	Enter a script execution timeout on the VM.

After you save the script, it appears under **Custom Script**. You can edit or delete scripts by clicking the **Edit** or **Delete** options from the vertical ellipsis against the custom script you added. After the scripts have been added and saved, click the drop-down arrow against **Custom Script**, to view the list of scripts and their status.

Note:

- The custom script must throw all errors in the format `ERROR|<Error_message>` for the error propagation to work. If the script does not throw an error in the given format, VCF Operations displays an error message `Unable to parse the error message. Please check the endpoint` in the user interface. This is by design, until cloud proxy propagates the exact error message.
- The bash script must start with `shebang (#!/bin/bash).`

Metrics Tab

When data is collected successfully, you can view the script as a metric for the target machine, in the **Metrics** tab. The script metrics are created under an object called `Custom Script` which is a single object per target machine. All the metrics from the scripts for the target machine are placed under that `Custom Script` object that contains all the custom scripts you have created. You can view the output for the specific metric. The metric name under the `Scripts` folder is the display name that the user specifies while creating the script configuration. For example, if you set the display name as `Python script`, then a metric is created with the name **Python script** if data is collected successfully.

Activate Remote Checks

You can activate remote checks such as ICMP Check, UDP Check, TCP Check, and HTTP Check.

- ICMP check sends a ping from the target machine to a specified resource.
- UDP check performs a UDP based connection check from the target machine to a specified resource.
- TCP check performs a connection check from the target machine to the port of a specified resource.
- HTTP check performs REST API calls from the target machine to a specified URL.

1. From the **Manage Telegraf Agents** page, filter by **Agent Status** > **Agent Running**.
2. Expand the drop-down arrow against the relevant target machine on which the agent is installed. You see the **Custom Monitoring** section.
3. Against the **ICMP Check**, **UDP Check**, **TCP Check**, or **HTTP Check** options, click the vertical ellipsis and then click **Add**.
4. From the dialog box that appears on the right side, you can activate and configure the remote checks to be monitored. For configuration information, see Configuring Supported Application Services.
5. Click **Save**.

Monitor Windows Services

After you install an agent on a target machine, you can monitor existing or custom Windows services that run on the target machine.

How to Monitor Windows Services

- From the **Manage Telegraf Agents** page, filter by **Agent Status** > **Agent Running**.
- From the data grid, expand the drop-down arrow in front of the relevant Windows target machine name on which the agent is installed. You see the **Custom Monitoring** section.
- Against the **Services** option, click the vertical ellipsis and then click **Add**.
- From the **Manage Service Activation** dialog box, you can add and configure the Windows services to be monitored.

Table 1692: Instance Settings and Other Options

Option	Descriptions
Status	Activate the monitoring of the Windows service.
Display Name	Add a suitable name for the Windows service. For new plugin activations, the VM name or the end point host name in the format <on *VM name*>, is automatically appended to the display name. For example, if the display name you enter is <system>, the VM name is automatically appended and the name is displayed as <system on *VM name*>. If the display name was <system on abcd>, after an upgrade, <abcd> is replaced with the VM name or end point host name. If the display name did not end with <on *text1*>, after an upgrade, <on *VM name*> is automatically appended to the existing display name. The following are invalid characters and must not be used in the name: <, **"**, >, and |.
Service Name	Enter a name of the Windows service you want to monitor.

Save the settings to add the Windows service. To edit or delete Windows services, click the **Edit** or **Delete** options from the vertical ellipsis against Windows service you added. After the services have been added and saved, click the drop-down arrow against **Services**, to view the list of Windows services and their status.

Metrics Tab

When data is collected successfully, you can view the metric from the **Manage Telegraf Agents** page, click the **Vertical Ellipsis** > **Go To Details** > **Metrics** tab. The metrics for the Windows service are created under an object called `Services` which is a single object per target machine.

Monitor Linux Processes

After you install an agent on a target machine, you can monitor existing or custom Linux processes that run on the machine.

How to Monitor Linux Services

- From the **Manage Telegraf Agents** page, filter by **Agent Status** > **Agent Running**.
- Expand the drop-down arrow against the relevant Linux machine on which the agent is installed. You see the **Custom Monitoring** section.
- Against the **Services** option, click the vertical ellipsis and then click **Add**.
- From the **Manage Service Activation** dialog box, you can add and configure the Linux services to be monitored.

Table 1693: Instance Settings and Other Options

Option	Description
Status	Activate or deactivate the monitoring of the Linux process.
Display Name	Add a suitable name for the Linux process you want to monitor. For new plugin activations, the VM name or end point host name in the format <on *VM name*>, is automatically appended to the display name. For example, if the display name you enter is <system>, the VM name is automatically appended and the name is displayed as <system on *VM name*>. If the display name was <system on abcd>, after an upgrade, <abcd> is replaced with the VM name or end point host name. If the display name did not end with <on *text1*>, after an upgrade, <on *VM name*> is automatically appended to the existing display name. The following are invalid characters and must not be used in the name: <, **"**, >, and |.
Filter Type	Select either **Executable Name**, **Regex Pattern**, or **Pid File** as the filter type from the drop-down menu.
Filter Value	The filter value could be a process executable name, a regex pattern, or a pid file absolute path.

Save the settings to add the Linux service. To edit or delete Linux services, click the **Edit** or **Delete** options from the vertical ellipsis against Linux service you added. After the services have been added and saved, click the drop down arrow against **Services**, to view the list of Linux services and their status.

Metrics Tab

When data is collected successfully, you can view the metric from the **Manage Telegraf Agents** page, click the **Vertical Ellipsis** > **Go To Details** > **Metrics** tab.

Deactivate an Application Service

You can deactivate an application service to stop monitoring the application service that is sending data to VCF Operations.

Prerequisite

- If plugin deactivation requires the location of a file (for example, client certificates for SSL Trust) on the target machine, the location and the files should have appropriate read permissions for the *arcuser* to access those files.

  Note:
  If the plugin displays a permission denied status, provide the *arcuser* with permissions to the file locations that you have specified during plugin activation.

Deactivate an Application Service

To deactivate a plugin to stop monitoring the application service that is sending data to VCF Operations, complete the following steps:

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, click **Manage Telegraf Agents**.
2. Filter by **Agent Status** > **Agent Running**.

3. Expand the drop-down arrow against the target machine on which the agent is installed. You see the **Services Discovered** section.
4. From the **Services Discovered** section, select a service that has been activated, click the vertical ellipsis and then click **Edit**.
5. Deactivate the application service from the dialog box that is displayed on the right side.
6. Click **Save**.

For information on activating an application service, see Activate an Application Service.

Summary of Discovered and Supported Operating Systems and Application Services

You can monitor application services and operating systems from VCF Operations and get insights into the services, processes, and infrastructure. You can view a summary of discovered operating systems and services, supported operating systems, and supported services.

To monitor and view applications and operating systems, from the left menu, click **Operate** > **Administration** > **Configurations**, and then click the **Application Monitoring: Telegraf** tile from the right panel.

Discovered Operating Systems and Services

You see the application services that are discovered on the end-point virtual machines where the agents are installed. From the **Discovered Operating Systems and Services** section in the **Monitor Applications** page, click the <discovered> link under the name of the application service to perform lifecycle management actions for the agent and application services. For more information, see Additional Operations from the Manage Telegraf Agents Page .

Supported Operating Systems

You see a list of supported operating systems for which VCF Operations collects metrics. For more information, see Supported Platforms.

Supported Services

You see a list of supported services for which VCF Operations collects metrics. For more information, see Supported Application Services .

Metrics Collected

Metrics are collected for operating systems, application services, remote checks, Linux processes, and Windows services.

Operating System Metrics

Metrics are collected for Linux and Windows operating systems.

Linux Platforms

The following metrics are collected for Linux operating systems:

Table 1694: Metrics for Linux

Metric	Metric Category	KPI
<Instance name>| Usage Idle	CPU	False
<Instance name>| Usage IO-Wait	CPU	False
<Instance name>|Time Active	CPU	True
<Instance name>|Time Guest	CPU	False
<Instance name>|Time Guest Nice	CPU	False
<Instance name>|Time Idle	CPU	False

Metric	Metric Category	KPI
<Instance name>|Time IO-Wait	CPU	False
<Instance name>|Time IRQ	CPU	True
<Instance name>|Time Nice	CPU	False
<Instance name>|Time Soft IRQ	CPU	True
<Instance name>|Time Steal	CPU	False
<Instance name>|Time System	CPU	False
<Instance name>|Time User	CPU	True
<Instance name>|Usage Active (%)	CPU	True
<Instance name>|Usage Guest (%)	CPU	False
<Instance name>|Usage Guest Nice (%)	CPU	False
<Instance name>|Usage IRQ (%)	CPU	True
<Instance name>|Usage Nice (%)	CPU	False
<Instance name>|Usage Soft IRQ (%)	CPU	True
<Instance name>|Usage Steal (%)	CPU	False
<Instance name>|Usage System (%)	CPU	True
<Instance name>|Usage User (%)	CPU	True
CPU Load1 (%)	CPU Load	False
CPU Load15 (%)	CPU Load	False
CPU Load5 (%)	CPU Load	False
<Instance name>|IO Time	Disk IO	False
<Instance name>|Read Time	Disk IO	False
<Instance name>|Reads	Disk IO	False
<Instance name>|Write Time	Disk IO	False
<Instance name>|Writes	Disk IO	False
<Instance name>|Disk Free	Disk	False
<Instance name>|Disk Total	Disk	False
<Instance name>|Disk Used (%)	Disk	False
Cached	Memory	False
Free	Memory	False
Inactive	Memory	False
Total	Memory	True
Used	Memory	True
Used Percent	Memory	True
Blocked	Processes	True
Dead	Processes	False
Running	Processes	False
Sleeping	Processes	False
Stopped	Processes	False
Zombies	Processes	False
Free	Swap	False

Metric	Metric Category	KPI
In	Swap	False
Out	Swap	False
Total	Swap	True
Used	Swap	True
Used Percent	Swap	True
Telegraf Availability	None	False

Windows Platforms

The following metrics are collected for Windows operating systems:

Table 1695: Metrics for Windows

Metric	Metric Category	KPI
Idle Time	CPU	False
Interrupt Time	CPU	False
Interrupts persec	CPU	True
Privileged Time	CPU	False
Processor Time	CPU	False
User Time	CPU	False
DPC Time (%)	CPU	False
Usage Guest (%)	CPU	False
Usage System (%)	CPU	False
Usage User (%)	CPU	False
Avg. Disk Bytes Read	Disk	False
Avg. Disk sec Read	Disk	False
Avg. Disk sec Write	Disk	False
Avg. Disk Write Queue Length	Disk	False
Avg. Disk Read Queue Length	Disk	False
Disk Read Time	Disk	False
Disk Write Time	Disk	False
Free Megabytes	Disk	False
Free Space	Disk	False
Idle Time	Disk	False
Split IO persec	Disk	False
Available Bytes	Memory	True
Cache Bytes	Memory	False
Cache Faults persec	Memory	False
Committed Bytes	Memory	True
Demand Zero Faults persec	Memory	False
Page Faults persec	Memory	True
Pages persec	Memory	False

Metric	Metric Category	KPI
Pool Nonpaged Bytes	Memory	True
Pool Paged Bytes	Memory	False
Transition Faults persec	Memory	False
Total (bytes)	Memory	False
Used (bytes)	Memory	False
Used Percent(%)	Memory	False
Bytes Received persec	Network	False
Bytes Sent persec	Network	False
Packets Outbound Discarded	Network	False
Packets Outbound Errors	Network	False
Packets Received Discarded	Network	False
Packets Received Errors	Network	False
Packets Received persec	Network	False
Packets Sent persec	Network	False
Elapsed Time	Process	False
Handle Count	Process	False
IO Read Bytes persec	Process	False
IO Read Operations persec	Process	False
IO Write Bytes persec	Process	False
IO Write Operations persec	Process	False
Privileged Time	Process	False
Processor Time	Process	False
Thread Count	Process	False
User Time	Process	False
Context Switches persec	System	False
Processes	System	False
Processor Queue Length	System	False
System Calls persec	System	False
System Up Time	System	False
Threads	System	False
Used Percent (%)	Swap	False
Total (bytes)	Swap	False
Telegraf Availability	None	False

Application Service Metrics

Metrics are collected for 23 application services.

Here is the list of application services:

- Active Directory
- Apache HTTPD
- Microsoft SQL Server

- PostgreQL
- Microsoft IIS
- MS Exchange
- JBoss Server
- RabbitMQ
- MySQL
- NGINX
- SharePoint
- Oracle WebLogic Server
- Pivotal TC Server
- ActiveMQ
- Apache HTTPD
- Oracle DB
- Cassandra
- HyperV
- MongoDB
- Riak KV
- Network Time Protocol
- WebSphere
- Java Application

Active Directory Metrics

Metrics are collected for the Active Directory application service.

Table 1696: Active Directory Metrics

Metric Name	Category	KPI
Database Cache % Hit (%)	Active Directory Database	True
Database Cache Page Faults/sec	Active Directory Database	True
Database Cache Size	Active Directory Database	False
Data Lookups	Active Directory DFS Replication	False
Database Commits	Active Directory DFS Replication	True
Avg Response Time	Active Directory DFSN	True
Requests Failed	Active Directory DFSN	False
Requests Processed	Active Directory DFSN	False
Dynamic Update Received	Active Directory DNS	False
Dynamic Update Rejected	Active Directory DNS	False
Recursive Queries	Active Directory DNS	False
Recursive Queries Failure	Active Directory DNS	False
Secure Update Failure	Active Directory DNS	False
Total Query Received	Active Directory DNS	True
Total Response Sent	Active Directory DNS	True
Digest Authentications	Active Directory Security System-Wide Statistics	True
Kerberos Authentications	Active Directory Security System-Wide Statistics	True
NTLM Authentications	Active Directory Security System-Wide Statistics	True

Metric Name	Category	KPI
Directory Services:<InstanceName>|Base Searches persec	Active Directory Services	False
Directory Services:<InstanceName>|Database adds persec	Active Directory Services	False
Directory Services:<InstanceName>|Database deletes persec	Active Directory Services	False
Directory Services<InstanceName>|Database modifys/sec	Active Directory Services	False
Directory Services<InstanceName>|Database recycles/sec	Active Directory Services	False
Directory Services<InstanceName>|DRA Inbound Bytes Total/sec	Active Directory Services	False
Directory Services<InstanceName>|DRA Inbound Objects/sec	Active Directory Services	False
Directory Services<InstanceName>|DRA Outbound Bytes Total/sec	Active Directory Services	False
Directory Services<InstanceName>|DRA Outbound Objects/sec	Active Directory Services	False
Directory Services<InstanceName>|DRA Pending Replication Operations	Active Directory Services	False
Directory Services<InstanceName>|DRA Pending Replication Synchronizations	Active Directory Services	False
Directory Services<InstanceName>|DRA Sync Requests Made	Active Directory Services	False
Directory Services<InstanceName>|DRA Sync Requests Successful	Active Directory Services	False
Directory Services<InstanceName>|DS Client Binds/sec	Active Directory Services	True
Directory Services<InstanceName>|DS Directory Reads/sec	Active Directory Services	False
Directory Services<InstanceName>|DS Directory Searches/sec	Active Directory Services	True
Directory Services<InstanceName>|DS Server Binds/sec	Active Directory Services	True
Directory Services<InstanceName>|DS Threads in Use	Active Directory Services	True
Directory Services:<InstanceName>|LDAP Active Threads	Active Directory Services	False
Directory Services:<InstanceName>|LDAP Client Sessions	Active Directory Services	True
Directory Services<InstanceName>|LDAP Closed Connections/sec	Active Directory Services	False
Directory Services<InstanceName>|LDAP New Connections/sec	Active Directory Services	True
Directory Services<InstanceName>|LDAP Searches/sec	Active Directory Services	True

Metric Name	Category	KPI
Directory Services<InstanceName>|LDAP Successful Binds/sec	Active Directory Services	False
Directory Services<InstanceName>|LDAP UDP operations/sec	Active Directory Services	False
Directory Services:<InstanceName>|LDAP Writes/sec	Active Directory Services	False
Application Availability	Active Directory	False

Tomcat Server

Metrics are collected for the Tomcat Server application service.

Table 1697: Tomcat Server

Metric Name	Category	KPI
Buffer Pool<InstanceName>|Count	Tomcat Server	False
Buffer Pool<InstanceName>|Memory Used	Tomcat Server	False
Buffer Pool<InstanceName>|Total Capacity	Tomcat Server	False
Class Loading|Loaded Class Count	Tomcat Server	False
Class Loading|Total Loaded Class Count	Tomcat Server	False
Class Loading|Unloaded Class Count	Tomcat Server	False
File Descriptor Usage|Max File Descriptor Count	Tomcat Server	False
File Descriptor Usage|Open File Descriptor Count	Tomcat Server	False
Garbage Collection:<InstanceName>|Total Collection Count	Tomcat Server	False
Garbage Collection:<InstanceName>|Total Collection Time	Tomcat Server	True
JVM Memory|Heap Memory Usage|Committed Memory	Tomcat Server	False
JVM Memory|Heap Memory Usage|Initial Memory	Tomcat Server	False
JVM Memory|Heap Memory Usage|Maximum Memory	Tomcat Server	False
JVM Memory|Heap Memory Usage|Used Memory	Tomcat Server	False
JVM Memory|Non Heap Memory Usage|Committed Memory	Tomcat Server	False
JVM Memory|Non Heap Memory Usage|Initial Memory	Tomcat Server	False
JVM Memory|Non Heap Memory Usage|Maximum Memory	Tomcat Server	False
JVM Memory|Non Heap Memory Usage|Used Memory	Tomcat Server	False
JVM Memory|Number of Object Pending Finalization Count	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Peak Usage|Committed Memory	Tomcat Server	False

Metric Name	Category	KPI
JVM Memory|Pool:<InstanceName>|Peak Usage| Initial Memory	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Peak Usage| Maximum Memory	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Peak Usage| Used Memory	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Usage| Committed Memory	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Usage|Initial Memory	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Usage| Maximum Memory	Tomcat Server	False
JVM Memory|Pool:<InstanceName>|Usage|Used Memory	Tomcat Server	False
Process CPU Usage (%)	Tomcat Server	True
System CPU Usage (%)	Tomcat Server	True
System Load Average (%)	Tomcat Server	True
Threading|Thread Count	Tomcat Server	False
Uptime	Tomcat Server	True
Application Availability	Tomcat Server	False
JSP Count	Tomcat Server Web Module	False
JSP Reload Count	Tomcat Server Web Module	False
JSP Unload Count	Tomcat Server Web Module	False
Servlet:<InstanceName>|Total Request Count	Tomcat Server Web Module	False
Servlet:<InstanceName>|Total Request Error Count	Tomcat Server Web Module	False
Servlet:<InstanceName>|Total Request Processing Time	Tomcat Server Web Module	False
Cache : Hit Count	Tomcat Server Web Module	False
Cache : Lookup Count	Tomcat Server Web Module	False
Current Thread Count	Tomcat Server Global Request Processor	True
Current Threads Busy	Tomcat Server Global Request Processor	True
errorRate	Tomcat Server Global Request Processor	False
Total Request Bytes Received	Tomcat Server Global Request Processor	False
Total Request Bytes Sent	Tomcat Server Global Request Processor	False
Total Request Count	Tomcat Server Global Request Processor	True
Total Request Error Count	Tomcat Server Global Request Processor	True
Total Request Processing Time	Tomcat Server Global Request Processor	False

Microsoft SQL Server Metrics

Metrics are collected for the Microsoft SQL Server application service.

Table 1698: MS SQL Metrics

Metric Name	Category	KPI
CPU<InstanceName>|CPU Usage (%)	Microsoft SQL Server	False
Database IO|Rows Reads Bytes/Sec	Microsoft SQL Server	False
Database IO|Rows Reads/Sec	Microsoft SQL Server	False
Database IO|Rows Writes Bytes/Sec	Microsoft SQL Server	False
Database IO|Rows Writes/Sec	Microsoft SQL Server	False
Performance|Access Methods|Full Scans per second	Microsoft SQL Server	False
Performance|Access Methods|Index Searches	Microsoft SQL Server	False
Performance|Access Methods|Page Splits per second	Microsoft SQL Server	False
Performance|Broker Activation|Stored Procedures Invoked per second	Microsoft SQL Server	False
Performance|Buffer Manager|Buffer cache hit ratio (%)	Microsoft SQL Server	True
Performance|Buffer Manager|Checkpoint Pages/sec	Microsoft SQL Server	True
Performance|Buffer Manager|Lazy writes per second	Microsoft SQL Server	True
Performance|Buffer Manager|Page life expectancy	Microsoft SQL Server	True
Performance|Buffer Manager|Page lookups per second	Microsoft SQL Server	False
Performance|Buffer Manager|Page reads per second	Microsoft SQL Server	False
Performance|Buffer Manager|Page writes per second	Microsoft SQL Server	False
Performance|Databases|Active Transactions	Microsoft SQL Server	True
Performance|Databases|Data File(s) Size	Microsoft SQL Server	True
Performance|Databases|Log Bytes Flushed/Sec	Microsoft SQL Server	False
Performance|Databases|Log File(s) Size	Microsoft SQL Server	False
Performance|Databases|Log File(s) Used Size	Microsoft SQL Server	False
Performance|Databases|Log Flush Wait Time	Microsoft SQL Server	False
Performance|Databases|Log Flushes per second	Microsoft SQL Server	False
Performance|Databases|Transactions per second	Microsoft SQL Server	False
Performance|Databases|Write Transactions per second	Microsoft SQL Server	False
Performance|Databases|XTP Memory Used	Microsoft SQL Server	False
Performance|General Statistics|Active temp Tables	Microsoft SQL Server	False
Performance|General Statistics|Logins per second	Microsoft SQL Server	False
Performance|General Statistics|Logouts per second	Microsoft SQL Server	False
Performance|General Statistics|Processes Blocked	Microsoft SQL Server	False
Performance|General Statistics|Temp Tables Creation Rate	Microsoft SQL Server	False
Performance|General Statistics|User Connections	Microsoft SQL Server	False
Performance|Locks|Average Wait Time	Microsoft SQL Server	False
Performance|Locks|Lock Requests per second	Microsoft SQL Server	False

Metric Name	Category	KPI
Performance|Locks|Lock Wait Time	Microsoft SQL Server	True
Performance|Locks|Lock Waits per second	Microsoft SQL Server	True
Performance|Locks|Number of Deadlocks per second	Microsoft SQL Server	True
Performance|Memory Manager|Connection Memory	Microsoft SQL Server	False
Performance|Memory Manager|Lock Memory	Microsoft SQL Server	False
Performance|Memory Manager|Log Pool Memory	Microsoft SQL Server	False
Performance|Memory Manager|Memory Grants Pending	Microsoft SQL Server	True
Performance|Memory Manager|SQL Cache Memory	Microsoft SQL Server	False
Performance|Memory Manager|Target Server Memory	Microsoft SQL Server	True
Performance|Memory Manager|Total Server Memory	Microsoft SQL Server	True
Performance|Resource Pool Stats|internal|Active memory grant amount	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|CPU Usage Percentage (%)	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|Disk Read Bytes per second	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|Disk Read IO	Microsoft SQL Server	False
Wait Stats:<InstanceName>|Wait Time (ms)	Microsoft SQL Server	False
Wait Stats<InstanceName>|Number of Waiting tasks (ms)	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|Disk Read IO Throttled Per Second	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|Disk Write Bytes per second (Bps)	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|Disk Write IO Throttled per second	Microsoft SQL Server	False
Performance|Resource Pool Stats|internal|Used Memory	Microsoft SQL Server	False
Performance|SQL Statistics | Batch Requests Per Second	Microsoft SQL Server	False
Performance|SQL Statistics | SQL Compilations per second	Microsoft SQL Server	False
Performance|SQL Statistics | SQL Re-Compilations per second	Microsoft SQL Server	False
Performance|Transactions | Free space in tempdb (KB)	Microsoft SQL Server	False
Performance|Transactions | Transactions	Microsoft SQL Server	False
Performance|Transactions | Version Store Size (KB)	Microsoft SQL Server	False
Performance|User Settable Counter | User Counter 0 to 10	Microsoft SQL Server	False

Metric Name	Category	KPI
Performance|Workload Group Stats|internal|Active Requests	Microsoft SQL Server	False
Performance|Workload Group Stats|internal|Blocked Tasks	Microsoft SQL Server	False
Performance|Workload Group Stats|internal|CpU Usage (%)	Microsoft SQL Server	False
Performance|Workload Group Stats|internal|Queued Requests	Microsoft SQL Server	False
Performance|Workload Group Stats|internal|Request Completed/sec	Microsoft SQL Server	False
Application Availability	Microsoft SQL Server	False

There are no metrics collected for Microsoft SQL Server Database.

PostgreSQL
Metrics are collected for the PostgreSQL application service.

Table 1699: PostgreSQL

Metric Name	Category	KPI
Buffers|Buffers Allocated	PostgreSQL	False
Buffers|Buffers Written by Backend	PostgreSQL	True
Buffers|Buffers Written by Background Writer	PostgreSQL	True
Buffers|Buffers Written During Checkpoints	PostgreSQL	True
Buffers|fsync Call Executed by Backend	PostgreSQL	False
Checkpoints|Checkpoints sync time	PostgreSQL	False
Checkpoints|Checkpoints write time	PostgreSQL	False
Checkpoints|Requested checkpoints performed count	PostgreSQL	False
Checkpoints|Scheduled checkpoints performed count	PostgreSQL	False
Clean scan stopped count	PostgreSQL	False
Application Availability	PostgreSQL	False
Disk Blocks|Blocks Cache Hits	PostgreSQL Database	False
Disk Blocks|Blocks Read	PostgreSQL Database	False
Disk Blocks|Blocks Read Time	PostgreSQL Database	False
Disk Blocks|Blocks Write Time	PostgreSQL Database	False
Statistics|Backends Connected	PostgreSQL Database	False
Statistics|Data Written by Queries	PostgreSQL Database	True
Statistics|Deadlocks Detected	PostgreSQL Database	True
Statistics|Queries Cancelled	PostgreSQL Database	True
Statistics|Temp Files Created by Queries	PostgreSQL Database	False
Transactions|Transactions Committed	PostgreSQL Database	True
Transactions|Transactions Rolled Back	PostgreSQL Database	True
Tuples|Tuples Deleted	PostgreSQL Database	True

Metric Name	Category	KPI
Tuples|Tuples Fetched	PostgreSQL Database	True
Tuples|Tuples Inserted	PostgreSQL Database	True
Tuples|Tuples Returned	PostgreSQL Database	True
Tuples|Tuples Updated	PostgreSQL Database	True

Microsoft IIS Metrics

Metrics are collected for the Microsoft IIS application service.

Table 1700: IIS Metrics

Metric Name	Category	KPI
HTTP Service Request Queues<InstanceName>AppPool|CurrentQueueSize	IIS HTTP Service Request Queues	True
HTTP Service Request Queues<InstanceName>AppPool|RejectedRequests	IIS HTTP Service Request Queues	False
Web Services<InstanceName> Web Site|Bytes Received	IIS Web Services	False
Web Services<InstanceName> Web Site|Bytes Sent/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Bytes Total/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Connection Attempts/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Current Connections	IIS Web Services	False
Web Services<InstanceName> Web Site|Get Requests/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Locked Errors/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Not Found Errors/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Post Requests/sec	IIS Web Services	False
Web Services<InstanceName> Web Site|Service Uptime	IIS Web Services	False
Web Services<InstanceName> Web Site|Total Bytes Sent	IIS Web Services	False
Web Services<InstanceName> Web Site|Total Get Requests	IIS Web Services	True
Web Services<InstanceName> Web Site|Total Post Requests	IIS Web Services	True
Web Services<InstanceName> Web Site|Total Put Requests	IIS Web Services	False
Current File Cache Memory Usage (bytes)	IIS Web Services Cache	False
File Cache Hits Percent (%)	IIS Web Services Cache	False
Kernel URI Cache Hits Percent (%)	IIS Web Services Cache	False

Metric Name	Category	KPI
Kernel URI Cache Misses	IIS Web Services Cache	False
Total Flushed URIs	IIS Web Services Cache	False
URI Cache Hits	IIS Web Services Cache	False
URI Cache Hits Percent (%)	IIS Web Services Cache	False
URI Cache Misses	IIS Web Services Cache	False
ASP.NET<InstanceName>|Application Restarts	IIS ASP.NET	True
ASP.NET<InstanceName>|Request Wait Time	IIS ASP.NET	True
ASP.NET<InstanceName>|Requests Current	IIS ASP.NET	True
ASP.NET<InstanceName>|Requests Queued	IIS ASP.NET	True
ASP.NET<InstanceName>|Requests Rejected	IIS ASP.NET	True
MS.NET<InstanceName>|Allocated Bytes/sec	MS.NET	True
MS.NET<InstanceName>|Current Queue Length	MS.NET	False
MS.NET<InstanceName>|Finalization Survivors	MS.NET	False
MS.NET<InstanceName>|Gen 0 Collections	MS.NET	False
MS.NET<InstanceName>|Gen 0 heap size	MS.NET	False
MS.NET<InstanceName>|Gen 1 Collections	MS.NET	False
MS.NET<InstanceName>|Gen 1 heap size	MS.NET	False
MS.NET<InstanceName>|Gen 2 Collections	MS.NET	False
MS.NET<InstanceName>|Gen 2 heap size	MS.NET	False
MS.NET<InstanceName>|IL Bytes Jitted / sec	MS.NET	False
MS.NET<InstanceName>|Induced GC	MS.NET	False
MS.NET<InstanceName>|Large Object Heap size	MS.NET	False
MS.NET<InstanceName>|No of current logical Threads	MS.NET	True
MS.NET<InstanceName>|No of current physical Threads	MS.NET	True
MS.NET<InstanceName>|No of current recognized threads	MS.NET	False
MS.NET<InstanceName>|No of Exceps Thrown / sec	MS.NET	True
MS.NET<InstanceName>|No of total recognized threads	MS.NET	False
MS.NET<InstanceName>|Percent Time in Jit	MS.NET	False
MS.NET<InstanceName>|Pinned Objects	MS.NET	False
MS.NET<InstanceName>|Stack Walk Depth	MS.NET	False
MS.NET<InstanceName>|Time in RT checks	MS.NET	False
MS.NET<InstanceName>|Time Loading	MS.NET	True
MS.NET<InstanceName>|Total No of Contentions	MS.NET	False
MS.NET<InstanceName>|Total Runtime Checks	MS.NET	True
Application Availability	Microsoft IIS	False

MS Exchange Metrics

Metrics are collected for the MS Exchange application service.

Table 1701: MS Exchange Metrics

Metric Name	Category	KPI
Active Manager Server|Active Manager Role	MS Exchange	False
Active Manager Server|Database State Info Writes per second	MS Exchange	False
Active Manager Server|GetServerForDatabase Server-Side Calls	MS Exchange	False
Active Manager Server|Server-Side Calls per second	MS Exchange	True
Active Manager Server|Total Number of Databases	MS Exchange	True
ActiveSync|Average Request Time	MS Exchange	True
ActiveSync|Current Requests	MS Exchange	False
ActiveSync|Mailbox Search Total	MS Exchange	False
ActiveSync|Ping Commands Pending	MS Exchange	False
ActiveSync|Requests per second	MS Exchange	True
ActiveSync|Sync Commands per second	MS Exchange	True
ASP.NET|Application Restarts	MS Exchange	False
ASP.NET|Request Wait Time	MS Exchange	True
ASP.NET|Worker Process Restarts	MS Exchange	False
Autodiscover Service|Requests per second	MS Exchange	True
Availability Service|Average Time to Process a Free Busy Request	MS Exchange	True
Outlook Web Access|Average Search Time	MS Exchange	True
Outlook Web Access|Requests per second	MS Exchange	False
Outlook Web Access|Current Unique Users	MS Exchange	False
Application Availability	MS Exchange	False
Performance|Database Cache Hit (%)	MS Exchange Database	False
Performance|Database Page Fault Stalls per second	MS Exchange Database	True
Performance|I/O Database Reads Average Latency	MS Exchange Database	True
Performance|I/O Database Writes Average Latency	MS Exchange Database	True
Performance|I/O Log Reads Average Latency	MS Exchange Database	False
Performance|I/O Log Writes Average Latency	MS Exchange Database	False
Performance|Log Record Stalls per second	MS Exchange Database	False
Performance|Log Threads Waiting	MS Exchange Database	False
Performance|I/O Database Reads Average Latency	MS Exchange Database Instance	False
Performance|I/O Database Writes Average Latency	MS Exchange Database Instance	False
Performance|Log Record Stalls per second	MS Exchange Database Instance	False
Performance|Log Threads Waiting	MS Exchange Database Instance	False
Performance|LDAP Read Time	MS Exchange Domain Controller	False
Performance|LDAP Search Time	MS Exchange Domain Controller	False
Performance|LDAP Searches Timed Out per minute	MS Exchange Domain Controller	False
Performance|Long Running LDAP Operations per minute	MS Exchange Domain Controller	False

Metric Name	Category	KPI
Performance|Connection Attempts per second	MS Exchange Web Server	True
Performance|Current Connections	MS Exchange Web Server	False
Performance|Other Request Methods per second	MS Exchange Web Server	False
Process|Handle Count	MS Exchange Windows Service	False
Process|Memory Allocated	MS Exchange Windows Service	False
Process|Processor Time (%)	MS Exchange Windows Service	True
Process|Thread Count	MS Exchange Windows Service	False
Process|Virtual Memory Used	MS Exchange Windows Service	False
Process|Working Set	MS Exchange Windows Service	False

JBoss Server Metrics

Metrics are collected for the JBoss Server application service.

Table 1702: JBoss Server Metrics

Metric Name	Category	KPI
Buffer Pool<InstanceName>|Count	Jboss Server	False
Buffer Pool<InstanceName>|Memory Used	Jboss Server	False
Buffer Pool<InstanceName>|Total Capacity	Jboss Server	False
Class Loading|Loaded Class Count	Jboss Server	False
Class Loading|Total Loaded Class Count	Jboss Server	False
Class Loading|Unloaded Class Count	Jboss Server	False
File Descriptor Usage|Max File Descriptor Count	Jboss Server	False
File Descriptor Usage|Open File Descriptor Count	Jboss Server	False
Http Listener<InstanceName>|Bytes Received	Jboss Server	False
Http Listener<InstanceName>|Bytes Sent	Jboss Server	False
Http Listener<InstanceName>|Error Count	Jboss Server	False
Http Listener<InstanceName>|Request Count	Jboss Server	False
Https Listener<InstanceName>|Bytes Received	Jboss Server	False
Https Listener<InstanceName>|Bytes Sent	Jboss Server	False
Https Listener<InstanceName>|Error Count	Jboss Server	False
Https Listener<InstanceName>|Request Count	Jboss Server	False
Process CPU Usage (%)	Jboss Server	False
System CPU Usage (%)	Jboss Server	False
System Load Average (%)	Jboss Server	False
Threading|Daemon Thread Count	Jboss Server	False
Threading|Peak Thread Count	Jboss Server	False
Threading|Thread Count	Jboss Server	False
Threading|Total Started Thread Count	Jboss Server	False
Uptime	Jboss Server	False
UTILIZATION|Heap Memory Usage	Jboss Server	False

Metric Name	Category	KPI
Application Availability	Jboss Server	False
Garbage Collection<InstanceName>|Total Collection Count	Jboss JVM Garbage Collector	False
Garbage Collection<InstanceName>|Total Collection Time	Jboss JVM Garbage Collector	False
JVM Memory|Heap Memory Usage|Committed Memory	Jboss JVM Memory	False
JVM Memory|Heap Memory Usage|Initial Memory	Jboss JVM Memory	False
JVM Memory|Heap Memory Usage|Maximum Memory	Jboss JVM Memory	False
JVM Memory|Heap Memory Usage|Used Memory	Jboss JVM Memory	True
JVM Memory|Non Heap Memory Usage|Committed Memory	Jboss JVM Memory	False
JVM Memory|Non Heap Memory Usage|Initial Memory	Jboss JVM Memory	False
JVM Memory|Non Heap Memory Usage|Maximum Memory	Jboss JVM Memory	False
JVM Memory|Non Heap Memory Usage|Used Memory	Jboss JVM Memory	False
JVM Memory|Object Pending Finalization Count	Jboss JVM Memory	True
UTILIZATION|Active Count	Jboss Datasource Pool	False
UTILIZATION|Available Count	Jboss Datasource Pool	False
JVM Memory Pool<InstanceName>|Collection Usage|Committed Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Collection Usage|Initial Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Collection Usage|Used Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Collection Usage|Maximum Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Peak Usage| Committed Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Peak Usage| Initial Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Peak Usage| Maximum Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Peak Usage| Used Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Usage| Committed Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Usage|Initial Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Usage|Maximum Memory	Jboss JVM Memory Pool	False
JVM Memory Pool<InstanceName>|Usage|Used Memory	Jboss JVM Memory Pool	False

RabbitMQ Metrics

Metrics are collected for the RabbitMQ application service.

Table 1703: RabbitMQ Metrics

Metric Name	Category	KPI
CPU|Limit	RabbitMQ	False
CPU|Used	RabbitMQ	True
Disk|Free	RabbitMQ	False
Disk|Free limit	RabbitMQ	False
FileDescriptor|Total	RabbitMQ	False
FileDescriptor|Used	RabbitMQ	False
Memory|Limit	RabbitMQ	False
Memory|Used	RabbitMQ	True
Messages|Acked	RabbitMQ	False
Messages|Delivered	RabbitMQ	False
Messages|Delivered get	RabbitMQ	False
Messages|Published	RabbitMQ	False
Messages|Ready	RabbitMQ	False
Messages|Unacked	RabbitMQ	False
Socket|Limit	RabbitMQ	False
Socket|Used	RabbitMQ	True
UTILIZATION|Channels	RabbitMQ	True
UTILIZATION|Connections	RabbitMQ	True
UTILIZATION|Consumers	RabbitMQ	True
UTILIZATION|Exchanges	RabbitMQ	True
UTILIZATION|Messages	RabbitMQ	True
UTILIZATION|Queues	RabbitMQ	True
Application Availability	RabbitMQ	False
Messages|Publish in	RabbitMQ Exchange	False
Messages|Publish out	RabbitMQ Exchange	False
Consumer Utilisation	RabbitMQ Queue	False
Consumers	RabbitMQ Queue	False
Memory	RabbitMQ Queue	False
Messages|Ack	RabbitMQ Queue	False
Messages|Ack rate	RabbitMQ Queue	False
Messages|Deliver	RabbitMQ Queue	False
Messages|Deliver get	RabbitMQ Queue	False
Messages|Persist	RabbitMQ Queue	False
Messages|Publish	RabbitMQ Queue	False
Messages|Publish rate	RabbitMQ Queue	False
Messages|Ram	RabbitMQ Queue	False
Messages|Ready	RabbitMQ Queue	False

Metric Name	Category	KPI
Messages|Redeliver	RabbitMQ Queue	False
Messages|Redeliver rate	RabbitMQ Queue	False
Messages|Space	RabbitMQ Queue	False
Messages|Unack	RabbitMQ Queue	False
Messages|Unacked	RabbitMQ Queue	False
Messages	RabbitMQ Queue	False

There are no metrics collected for RabbitMQ Virtual Host.

MySQL Metrics

Metrics are collected for the MySQL application service.

Table 1704: MySQL Metrics

Metric Name	Category	KPI
Aborted connection count	MySQL	True
Connection count	MySQL	True
Event wait average time	MySQL	False
Event wait count	MySQL	False
Binary Files|Binary Files Count	MySQL	False
Binary Files|Binary Size Bytes	MySQL	False
Global Status|Aborted Clients	MySQL	False
Global Status|Binlog Cache Disk Use	MySQL	False
Global Status|Bytes Received	MySQL	False
Global Status|Bytes Sent	MySQL	False
Global Status|Connection Errors Accept	MySQL	False
Global Status|Connection Errors Internal	MySQL	False
Global Status|Connection Errors Max Connections	MySQL	False
Global Status|Queries	MySQL	False
Global Status|Threads Cached	MySQL	False
Global Status|Threads Connected	MySQL	False
Global Status|Threads Running	MySQL	False
Global Status|Uptime	MySQL	False
Global Variables|Delayed Insert Limit	MySQL	False
Global Variables|Delayed Insert Timeout	MySQL	False
Global Variables|Delayed Queue Size	MySQL	False
Global Variables|Max Connect Errors	MySQL	False
Global Variables|Max Connections	MySQL	False
Global Variables|Max Delayed Threads	MySQL	False
Global Variables|Max Error Count	MySQL	False
InnoDB|All deadlock count	MySQL	False
InnoDB|Buffer Pool Bytes Data	MySQL	False

Metric Name	Category	KPI
InnoDB|Buffer Pool Bytes Data	MySQL	False
InnoDB|Buffer Pool Bytes Dirty	MySQL	False
InnoDB|Buffer Pool Dump Status	MySQL	False
InnoDB|Buffer Pool Load Status	MySQL	False
InnoDB|Buffer Pool Pages Data	MySQL	False
InnoDB|Buffer Pool Pages Dirty	MySQL	False
InnoDB|Buffer Pool Pages Flushed	MySQL	False
InnoDB|Buffer pool size	MySQL	True
InnoDB|Checksums	MySQL	False
InnoDB|Open file count	MySQL	False
InnoDB|Row lock average time	MySQL	False
InnoDB|Row lock current waits	MySQL	False
InnoDB|Row lock maximum time	MySQL	False
InnoDB|Row lock time	MySQL	False
InnoDB|Row lock waits	MySQL	True
InnoDB|Table lock count	MySQL	False
Performance Table IO Waits|IO Waits Total Delete	MySQL	False
Performance Table IO Waits|IO Waits Total Fetch	MySQL	False
Performance Table IO Waits|IO Waits Total Insert	MySQL	False
Performance Table IO Waits|IO Waits Total Update	MySQL	False
Process List|Connections	MySQL	False
Application Availability	MySQL	False
IO waits average time	MySQL Database	False
IO waits count	MySQL Database	True
Read high priority average time	MySQL Database	False
Read high priority count	MySQL Database	False
Write concurrent insert average time	MySQL Database	False
Write concurrent insert count	MySQL Database	False

NGINX Metrics

Metrics are collected for the NGINX application service.

Table 1705: NGINX Metrics

Metric Name	Category	KPI
HTTP Status Info|Accepts	Nginx	True
HTTP Status Info|Active connections	Nginx	False
HTTP Status Info|Handled	Nginx	True
HTTP Status Info|Reading	Nginx	False
HTTP Status Info|Requests	Nginx	False
HTTP Status Info|Waiting	Nginx	True

Metric Name	Category	KPI
HTTP Status Info|Writing	Nginx	False
Application Availability	Nginx	False

SharePoint Metrics

Metrics are collected for the SharePoint Server application service.

Table 1706: SharePoint Server Metrics

Metric Name	Category	KPI
Sharepoint Foundation|Active Threads	SharePoint Server	True
Sharepoint Foundation|Current Page Requests	SharePoint Server	False
Sharepoint Foundation|Executing SQL Queries	SharePoint Server	False
Sharepoint Foundation|Executing Time/Page Request	SharePoint Server	True
Sharepoint Foundation|Incoming Page Requests Rate	SharePoint Server	False
Sharepoint Foundation|Object Cache Hit Count	SharePoint Server	False
Sharepoint Foundation|Reject Page Requests Rate	SharePoint Server	False
Sharepoint Foundation|Responded Page Requests Rate	SharePoint Server	True
SQL query executing time	SharePoint Server	False
Application Availability	SharePoint Server	False
Network|Received Data Rate	SharePoint Web Server	True
Network|Sent Data Rate	SharePoint Web Server	True
Process|Processor Time (%)	SharePoint Windows Service	False
Process|Threads	SharePoint Windows Service	False

Oracle WebLogic Server Metrics

Metrics are collected for the Oracle WebLogic Server application service.

Table 1707: Oracle WebLogic Server Metrics

Metric Name	Category	KPI
UTILIZATION|Process Cpu Load	Oracle WebLogic Server	True
UTILIZATION|System Cpu Load	Oracle WebLogic Server	False
UTILIZATION|System Load Average	Oracle WebLogic Server	False
Application Availability	Oracle WebLogic Server	False
UTILIZATION|Collection Time	Weblogic Garbage Collector	True
UTILIZATION|Connections HighCount	Weblogic JMS Runtime	True
UTILIZATION|JMS Servers TotalCount	Weblogic JMS Runtime	False
UTILIZATION|Active Total Count Used	Weblogic JTA Runtime	False
UTILIZATION|Active Transactions TotalCount	Weblogic JTA Runtime	False
UTILIZATION|Transaction Abandoned TotalCount	Weblogic JTA Runtime	True

Metric Name	Category	KPI
UTILIZATION|Transaction RolledBack App TotalCount	Weblogic JTA Runtime	True
UTILIZATION|Heap Memory Usage	Weblogic JVM Memory	True
UTILIZATION|Non Heap Memory Usage	Weblogic JVM Memory	False
UTILIZATION|Peak Usage	Weblogic JVM Memory Pool	True
UTILIZATION|Usage	Weblogic JVM Memory Pool	False
UTILIZATION|UpTime	Weblogic JVM Runtime	False

Pivotal TC Server Metrics

Metrics are collected for the Pivotal TC Server application service.

Table 1708: Pivotal TC Server Metrics

Metric Name	Category	KPI
Buffer Pool<InstanceName>|Count	Pivotal TC Server	False
Buffer Pool<InstanceName>|Memory Used	Pivotal TC Server	False
Buffer Pool<InstanceName>|Total Capacity	Pivotal TC Server	False
Class Loading|Loaded Class Count	Pivotal TC Server	False
Class Loading|Total Loaded Class Count	Pivotal TC Server	False
Class Loading|Unloaded Class Count	Pivotal TC Server	False
File Descriptor Usage|Max File Descriptor Count	Pivotal TC Server	False
File Descriptor Usage|Open File Descriptor Count	Pivotal TC Server	False
Garbage Collection:<InstanceName>|Total Collection Count	Pivotal TC Server	False
Garbage Collection:<InstanceName>|Total Collection Time	Pivotal TC Server	False
Process CPU Usage (%)	Pivotal TC Server	True
JVM Memory|Heap Memory Usage|Committed Memory	Pivotal TC Server	True
JVM Memory|Heap Memory Usage|Initial Memory	Pivotal TC Server	False
JVM Memory|Heap Memory Usage|Maximum Memory	Pivotal TC Server	False
JVM Memory|Heap Memory Usage|Used Memory	Pivotal TC Server	True
JVM Memory|Non Heap Memory Usage|Committed Memory	Pivotal TC Server	True
JVM Memory|Non Heap Memory Usage|Initial Memory	Pivotal TC Server	False
JVM Memory|Non Heap Memory Usage|Maximum Memory	Pivotal TC Server	False
JVM Memory|Non Heap Memory Usage|Used Memory	Pivotal TC Server	True
JVM Memory|Number of Object Pending Finalization Count	Pivotal TC Server	True

Metric Name	Category	KPI
JVM Memory|Pool:<InstanceName>|Peak Usage| Committed Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Peak Usage| Initial Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Peak Usage| Maximum Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Peak Usage| Used Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Usage| Committed Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Usage|Initial Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Usage| Maximum Memory	Pivotal TC Server	False
JVM Memory|Pool:<InstanceName>|Usage|Used Memory	Pivotal TC Server	False
Process CPU Usage (%)	Pivotal TC Server	True
System CPU Usage (%)	Pivotal TC Server	True
Uptime	Pivotal TC Server	True
Threading|Thread Count	Pivotal TC Server	False
System Load Average	Pivotal TC Server	False
Application Availability	Pivotal TC Server	False
Current Thread Count	Pivotal TC Server Thread Pool	False
Current Threads Busy	Pivotal TC Server Thread Pool	True
Total Request Bytes Received	Pivotal TC Server Thread Pool	False
Total Request Bytes Sent	Pivotal TC Server Thread Pool	False
Total Request Count	Pivotal TC Server Thread Pool	True
Total Request Error Count	Pivotal TC Server Thread Pool	True
Total Request Processing Time	Pivotal TC Server Thread Pool	True
JSP Count	Pivotal TC Server Web Module	False
JSP Reload Count	Pivotal TC Server Web Module	False
JSP Unload Count	Pivotal TC Server Web Module	False

ActiveMQ Metrics

Metrics are collected for the ActiveMQ application service.

Table 1709: ActiveMQ Metrics

Metric Name	Category	KPI
Buffer Pool<InstanceName>| Count	Active MQ	False
Buffer Pool<InstanceName>| Memory Used	Active MQ	False

Metric Name	Category	KPI
Buffer Pool<InstanceName>|Total Capacity	Active MQ	False
Class Loading|Loaded Class Count	Active MQ	False
Class Loading|Unloaded Class Count	Active MQ	False
Class Loading|Total Loaded Class Count	Active MQ	False
File Descriptor Usage|Max File Descriptor Count	Active MQ	False
File Descriptor Usage|Open File Descriptor Count	Active MQ	False
Garbage Collection<InstanceName>|Total Collection Count	Active MQ	False
Garbage Collection<InstanceName>|Total Collection Time	Active MQ	False
JVM Memory Pool<InstanceName>|Peak Usage|Committed Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Peak Usage|Initial Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Peak Usage|Maximum Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Peak Usage|Used Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Usage|Committed Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Usage|Initial Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Usage|Maximum Memory	Active MQ	False
JVM Memory Pool<InstanceName>|Usage|Used Memory	Active MQ	False
Application Availability	Active MQ	False
Threading|Thread Count	Active MQ	False
Uptime	Active MQ	False
UTILIZATION|Process CpuLoad	Active MQ	False
UTILIZATION|Memory Limit	ActiveMQ Broker	True

Metric Name	Category	KPI
UTILIZATION|Memory Percent Usage (%)	ActiveMQ Broker	True
UTILIZATION|Store Limit	ActiveMQ Broker	False
UTILIZATION|Store Percent Usage (%)	ActiveMQ Broker	False
UTILIZATION|Temp Limit	ActiveMQ Broker	False
UTILIZATION|Temp Percent Usage (%)	ActiveMQ Broker	False
UTILIZATION|Total Consumer Count	ActiveMQ Broker	True
UTILIZATION|Total Dequeue Count	ActiveMQ Broker	True
UTILIZATION|Total Enqueue Count	ActiveMQ Broker	True
UTILIZATION|Total Message Count	ActiveMQ Broker	True
JVM Memory|Heap Memory Usage|Initial Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Heap Memory Usage|Committed Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Heap Memory Usage|Maximum Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Heap Memory Usage|Used Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Non Heap Memory Usage|Committed Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Non Heap Memory Usage|Initial Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Non Heap Memory Usage|Maximum Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Non Heap Memory Usage|Used Memory	ActiveMQ JVM Memory Usage	False
JVM Memory|Object Pending FinalizationCount	ActiveMQ JVM Memory Usage	False
UTILIZATION|Process CpuLoad	ActiveMQ OS	False
UTILIZATION|System Cpu Load	ActiveMQ OS	False
UTILIZATION|Consumer Count	ActiveMQ Topic	True
UTILIZATION|Dequeue Count	ActiveMQ Topic	True
UTILIZATION|Enqueue Count	ActiveMQ Topic	True
UTILIZATION|Queue Size	ActiveMQ Topic	True
UTILIZATION|Producer Count	ActiveMQ Topic	False

Apache HTTPD Metrics

Metrics are collected for the Apache HTTPD application service.

Note:
Metrics are collected for the Events MPM. Metrics are not collected for the other MPMs.

Table 1710: Apache HTTPD Metrics

Metric Name	Category	KPI
UTILIZATION|Busy Workers	Apache HTTPD	True
UTILIZATION|Bytes Per Req	Apache HTTPD	False
UTILIZATION|Bytes Per Sec	Apache HTTPD	False
UTILIZATION|CPU Load	Apache HTTPD	True
UTILIZATION|CPU User	Apache HTTPD	False
UTILIZATION|Idle Workers	Apache HTTPD	True
UTILIZATION|Request Per Sec	Apache HTTPD	True
UTILIZATION|SCBoard Closing	Apache HTTPD	False
UTILIZATION|SCBoard DNS Lookup	Apache HTTPD	False
UTILIZATION|SCBoard Finishing	Apache HTTPD	False
UTILIZATION|SCBoard Idle Cleanup	Apache HTTPD	False
UTILIZATION|SCBoard Keep Alive	Apache HTTPD	False
UTILIZATION|SCBoard Logging	Apache HTTPD	False
UTILIZATION|SCBoard Open	Apache HTTPD	False
UTILIZATION|SCBoard Reading	Apache HTTPD	False
UTILIZATION|SCBoard Sending	Apache HTTPD	False
UTILIZATION|SCBoard Starting	Apache HTTPD	False
UTILIZATION|SCBoard Waiting	Apache HTTPD	False
UTILIZATION|Total Accesses	Apache HTTPD	False
UTILIZATION|Total Bytes	Apache HTTPD	True
UTILIZATION|Total Connections	Apache HTTPD	False
UTILIZATION|Uptime	Apache HTTPD	True
UTILIZATION|Asynchronous Closing Connections	Apache HTTPD	False
UTILIZATION|Asynchronous Keep Alive Connections	Apache HTTPD	False
UTILIZATION|Asynchronous Writing Connections	Apache HTTPD	False
UTILIZATION|ServerUptimeSeconds	Apache HTTPD	False
UTILIZATION|Load1	Apache HTTPD	False
UTILIZATION|Load5	Apache HTTPD	False
UTILIZATION|ParentServerConfigGeneration	Apache HTTPD	False
UTILIZATION|ParentServerMPMGeneration	Apache HTTPD	False
Application Availability	Apache HTTPD	False

Oracle DB Metrics

Metrics are collected for the Oracle DB application service.

Oracle DB cannot be activated on Linux platforms.

Table 1711: Oracle DB Metrics

Metric Name	Category	KPI
Utilization|Active Sessions	OracleDB	True
Utilization|Buffer CacheHit Ratio	OracleDB	False
Utilization|Cursor CacheHit Ratio	OracleDB	False
Utilization|Database Wait Time	OracleDB	False
Utilization|Disk Sort persec	OracleDB	False
Utilization|Enqueue Timeouts Persec	OracleDB	False
Utilization|Global Cache Blocks Corrupted	OracleDB	False
Utilization|Global Cache Blocks Lost	OracleDB	False
Utilization|Library CacheHit Ratio	OracleDB	False
Utilization|Logon persec	OracleDB	True
Utilization|Memory Sorts Ratio	OracleDB	True
Utilization|Rows persort	OracleDB	False
Utilization|Service Response Time	OracleDB	False
Utilization|Session Count	OracleDB	True
Utilization|Session Limit	OracleDB	False
Utilization|Shared Pool Free	OracleDB	False
Utilization|Temp Space Used	OracleDB	False
Utilization|Total Sorts persec	OracleDB	False
Utilization|Physical Read Bytes Persc	OracleDB	False
Utilization|Physical Read IO Requests Persc	OracleDB	False
Utilization|Physical Read Total Bytes Persec	OracleDB	False
Utilization|Physical Reads Persec	OracleDB	True
Utilization|Physical Reads Per Txn	OracleDB	False
Utilization|Physical Write Bytes Persc	OracleDB	False
Utilization|Physical Write IO Requests Persc	OracleDB	False
Utilization|Physical Write Total Bytes Persc	OracleDB	False
Utilization|Physical Writes Persc	OracleDB	True
Utilization|Physical Writes Per Txn	OracleDB	False
Utilization|User Commits Percentage	OracleDB	False
Utilization|User Commits Persc	OracleDB	False
Utilization|User Rollbacks Percentage	OracleDB	False
Utilization|User Rollbacks persec	OracleDB	True
Utilization|User Transaction Persec	OracleDB	False
Utilization|Database Time Persc	OracleDB	False
Application Availability	Oracle DB	False

Cassandra Metrics

Metrics are collected for the Cassandra application service.

Table 1712: Cassandra Metrics

Metric Name	Category	KPI
Cache<InstanceName>|Capacity	Cassandra	False
Cache<InstanceName>|Entries	Cassandra	True
Cache<InstanceName>|HitRate	Cassandra	True
Cache<InstanceName>|Requests	Cassandra	True
Cache<InstanceName>|Size	Cassandra	False
ClientRequest<InstanceName>|Failures	Cassandra	False
ClientRequest<InstanceName>|Latency	Cassandra	False
ClientRequest<InstanceName>|Timeouts	Cassandra	False
ClientRequest<InstanceName>|Total Latency	Cassandra	False
ClientRequest<InstanceName>|Unavailables	Cassandra	False
CommitLog|Pending Tasks	Cassandra	False
CommitLog|Total Commit Log Size	Cassandra	False
Compaction|Bytes Compacted	Cassandra	False
Compaction|Completed Tasks	Cassandra	False
Compaction|Pending Tasks	Cassandra	False
Compaction|Total Compactions Completed	Cassandra	False
Connected Native Clients	Cassandra	False
HeapMemoryUsage|committed	Cassandra	False
HeapMemoryUsage|init	Cassandra	False
HeapMemoryUsage|max	Cassandra	False
HeapMemoryUsage|used	Cassandra	False
NonHeapMemoryUsage|committed	Cassandra	False
NonHeapMemoryUsage|init	Cassandra	False
NonHeapMemoryUsage|max	Cassandra	False
NonHeapMemoryUsage|used	Cassandra	False
ObjectPendingFinalizationCount	Cassandra	False
Storage|Exceptions Count	Cassandra	False
Storage|Load Count	Cassandra	False
Table<InstanceName>|Coordinator Read Latency	Cassandra	False
Table<InstanceName>|Live Diskspace Used	Cassandra	False
Table<InstanceName>|Read Latency	Cassandra	False
Table<InstanceName>|Total Diskspace Used	Cassandra	False
Table<InstanceName>|Total Read Latency	Cassandra	False
Table<InstanceName>|Total Write Latency	Cassandra	False
Table<InstanceName>|Write Latency	Cassandra	False
ThreadPools<InstanceName>|Active Tasks	Cassandra	False
ThreadPools<InstanceName>|Currently Blocked Tasks	Cassandra	False
ThreadPools<InstanceName>|Pending Tasks	Cassandra	False

Metric Name	Category	KPI
Application Availability	Cassandra	False

HyperV Metrics
Metrics are collected for the HyperV application service.

Table 1713: HyperV Metrics

Metric Name	Category	KPI
VM:Hyper-V Virtual Machine Health Summary|Health Critical	HyperV	False
VM<instanceName>|Physical Memory	HyperV	False
VM<instanceName>Hv VP 0|Total Run Time	HyperV	False
VM<instanceName>|Bytes Received	HyperV	False
VM<instanceName>|Bytes Sent	HyperV	False
VM<instanceName>|Error Count	HyperV	False
VM<instanceName>|Latency	HyperV	False
VM<instanceName>|Queue Length	HyperV	False
VM<instanceName>|Throughput	HyperV	False
CPU<instanceName>|Idle Time	HyperV	True
CPU<instanceName>|Processor Time	HyperV	True
CPU<instanceName>|User Time	HyperV	True
Disk<instanceName>|Avg Disk Queue Length	HyperV	False
Disk<instanceName>|Idle Time	HyperV	False
Disk<instanceName>|Read Time	HyperV	True
Disk<instanceName>|Write Time	HyperV	True
Process<instanceName>|Private Bytes	HyperV	False
Process<instanceName>|Processor Time	HyperV	False
Process<instanceName>|Thread Count	HyperV	False
Process<instanceName>|User Time	HyperV	False
System|Processes	HyperV	False
System|Processor Queue Length	HyperV	False
System|System UpTime	HyperV	False
Memory|Available Bytes	HyperV	False
Memory|Cache Bytes	HyperV	False
Memory|Cache Faults	HyperV	False
Memory|Pages	HyperV	False
Network<instanceName>|Packets Outbound Error	HyperV	False
Network<instanceName>|Packets Received Error	HyperV	False
Application Availability	HyperV	False

MongoDB Metrics
Metrics are collected for the MongoDB application service.

Table 1714: MongoDB Metrics

Metric Name	Category	KPI
UTILIZATION|Active Reads	MongoDB	True
UTILIZATION|Active Writes	MongoDB	True
UTILIZATION|Connections Available	MongoDB	False
UTILIZATION|Connections Total Created	MongoDB	False
UTILIZATION|Current Connections	MongoDB	True
UTILIZATION|Cursor Timed Out	MongoDB	True
UTILIZATION|Deletes Per Sec	MongoDB	False
UTILIZATION|Document Inserted	MongoDB	False
UTILIZATION|Document Deleted	MongoDB	False
UTILIZATION|Flushes Per Sec	MongoDB	False
UTILIZATION|Inserts Per Sec	MongoDB	False
UTILIZATION|Net Input Bytes	MongoDB	False
UTILIZATION|Open Connections	MongoDB	True
UTILIZATION|Page Faults Per Second	MongoDB	False
UTILIZATION|Net Output Bytes	MongoDB	False
UTILIZATION|Queries Per Sec	MongoDB	False
UTILIZATION|Queued Reads	MongoDB	True
UTILIZATION|Queued Writes	MongoDB	True
UTILIZATION|Total Available	MongoDB	False
UTILIZATION|Total Deletes Per Sec	MongoDB	False
UTILIZATION|Total Passes Per Sec	MongoDB	False
UTILIZATION|Total Refreshing	MongoDB	False
UTILIZATION|Updates Per Sec	MongoDB	False
UTILIZATION|Volume Size MB	MongoDB	False
Application Availability	MongoDB	False
UTILIZATION|Collection Stats	MongoDB DataBases	False
UTILIZATION|Data Index Stats	MongoDB DataBases	True
UTILIZATION|Data Indexes	MongoDB DataBases	False
UTILIZATION|Data Size Stats	MongoDB DataBases	True
UTILIZATION|Average Object Size stats	MongoDB DataBases	False
UTILIZATION|Num Extents Stats	MongoDB DataBases	False

Riak KV Metrics

Metrics are collected for the Riak KV application service.

Table 1715: Riak KV Metrics

Metric Name	Category	KPI
UTILIZATION|CPU Average	Riak KV	False
UTILIZATION|Memory Processes	Riak KV	False

Metric Name	Category	KPI
UTILIZATION|Memory Total	Riak KV	False
UTILIZATION|Node GETs	Riak KV	True
UTILIZATION|Node GETs Total	Riak KV	False
UTILIZATION|Node PUTs	Riak KV	True
UTILIZATION|Node PUTs Total	Riak KV	False
UTILIZATION|PBC Active	Riak KV	True
UTILIZATION|PBC Connects	Riak KV	True
UTILIZATION|Read Repairs	Riak KV	True
UTILIZATION|vNODE Index Reads	Riak KV	True
UTILIZATION|vNODE Index Writes	Riak KV	True
Application Availability	Riak KV	False

Network Time Protocol Metrics

Metrics are collected for the Network Time Protocol application service.

Table 1716: Network Time Protocol Metrics

Metric Name	Category	KPI
NTPD |delay	Network Time Protocol	True
NTPD | jitter	Network Time Protocol	True
NTPD | offset	Network Time Protocol	True
NTPD | poll	Network Time Protocol	False
NTPD | reach	Network Time Protocol	True
NTPD | when	Network Time Protocol	False
Application Availability	Network Time Protocol	False

WebSphere Metrics

Metrics are collected for the WebSphere application service.

Table 1717: WebSphere Metrics

Metric Name	Category	KPI
Thread Pool|Active Count| Current	Thread Pool	False
Thread Pool|Active Count|High	Thread Pool	False
Thread Pool|Active Count|Low	Thread Pool	False
Thread Pool|Active Count|Lower	Thread Pool	False
Thread Pool|Active Count|Upper	Thread Pool	False
JDBC|Close Count	JDBC	False
JDBC|Create Count	JDBC	False
JDBC|JDBC Pool Size|Average	JDBC	False
JDBC|JDBC Pool Size|Current	JDBC	False
JDBC|JDBC Pool Size|Lower	JDBC	False

Metric Name	Category	KPI
JDBC|JDBC Pool Size|Upper	JDBC	False
Garbage Collection<InstanceName>|Total Collection Count	WebSphere	False
Garbage Collection<InstanceName>|Total Collection Time	WebSphere	False
JVM Memory|Heap Memory Usage|Committed Memory	WebSphere	False
JVM Memory|Heap Memory Usage|Initial Memory	WebSphere	False
JVM Memory|Heap Memory Usage|Maximum Memory	WebSphere	False
JVM Memory|Heap Memory Usage|Used Memory	WebSphere	False
JVM Memory|Non Heap Memory Usage|Committed Memory	WebSphere	False
JVM Memory|Non Heap Memory Usage|Initial Memory	WebSphere	False
JVM Memory|Non Heap Memory Usage|Maximum Memory	WebSphere	False
JVM Memory|Non Heap Memory Usage|Used Memory	WebSphere	False
JVM Memory|Number of Object Pending Finalization Count	WebSphere	False
JVM Memory| Pool<InstanceName>|Peak Usage|Committed Memory	WebSphere	False
JVM Memory| Pool<InstanceName>|Peak Usage|Initial Memory	WebSphere	False
JVM Memory| Pool<InstanceName>|Peak Usage|Maximum Memory	WebSphere	False
JVM Memory| Pool<InstanceName>|Peak Usage|Used Memory	WebSphere	False
JVM Memory| Pool<InstanceName>|Usage| Committed Memory	WebSphere	False
JVM Memory| Pool<InstanceName>|Usage| Initial Memory	WebSphere	False
JVM Memory| Pool<InstanceName>|Usage| Maximum Memory	WebSphere	False

Metric Name	Category	KPI
JVM Memory|Pool<InstanceName>|Usage|Used Memory	WebSphere	False
Process Cpu Load	WebSphere	False
System Cpu Load	WebSphere	False
System Load Average	WebSphere	False
Application Availability	WebSphere	False

Java Application Metrics

Metrics are collected for the Java application service.

Table 1718: Java Application Metrics

Metric Name	Category	KPI
Buffer Pool<InstanceName>|Count	Java Application	False
Buffer Pool<InstanceName>|Memory Used	Java Application	False
Buffer Pool<InstanceName>|Total Capacity	Java Application	False
Class Loading|Loaded Class Count	Java Application	True
Class Loading|Total Loaded Class Count	Java Application	False
Class Loading|Unloaded Class Count	Java Application	False
Garbage Collection<InstanceName>|Total Collection Count	Java Application	False
Garbage Collection<InstanceName>|Total Collection Time	Java Application	False
JVM Memory|Heap Memory Usage|Committed Memory	Java Application	False
JVM Memory|Heap Memory Usage|Initial Memory	Java Application	False
JVM Memory|Heap Memory Usage|Maximum Memory	Java Application	False
JVM Memory|Heap Memory Usage|Used Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Peak Usage|Committed Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Peak Usage|Initial Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Peak Usage|Maximum Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Peak Usage|Used Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Usage|Committed Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Usage|Initial Memory	Java Application	False
JVM Memory|JVM Memory Pool<InstanceName>|Usage|Maximum Memory	Java Application	False

Metric Name	Category	KPI
JVM Memory|JVM Memory Pool<InstanceName>| Usage|Used Memory	Java Application	False
JVM Memory|Non Heap Memory Usage|Committed Memory	Java Application	False
JVM Memory|Non Heap Memory Usage|Initial Memory	Java Application	False
JVM Memory|Non Heap Memory Usage|Maximum Memory	Java Application	False
JVM Memory|Non Heap Memory Usage|Used Memory	Java Application	False
JVM Memory|Object Pending Finalization Count	Java Application	False
Uptime	Java Application	True
Threading|Thread Count	Java Application	True
Process CPU Usage %	Java Application	False
System CPU Usage %	Java Application	False
System Load Average %	Java Application	False

Remote Check Metrics

Metrics are collected for object types such as HTTP, ICMP, TCP, and UDP.

HTTP Metrics
VCF Operations discovers metrics for HTTP remote checks.

HTTP Metrics

Table 1719: HTTP Metrics

Metric Name	KPI
Availability	False
Content Length	False
Response Code	False
Response Time	True
Result Code	False

ICMP Metrics
VCF Operations discovers metrics for the ICMP object type.

Table 1720: ICMP Metrics

Metric Name	KPI
Availability	False
Average Response Time	True
Packet Loss (%)	False
Packets Received	False
Packets Transmitted	False

Metric Name	KPI
Result Code	False

TCP Metrics

VCF Operations discovers metrics for the TCP object type.

Table 1721: TCP Metrics

Metric Name	KPI
Availability	False
Response Time	True
Result Code	False

UDP Metrics

VCF Operations discovers metrics for the UDP object type.

Table 1722: UDP Metrics

Metric Name	KPI
Availability	False
Response Time	True
Result Code	False

Linux Process Metrics

Metrics are collected for Linux services.

Table 1723: Linux Process Metrics

Metric Name	Category	KPI
AVAILABILITY|Resource Availability	Processes	False
UTILIZATION|Memory Usage (%)	Processes	False
UTILIZATION|CPU Usage (%)	Processes	False
UTILIZATION|Number of Processes	Processes	False

Windows Service Metrics

Metrics are collected for Windows services.

Table 1724: Windows Service Metrics

Metric Name	Category	KPI
AVAILABILITY|Resource Availability	Services	False
UTILIZATION|Memory Usage(%)	Services	False
UTILIZATION|CPU Usage(%)	Services	False

OS and Application Monitoring Properties

Properties are collected for operating systems, application services, remote checks, Linux processes, and Windows services which can be used to create reports, views, and dashboards.

Guest Information Properties

VCF Operations displays the following guest information properties for all objects created by the OS and Application Monitoring management pack.

- Guest Info
  - Hostname
  - IP
  - OS Name
  - OS Version
  - Telegraf Version

Troubleshooting

Troubleshooting Agent Installation

Agent Install Failure Because of the vCenter User Permissions

Guest operation privileges are required to install agents on end-point virtual machines.

Agent installation fails with the following error message if there are no guest operation privileges:

```
vCenter adapter user is missing either of the following guest operations privileges –
 execute, modify, query
```

1. Verify that you have configured a vCenter adapter.
2. The vCenter user account with which the vCenter adapter is configured in VCF Operations, should have the following permissions: **Guest operation modifications**, **Guest operation program execution**, and **Guest operation queries**.

Agent Install Failure Because NTP is Not in Sync

After you install or upgrade to the latest version of cloud proxy, you must set up accurate timekeeping as part of the deployment. If the time settings between cloud proxy and VCF Operations are not synchronized, you face agent installation and metric collection issues. Ensure time synchronization between the endpoint VMs, vCenter, ESX Hosts, cloud proxy and VCF Operations using the Network Time Protocol (NTP).

- Agent installation fails
- Log in to cloud proxy and run the following command to stop the NTP daemon:

  ```
  systemctl stop ntpd
  ```
- Run the following command to update the time immediately from an NTP server:

  ```
  ntpdate time.vmware.com
  ```

  Note:
  Replace `time.vmware.com` with a suitable time server setting. You can use the FQDN or IP of the time server.
- Enter the following command to start the NTP daemon:

  ```
  systemctl start ntpd
  ```

  Note:
  The system time takes about five minutes to sync with the NTP server time.

Agent Install Fails on a Linux End Point

Install of an agent on a Linux end point fails for a non-root user with a specific set of privileges.

Agent installation fails with the following error if the `tty` command is not added:

```
Bootstrap Failed for VM <VM ID> with error message:{ "status":"FAILED", "data":
[ { "status":"FAILED", "message":"Failed - install - passwordless sudo access is required
 for the user <Install Username> on the command mkdir. [sudo: sorry, you must have a tty
 to run sudo]", "stage":"0" } ], "currentstage":"0", "totalstages":"0" }
```

The following are minimal necessary permissions of the user to install agents and should be mentioned in the `sudoers` file:

- The following are minimal necessary permissions of the user to install agents and should be mentioned in the `sudoers` file. For example, for a user called **telegrafinstall**, you can find the sudoers file in the **/etc/sudoers** file or in the folder **/etc/sudoers.d/:**. Add these lines to `/etc/sudoers`, if you have not added them.

  ```
  Defaults:telegrafinstall !requiretty

  Cmnd_Alias ARC_INSTALL_USER_COMMANDS=/usr/bin/cp*,/bin/cp*,/usr/bin/mkdir*,/bin/mkdir*,/usr/bin/chmod*,/
  bin/chmod*,/opt/vmware/ucp/bootstrap/uaf-bootstrap.sh,/opt/vmware/ucp/ucp-minion/bin/ucp-minion.sh

  telegrafinstall ALL=(ALL)NOPASSWD: ARC_INSTALL_USER_COMMANDS
  ```

Agent Install Fails on Windows When UAC is Deactivated

Install of the agent fails even when UAC is deactivated.

Make sure UAC (previously known as LUA) was deactivated on Windows in the following way:

a) In the registry path `HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System`, set the value for the key `EnableLUA` to `0`.
b) You must reboot the machine for the changes to take effect.

Agent Install Fails on Windows with a Permission Denied Error

In Windows, during bootstrap, when the Telegraf folder is renamed to `ucp-telegraf`, it can result in a failure because of a permission error.

Sometimes, there are certain antiviruses running, which prevent the application from renaming or modifying the directory or files. In such a situation, the following error message is displayed:

```
Install telegraf [unable to install telegraf due to system error : [WinError 5] Access is denied:
'C:\\VMware\\UCP\\ucp-telegraf']"}].
```

Deactivate the antivirus and then proceed with bootstrapping.

Agent Install Does Not Progress

During agent install, new tasks do not progress beyond the Starting phase, from recent tasks. Adapter logs are not written.

Verify that the adapter instance in cloud proxy is in a Data Collecting state. If not, restart the adapter instance from the user interface. In the VCF Operations navigation bar at the top, click **Home** and in the left navigation pane, select **Administration** > **Configurations**, and then click the **Inventory Management** tile. Navigate to **VCF Operations Application Management Adapter Instance**. From the **Objects** tab in the right pane, select the adapter instance and click **Stop Collecting** and then **Start Collecting**.

Agent Install Fails Without an Error Message

Agent install fails without any error message in the user interface.

The `uaf_bootstrap.log` at the endpoint VM displays the following log message: `'findstr' is not recognized as an internal or external command.`

This happens because `C:\Windows\System32` is not available in the environment variable *PATH*.

Add `C:\Windows\System32` to the environment variable *PATH*.

Script Download Fails on a Windows Platform

When a file is downloaded from cloud proxy to a Windows end point, it could fail due to security protocols.

Script download fails on a Windows platform with the following message:

```
The request was aborted: Could not create SSL/TLS secure channel.
```

There are three kinds of PowerShell scripts hosted in cloud proxy that can be downloaded and executed at the Windows end-point VMs for different purposes:

- To install custom Telegraf using a script (download.ps1).
- To install custom Telegraf on a physical server (unmanagedagent_setup_sample.ps1).
- To configure open source Telegraf on managed or unmanaged VMs (open_source_telegraf_monitor.ps1).

Ignore the `ServerCertificateValidationCallback` using the following command.

```
if (-not ([System.Management.Automation.PSTypeName]'ServerCertificateValidationCallback').Type)
{
$certCallback = @"
    using System;
    using System.Net;
    using System.Net.Security;
    using System.Security.Cryptography.X509Certificates;
    public class ServerCertificateValidationCallback
    {
        public static void Ignore()
        {
            if(ServicePointManager.ServerCertificateValidationCallback ==null)
            {
                ServicePointManager.ServerCertificateValidationCallback +=
                    delegate
                    (
                        Object obj,
                        X509Certificate certificate,
                        X509Chain chain,
                        SslPolicyErrors errors
                    )
                    {
                        return true;
                    };
            }
        }
    }
"@
    Add-Type $certCallback
 }
[ServerCertificateValidationCallback]::Ignore()
[Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
```

After downloading and executing the required script, `ServerCertificateValidationCallback` can be activated.

Troubleshooting Plugin Related Failures

Unable to Activate a Plugin

Unable to activate a plugin with the same fields until the plugin configuration is deleted.

An error message is displayed in the user interface of VCF Operations that states the following:

```
Failed to update resource: Resource with same key already exists
```

Manually delete the existing plugin configuration and then continue with the activation of the plugin. If the problem persists, delete the corresponding resource from the inventory.

RabbitMQ Plugin Error

You might receive an error while monitoring the RabbitMQ plugin.

While monitoring the RabbitMQ plugin, an error might occur if you upgrade to VCF Operations, which has a Telegraf version of 1.19. The following message is displayed:

```
getting "/api/federation-links" failed: 404 Not Found
```

The Federation plugins in the RabbitMQ setup are not activated.

Activate the Federation plugins in the RabbitMQ setup by running the following commands in the RabbitMQ VM:

```
rabbitmq-plugins enable rabbitmq_federation

rabbitmq-plugins enable rabbitmq_federation_management
```

Troubleshooting Metric Collection

Troubleshoot Agent Installation and Metric Collection Issues

If the time settings between cloud proxy and VCF Operations are not synchronized, you might face agent installation and metric collection issues. Eventually, you might not see any metrics in the VCF Operations dashboards.

You might notice the following issues in VCF Operations:

- You cannot install an agent in the Windows and Linux target VMs.

Time synchronization is a prerequisite of the TLS/SSO communication between client and server.

If the VCF Operations and cloud proxy are not time synchronized, the test connection fails while configuring cloud proxy in VCF Operations.

If the Windows and Linux target VMs are not time synchronized with VCF Operations, communication between cloud proxy and agents will break after installing the agents. Hence monitored metrics are not sent to VCF Operations. Alternatively, stop and restart the agent to resolve this issue.

1. Check the VCF Operations support bundle in the following path: `COLLECTOR/adapters/APPOSUCPAdapter/` for errors.
2. Check the cloud proxy support bundle, *ucpapi.log*, for errors.
3. Ensure time synchronization between cloud proxy, VCF Operations and the Windows and Linux target VMs.
4. To start and restart the agent, see Additional Operations from the Manage Telegraf Agents Page .

Troubleshooting Content Upgrade for an End Point

Content upgrade for an end point fails with the following error:

```
Timeout Error. Please retry the action after some time.
```

Sometimes content upgrade for an end point fails because of a timeout in the cloud proxy.

Retrigger content upgrade for the end point to resolve the issue.

Telegraf Agent Related Actions Fail after Cloud Proxy is Restarted or Upgraded

After an upgrade of VCF Operations and cloud proxy from 8.4 to a later release or if you restart cloud proxy, Telegraf agent management related actions could fail with the error message `Connect to Salt Master`.

The saltmaster docker container inside cloud proxy does not run properly causing all Telegraf related actions to fail.

1. SSH to the cloud proxy VM.
2. Verify if the salt process count by command is: `ps -ef | grep salt`.
3. If the salt process count is less than 15, run the following command: `/rpm-content/ucp/subsequentboot.sh`.
   You can view the log from the following location: `/opt/vmware/var/log/ucp-subsequentboot`.

Troubleshooting Using Support Bundles

Support bundles are required to troubleshoot problems that are related to application monitoring. For Linux and Windows end-point VMs, run the specified command and access the support bundle.

For End-Point VMs

1. Log in to the end point.
2. Run the following commands based on the end-point VM's operating system type:
   For Linux End Point VMs
   ```
   /opt/vmware/ucp/ucp-minion/bin/ucp-minion.sh --config /opt/vmware/ucp/salt-minion/etc/salt/grains --action
    gen_support_bundle --log_level INFO
   ```
   The support bundle is generated and placed as a ZIP file in the `/opt/vmware/ucp/support-bundle-endpoints/` directory.
   For Windows End Point VMs
   ```
   C:\VMware\UCP\ucp-minion\bin\ucp-minion.bat --config C:\VMware\UCP\salt\conf\grains --action
    gen_support_bundle --log_level INFO
   ```
   The support bundle is generated and placed as a ZIP file in the `%SystemDrive%\VMware\UCP\support-bundle-endpoints\` directory.

Objects Do Not Receive Data from the Adapter Instance

In case of a HA failover, you see the following alert: `Objects are not receiving data from the adapter instance`. This is an adapter-specific alert that does not determine whether the object is receiving data from any of the adapters. It is recommended that you deactivate this alert.

1. Create a new policy that inherits the AppOS adapter instance's pre-applied policy. In the VCF Operations navigation bar at the top, click **Home** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Policy Definiton** tile.
2. Click **Add** to add a policy, or select the policy and click **Edit Policy** to edit an existing policy.
   You can add and edit policies and remove certain policies. You can use the Base Settings policy or the Default Policy as the root policy for the settings in other policies that you create. You can set any policy to be the default policy.

3. In the **Create Policies** workspace, assign a name to the policy and enter the description.
   a) Give the policy a meaningful name and description so that all users know the purpose of the policy.
   b) From the **Inherit From** drop-down option, select the policy to use as a baseline to define the settings for your new local policy.
4. Click **Create Policy**. The **Create Policies** workspace provides options to customize your policy.
5. Click the **Alerts and Symptoms** tile and search for the alert called `Objects are not receiving data from adapter instances` from the policy created in step 1.
6. Click on the alert and deactivate it.
7. Apply the policy that is created in step 1 to the AppOS adapter instances.

Monitoring Applications Using Open Source Telegraf

You can use open source Telegraf to monitor application services and operating systems on a Linux platform or on a Windows platform.

Monitoring Application Services and Operating Systems using Open Source Telegraf

You can use open source Telegraf that runs on a vCenter VM and physical servers to send metrics to VCF Operations using a helper script or by providing specific configurations required on the end point to post metrics to cloud proxy. The helper script adds necessary configurations in the Telegraf configuration that is directly identified by VCF Operations.

You can monitor application services and operating systems that are supported by VCF Operations and you can also monitor unsupported application services. For a list of supported application services, see Supported Application Services . Supported application services (curated plugins) are supported by application monitoring in VCF Operations.

Open source Telegraf is supported on Linux and Windows platforms.

Notice: VMware provides support only for the installation of open source Telegraf and its intial configuration that is required to point to cloud proxy. After the initial configuration is complete, and we have made sure that the metrics are received consistently for a couple of end points, and there are no common patterns of failures observed on VCF Operations, no further support is offered. Any other queries on the agent has to go through the Telegraf open source community. Issues related to Telegraf can be raised through: https://github.com/influxdata/telegraf/issues.

Points to Note

As you can monitor both supported and unsupported application services using open source Telegraf, keep in mind the following points.

- Port Information
  In VCF 9.1, port 8443 is deprecated and will reach end of support in a future version. It is recommended that you use port 443 so that the end point VM/physical server has access to port 443 of an individual cloud proxy or cloud proxies that are a part of a collector group.
  - As a new user, port 443 is enabled by default so that the end point VM/physical server must have access to port 443 of an individual cloud proxy or cloud proxies that are a part of a collector group.
  - For users who upgrade to VCF 9.1, it is recommended that you reconfigure your open source Telegraf workloads to use port 443 instead of ports 8443. Complete the following steps.
    - Reconfigure the collector group. As an existing user, to use port 443 for cloud proxies that are part of a HA enabled collector group, reconfigure the collector group by using the **Retry Collector Group Configuration** option. For more information, see Configuring Collector Groups.
- Mutual authentication
  - In VCF 9.1 and later versions, new cloud proxies and collector group configurations are mutually authenticated. There will be no change in the behavior of existing configurations.

  - For individual cloud proxies and cloud proxies that are a part of a non-HA collector group, mutual authentication is activated post upgrade to VCF 9.1.
  - For collector groups that have HA activated, mutual athentication is activated after you trigger the **Reconfigure Collector Group** action or **Add/Remove** the cloud proxy from the HA-activated collector group.
- Load Balancing
  - In VCF 9.1 and later versions, for load balancing, use application monitoring HA activated collector groups with activated load balancing. You can activate load balancing in the collector group settings. For more information, see Adding a Collector Group. Expect a collector group HA failover for about 20 minutes.
  - In VCF 9.1 and later versions, re-balancing is now available for application monitoring HA activated collector groups using open source Telegraf. For more information about re-balancing, see Configuring Collector Groups.
- There are no alerts generated for non-supported application services.
- You cannot add additional metrics for supported application services.
- Application service objects that are not supported are named as follows: *<application service name_Generic>*
- Metrics for unsupported application services are displayed without proper categorization and appears as is from Telegraf.
- Metrics for unsupported application services are not localized and are provided only in English.
- If there are multiple instances of unsupported application services, to distinguish between them in the user interface, add the tag 'identifier' and provide a unique tag value in the `telegraf.conf` file. Example, [inputs.mongodb.tags] identifier="1".
- If multiple instances of the same supported application service is monitored by open source Telegraf, in the user interface, the display name is the same for all instances of the application service. To differentiate between the instances of the same application service, edit the display name from the user interface and provide a specific name.
- Inactivated application services with a managed agent are displayed as activated in the **Manage Telegraf Agents** page. Uninstall managed agents before you use open source Telegraf.
- Telegraf version 1.19.0 is not supported when you use open source Telegraf.

Install and Configure Open Source Telegraf

Install and configure open source Telegraf to monitor your applications.

Telegraf uses input plugins (where the metrics come from) and output plugins (where the metrics go) in the configuration files. You can see all the supported plugins at Plugin Directory. Input and output plugins should be written in the Telegraf configuration file and configuration directory. The Telegraf configuration file (telegraf.conf) lists all available Telegraf plugins.

Telegraf uses the `--config` flag to specify the configuration file location and `--config-directory` flag to include files ending with `.conf` in the specified directory. On most Linux systems, the default locations are `/etc/telegraf/telegraf.conf` for the main configuration file and `/etc/telegraf/telegraf.d` for the directory of configuration files. For Windows platforms, it will be in the location where the `telegraf zip` is extracted.

For more information, see the Telegraf documentation, Get Started and Configuration Options. After you have downloaded and installed Telegraf, you are ready to begin collecting and sending data. To collect and send data Telegraf should be configured. Follow the steps below:

1. Install open source Telegraf on the end point. If you have an instance installed, you can skip this step. To download and install a new instance of Telegraf, see the official documentation and search for the corresponding OS version from https://www.influxdata.com/time-series-platform/telegraf/ and https://portal.influxdata.com/downloads/.

   On a Windows platform only, after downloading and extracting Telegraf files, besides `telegraf.exe` and `telegraf.conf` files, create a folder with the name `telegraf.d` that you use to run the helper script and monitor applications.

2. Run the helper script to configure Telegraf to send data to cloud proxy.

   For more information, see Monitoring Applications using Open Source Telegraf on a Linux Platform and Monitoring Applications using Open Source Telegraf on a Windows Platform.

   After you run the helper script, a `cloudproxy-http.conf` file is created and the output plugin is added to the file with the properties necessary for sending data and the input plugins are updated for OS metrics. For OS

metrics configuration template, see Telegraf Configuration Details for Operating Systems. For information about the `cloudproxy-http.conf` configuration template, see Sample Configurations.

3. Using open source Telegraf, you can collect metrics from different types of application services.
   - If an application service (curated plugin) that is supported by Application Monitoring in VCF Operations is running on the end point and you want to monitor it, update the Telegraf configuration file or directory with necessary inputs for Telegraf.
     For a list of supported application services (curated plugins), see Supported Application Services . For the list of configurations, see Telegraf Configuration Details for Supported Application Services.
   - For unsupported (non-curated plugins) application services, update the Telegraf configuration file or directory with the necessary inputs for Telegraf. For more information, see the Telegraf documentation Plugin Directory. For example, if you want to collect data for Ethernet device stats (device name is eth0) you can create the `ethools.conf` file in the Telegraf configuration directory and add the following content (for more information, see the documentation about Ethtool Input Plugin):

     ```
     [[inputs.ethtool]]
     # List of interfaces to pull metrics for
      interface_include = ["eth0"]
     ```

   - Whenever the `telegraf-utils` script is executed with Open Source Telegraf, the `cloudproxy-http.conf` file will be overwritten in the `telegraf.d` folder. You must maintain a different config file (for example, `postgres.conf`) for other custom input plugins to retain the configuration.
4. Start or restart Telegraf.

Monitoring Applications using Open Source Telegraf on a Linux Platform

Use the helper script to monitor applications and operating systems on a Linux platform using open source Telegraf.

- Install the jq package. For more information, see the official documentation for jq from https://stedolan.github.io/jq/download/.
- Ensure that VMTools version >=10.2 is installed, if a vCenter VM is monitored by VCF Operations. `vmware-toolbox-cmd` is necessary only for vCenter VMs.
- Ensure that you have permissions to run the following commands and use the following packages:
  - Commands: /bin/bash, awk, sed, vmware-toolbox-cmd, jq, curl, uuidgen.
  - Packages: coreutils (chmod, chown, cat), net-tools (ip, /dev/tcp, curl, wget)
- Only IPv4 is supported at present for cloud proxy.
- Ensure that cloud proxy is up and online in VCF Operations. In the case of collector groups, ensure that at least one cloud proxy in the application monitoring high availability activated collector group is up and online in VCF Operations.
- Verify that unzip is at 6.0-20.el7 or above.
- The uuidgen package must exist on the endpoint VM/physical server
- The endpoint VM/physical server must have access to port 443 of cloud proxy or the virtual IP of the application monitoring high availability activated collector group.

The helper script is tested only on the following operating systems:

- CentOS 8.x
- RHEL 8.x
- SUSE 15.x
- OEL 8.x
- Ubuntu 16.x, Ubuntu 18.x, Ubuntu 20.x, and Ubuntu 22.x
- VMware Photon Linux

1. Download the helper script from cloud proxy located at `https://<CloudProxy-IP>/downloads/salt/telegraf-utils.sh`.

   Note:
   Use the relevant cloud proxy **IP address** for <CloudProxy-IP> in the preceding commands and location specified. For application monitoring high availability activated collector groups, provide the virtual IP.

2. Navigate to the directory where the script is downloaded.
3. Activate execution permission of the script for Linux VM.

```
chmod +x telegraf-utils.sh
```

4. Run the helper script to update Telegraf configurations.

```
telegraf-utils.sh  opensource -c cloud_proxy_ip_or_collector_group_name -t token -d telegraf_conf_dir -e
 telegraf_bin_path -v vmwareariaoperations_ip_or_fqdn -k skip_certificate -h certificate_path

Description of arguments:
      -c : [REQUIRED] cloud_proxy_ip_or_collector_group_name - Cloud Proxy IP or Collector Group Name
       -t : [REQUIRED] token - Auth Token of the user/account. For getting a new token:
          (https://<VMwareAriaOperations_IP>/suite-api/ or curl -ks -X POST https://
<VMwareAriaOperations_IP>/suite-api/api/auth/token/acquire -H \"Content-Type: application/json\"
 -H \"Accept: application/json\" -d \"{\"username\":\"<VMwareAriaOperations_USER>\",\"password\":
\"<VMwareAriaOperations_USER_PASSWORD>\"}\")
      -d : [REQUIRED] telegraf_conf_dir - Telegraf configuration directory. ex: /etc/telegraf/telegraf.d
      -e : [REQUIRED] telegraf_bin_path - Path of telegraf executable ex: /usr/bin/telegraf
      -v : [REQUIRED] vmwareariaoperations_ip_or_fqdn - IP/FQDN of VCF Operations
 master node and required for on-prem.Example: /bin/bash telegraf-utils.sh opensource -t
 gi7lwabjnvdfiawt4watzksuol8sywrjvg8kabh31mx9x1guepgyhycyx61dqrpq -v 10.10.10.100 -c 10.10.10.101 -d /etc/
telegraf/telegraf.d -e /usr/bin/telegraf
     -k : [MANDATORY]  skip_certificate - To skip certificate verification when downloading  certificate
 bundle from VCF Operations. Accepted values : {1,0}.  1 indicates skip and 0 indicates provide CA
 certificate part of -h argument.
      -h : [CONDITIONAL]  certificate_path - Path of the certificate used for SSL verification. This is
 mandatory field if skip_certificate is set to 0.
```

   Note:
   After you run the helper script, ensure that the respective configurations are set correctly in the given config directory (`-d` option) path with the name `cloudproxy-http.conf`. See Sample Configurations for more details. For managed vCenter VMs you might see unmanaged configurations, because of one of the following reasons:

   - vCenter VM details are not available in VCF Operations by the vCenter adapter. Wait for a minimum of one to two collection cycles after configuring the VCF OperationsvCenter cloud accounts.
   - An incorrect `AUTHENTIFICATION_TOKEN or vROps_IP`.

   Note:
   By default, the InfluxDB output plugin is active in the `telegraf.conf` file and data is sent to the influxdb server so that you do not get multiple warning messages in the logs about the lack of configured influxdb server comment the "[[outputs.influxdb]]" line. The following warning message is displayed:`W! [outputs.influxdb] When writing to [http://localhost:8086]: database "telegraf" creation failed: Post "http://localhost:8086/query": dial tcp [::1]:8086: connect: connection refused`

   Example: `#[[outputs.influxdb]]`

   Note:
   Ensure that the input plugins in the `telegraf.conf` file are related to the corresponding operating system. See Telegraf Configuration Details for Operating Systems.

5. Restart the Telegraf service.

```
systemctl restart telegraf
or
/usr/bin/telegraf -config /etc/telegraf/telegraf.conf -config-directory /etc/telegraf/telegraf.d
```

Managed VM object hierarchy:

- If a vCenter VM is monitored by VCF Operations, then the operating system and application objects fall under the respective **VM** > **OS object** > **'application service' instance**.

Machines not monitored by VCF Operations: If a vCenter VM or a physical server is not monitored by VCF Operations, then the operating system and application objects fall under **Infrastructure Operations** > **Configurations** > **Inventory Management** > **Endpoint** > **OS Object** > **'application service' instance**.

View Data Collected: Data is collected and appears in the **Manage Telegraf Agents** page (Operate > Workloads > Applications). The page is updated after some time. To view the details, you can filter by **Agent Type** > **Open source agent** from the **Manage Telegraf Agents** page .

Monitoring Applications using Open Source Telegraf on a Windows Platform

Use the helper script to monitor applications and operating systems on a Windows platform using open source Telegraf.

- Verify that Windows PowerShell is at 4.0 or above.
- Ensure that VMTools version >=10.2 if a vCenter VM is monitored by VCF Operations. `vmware-toolbox-cmd` is necessary only for vCenter VMs.
- Only IPv4 is supported at present for cloud proxy.
- Ensure that cloud proxy is up and online in VCF Operations. In the case of collector groups, ensure that at least one cloud proxy in the application monitoring high availability activated collector group is up and online in VCF Operations.
- The endpoint VM/physical server must have access to port 443 of cloud proxy or the virtual IP of the application monitoring high availability activated collector group.

The helper script is tested only on Windows Server 2012, 2012 R2, 2016, 2019, and 2022.

1. Download the helper script from cloud proxy located at `https://<CloudProxy-IP>/downloads/salt/telegraf-utils.ps1`.

   Note:
   Use the relevant cloud proxy **IP address** for <CloudProxy-IP> in the preceding commands and location specified. For application monitoring high availability activated collector groups, provide the virtual IP.

   If the script download fails with the following message: `The request was aborted: Could not create SSL/TLS secure channel`, follow the steps mentioned in Script Download Fails on a Windows Platform.

2. Navigate to the directory where the script is downloaded.
3. Run the helper script to update Telegraf configurations.

```
telegraf-utils.ps1  opensource -c cloud_proxy_ip_or_collector_group_name -t token -d telegraf_conf_dir -e
 telegraf_bin_path -v vmwareariaoperations_ip_or_fqdn -k skip_certificate -h certificate_path

Description of arguments:
      -c : [REQUIRED] cloud_proxy_ip_or_collector_group_name - Cloud Proxy IP or Collector Group Name
      -t : [REQUIRED] token - Auth Token of the user/account. For getting a new token:
           (https://<VMwareAriaOperations_IP>/suite-api/ or curl -ks -X POST https://
<VMwareAriaOperations_IP>/suite-api/api/auth/token/acquire -H \"Content-Type: application/json\"
 -H \"Accept: application/json\" -d \"{\"username\":\"<VMwareAriaOperations_USER>\",\"password\":
\"<VMwareAriaOperations_USER_PASSWORD>\"}\")
      -d : [REQUIRED] telegraf_conf_dir - Telegraf configuration directory and it is required argument.
 ex: C:\Telegraf\telegraf-1.20.4\telegraf.d
      -e : [REQUIRED] telegraf_bin_path - Path of telegraf executable ex: C:\Telegraf
\telegraf-1.20.4\telegraf.exe
      -v : [REQUIRED] vmwareariaoperations_ip_or_fqdn - IP/FQDN of VMware Aria Operations master
 node and required for on-prem.Example: .\telegraf-utils.ps1 opensource -c 10.10.10.101 -t
 41ef6601-6da4-4757-a51d-cbc08dd77355::4398b23d-e388-496e-ae91-bc04d5735345 -v 10.10.10.100 -d C:\Telegraf
\telegraf-1.20.4\telegraf.d -e C:\Telegraf\telegraf-1.20.4\telegraf.exe
      -k : [MANDATORY]  skip_certificate - To skip certificate verification when downloading certificate
 bundle from VMware Aria Operations. - Accepted values : {1,0}. 1 indicates skip and 0 indicates provide
 CA certificate part of -h argument
      -h : [CONDITIONAL]  certificate_path - Path of the certificate used for SSL Verification. This is
 mandatory field if skip_certificate is set to 0
```

Note:
After you run the helper script, ensure that the respective configurations are set correctly in the given config directory (`-d` option) path with the name `cloudproxy-http.conf`. See Sample Configurations for more details. For managed vCenter VMs you might see unmanaged configurations, because of one of the following reasons:

- vCenter VMs are not available in VCF Operations by the vCenter. Wait for a minimum of one to two collection cycles after configuring the vCenter cloud accounts.
- An incorrect `AUTHENTIFICATION_TOKEN or vROps_IP`.

Note:

- Do not use a space in the configuration path. Paths with spaces can be passed as a short name notation, such as `c:\PROGRA~1` for `c:\Program Files`.

Note:
Ensure that the input plugins in the `telegraf.conf` file are related to the corresponding operating system. See Telegraf Configuration Details for Operating Systems.

Note:

When you download the helper script from cloud proxy, http south configuration is updated with new cipher suites and known vulnerable suites are removed.

List of ciphers removed in 9.0:

'HIGH:!aNULL!ADH:!EXP:!MD5:!3DES:!CAMELLIA:!PSK:!SRP:!DH:!AES256-GCM-SHA384:!AES256-SHA256:!AES256-SHA:!AES128-GCM-SHA256:!AES128-SHA256:!AES128-SHA:@STRENGTH'

List of ciphers added in 9.0:

ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES128-GCM-SHA256:ECDHE-RSA-AES256-GCM-SHA384:TLS_AES_256_GCM_SHA384:TLS_AES_128_GCM_SHA256:TLS_CHACHA20_POLY1305_SHA256

4. Restart the Telegraf service.

```
telegraf.exe --config telegraf.conf --config-directory telegraf.d
```

Or you can make Telegraf a Windows service.

```
<Telegraf_executable_path> --config <Telegraf_config_file_path> --config-directory
 <Telegraf_config_directory_path> --service install net start telegraf
```

For example:

```
& 'C:\Telegraf\telegraf-1.20.4\telegraf.exe'  --config 'C:\Telegraf\telegraf-1.20.4\telegraf.conf' --
config-directory 'C:\Telegraf\telegraf-1.20.4\telegraf.d'  --service install
net start telegraf
```

Managed VM object hierarchy:

- If a vCenter VM is monitored by VCF Operations, then the operating system and application objects will fall under the respective **VM** > **OS object** > **'application service' instance**

Machines not monitored by VCF Operations: If a vCenter VM or a physical server is not monitored by VCF Operations, then the operating system and application objects will fall under **Infrastructure Operations** > **Configurations** > **Inventory Management** > **Endpoint** > **OS Object** > **'application service' instance**.

View Data Collected: Data is collected and appears in the **Manage Telegraf Agents** page. To view the details, you can filter by **Agent Type** > **Open source agent** from the **Manage Telegraf Agents** page.

Telegraf Configuration Details for Operating Systems

Linux Operating Systems

To collect Linux OS related metrics and properties as it is in the managed Telegraf agent installation flow and to have localized object types, names, alerts, symptoms, metrics and properties, and so on, the `telegraf.conf` file (in case of installed telegraf it is located in `/etc/telegraf/telegraf.conf`) should be customized and have the following content:

```
# Read metrics about cpu usage
[[inputs.cpu]]
  ## Whether to report per-cpu stats or not
  percpu = true
  ## Whether to report total system cpu stats or not
  totalcpu = true
  ## If true, collect raw CPU time metrics
  collect_cpu_time = true
  ## If true, compute and report the sum of all non-idle CPU states
  report_active = true

# Read metrics about memory usage
[[inputs.mem]]
  # no configuration

# Read metrics about system load & uptime
[[inputs.system]]
  ## Uncomment to remove deprecated metrics.
  # fielddrop = ["uptime_format"]

# Read metrics about network interface usage
[[inputs.net]]
  ## By default, telegraf gathers stats from any up interface (excluding loopback)
  ## Setting interfaces will tell it to gather these explicit interfaces,
  ## regardless of status.
  ##
 # interfaces = ["eth0"]

# Read metrics about swap memory usage
[[inputs.swap]]
  # no configuration

# Read metrics about disk usage by mount point
[[inputs.disk]]
  ## By default stats will be gathered for all mount points.
  ## Set mount_points will restrict the stats to only the specified mount points.
  # mount_points = ["/"]

  ## Ignore mount points by filesystem type.
  # ignore_fs = ["tmpfs", "devtmpfs", "devfs", "iso9660", "overlay", "aufs", "squashfs"]

# Get the number of processes and group them by status
[[inputs.processes]]
  # no configuration

# Read metrics about disk IO by device
[[inputs.diskio]]
  ## By default, telegraf will gather stats for all devices including
  ## disk partitions.
  ## Setting devices will restrict the stats to the specified devices.
  # devices = ["sda", "sdb", "vd*"]
  ## Uncomment the following line if you need disk serial numbers.
```

```
  # skip_serial_number = false
```

Windows Operating System

To collect Windows OS related metrics and properties as it is in a managed Telegraf agent installation flow and to have localized object types, names, alerts, symptoms, metrics and properties, and so on, the `telegraf.conf` file should be customized and have the following content.

For Windows OS, the `telegraf.conf` file default configurations were changed for Telegraf versions greater than or equal to 1.20.0 and includes Linux related input plugins, like `inputs.cpu, inputs.disk, inputs.diskio, inputs.kernel, inputs.mem, inputs.processes, inputs.swap, inputs.system`, and so on. They should be commented and those related to Windows should be uncommented.

```
[[inputs.win_perf_counters]]
PrintValid=true

[[inputs.win_perf_counters.object]]
  ObjectName = "Processor"
  Instances = ["*"]
  Counters = ["% Idle Time", "% Interrupt Time", "% Privileged Time", "% Processor Time", "% User Time",
 "Interrupts/sec", "% DPC Time"]
  Measurement = "win.cpu"
  IncludeTotal = true

[[inputs.win_perf_counters.object]]
  ObjectName = "LogicalDisk"
  Instances = ["*"]
  Counters = ["% Disk Read Time", "% Disk Write Time", "% Free Space", "% Idle Time", "Avg. Disk Bytes/Read",
 "Avg. Disk Bytes/Write", "Avg. Disk Queue Length", "Avg. Disk sec/Read", "Avg. Disk sec/Write", "Avg. Disk
 Write Queue Length", "Avg. Disk Read Queue Length", "Free Megabytes", "Split IO/Sec"]
  Measurement = "win.disk"

[[inputs.win_perf_counters.object]]
  ObjectName = "Memory"
  Counters = ["Available Bytes", "Cache Bytes", "Committed Bytes", "Cache Faults/sec", "Demand Zero Faults/
sec", "Page Faults/sec", "Pages/sec", "Transition Faults/sec", "Pool Nonpaged Bytes", "Pool Paged Bytes"]
  Instances = ["------"]
  Measurement = "win.mem"

[[inputs.win_perf_counters.object]]
  ObjectName = "Network Interface"
  Counters = ["Bytes Received/sec", "Bytes Sent/sec", "Packets Outbound Discarded", "Packets Outbound Errors",
 "Packets Received Discarded", "Packets Received Errors", "Packets Received/sec", "Packets Sent/sec",
 "Connections Established"]
  Instances = ["*"]
  Measurement = "win.net"
  IncludeTotal = true

[[inputs.win_perf_counters.object]]
  ObjectName = "Paging File"
  Counters = ["% Usage"]
  Instances = ["*"]
  Measurement = "win.paging"
  IncludeTotal = true

  [[inputs.win_perf_counters.object]]
    ObjectName = "Process"
```

```
    Counters = ["% Privileged Time", "% Processor Time", "% User Time", "Elapsed Time", "Handle Count", "IO
 Read Bytes/sec", "IO Read Operations/sec", "IO Write Bytes/sec", "IO Write Operations/sec", "Private Bytes",
 "Thread Count", "Virtual Bytes", "Working Set", "Working Set - Private"]
    Instances = ["_Total", "telegraf", "w3wp"]         # Replace this with a list of process names that you
 want to monitor. "_Total" is all processes combined
    Measurement = "win.process"

[[inputs.win_perf_counters.object]]
  ObjectName = "System"
  Counters = ["Context Switches/sec", "Processes", "Processor Queue Length", "System Calls/sec", "System Up
 Time", "Threads"]
  Instances = ["------"]
  Measurement = "win.system"

[[inputs.win_perf_counters.object]]
  ObjectName = "TCPv4"
  Counters = ["Connection Failures", "Connections Active", "Connections Established", "Connections Passive",
 "Connection Reset", "Segments Received/sec", "Segments Retransmitted/sec", "Segments Sent/sec"]
  Instances = ["------"]
  Measurement = "win.net.tcp"

[[inputs.win_perf_counters.object]]
  ObjectName = "TCPv6"
  Counters = ["Connection Failures", "Connections Active", "Connections Established", "Connections Passive",
 "Connection Reset", "Segments Received/sec", "Segments Retransmitted/sec", "Segments Sent/sec"]
  Instances = ["------"]
  Measurement = "win.net.tcp"

[[inputs.win_perf_counters.object]]
  ObjectName = "UDPv4"
  Counters = ["Datagrams No Port/sec", "Datagrams Received/Errors", "Datagrams Received/sec", "Datagrams Sent/
sec"]
  Instances = ["------"]
  Measurement = "win.net.udp"

[[inputs.win_perf_counters.object]]
  ObjectName = "UDPv6"
  Counters = ["Datagrams No Port/sec", "Datagrams Received/Errors", "Datagrams Received/sec", "Datagrams Sent/
sec"]
  Instances = ["------"]
  Measurement = "win.net.udp"
```

Sample Configurations

If you do not use the helper script, you must provide specific configurations required on the end point to post metrics to cloud proxy. Based on whether the vCenter VM is managed or unmanaged, the following configurations must be provided for open source Telegraf.You must provide correct values for the variables enclosed in <>.

If you do not use the helper script, you must download mandatory_tags.sh or mandatory_tags.bat and provide the path to the script that you have downloaded and the path to the Telegraf executable. For example, in the following code, you must provide the following: <Path to mandatory_tags.sh/bat> <path to telegraf executable>.

For Linux, download the mandatory_tags.sh from cloud proxy located at `https://<CP_IP>/downloads/salt/mandatory_tags.sh`.

For Windows, download the mandatory_tags.ps1 from cloud proxy located at `https://<CP_IP>/downloads/salt/mandatory_tags.ps1`.

Note:
For Windows and Linux, use the relevant cloud proxy IP address/FQDN for <CP_IP> in the preceding commands and location specified. For application monitoring high availability activated collector groups, provide the virtual IP

Managed vCenter VM

```
[agent]
  interval = "300s"
  round_interval = true
  metric_batch_size = 1000
  metric_buffer_limit = 2000
  collection_jitter = "0s"
  flush_interval = "60s"
  flush_jitter = "0s"
  precision = ""
  debug = false
  quiet = false
  logfile = ""
  hostname = "<VM_NAME/HOSTNAME>"
  omit_hostname = false

# Configuration for HTTP server to send metrics to
[[outputs.http]]
  url = "https://<CP_IP_or_FQDN>/opensource/default/metric"
  timeout = "5s"
  method = "POST"
  insecure_skip_verify = false
  tls_ca = "/etc/telegraf/telegraf.d/ca.pem"
  tls_cert = "/etc/telegraf/telegraf.d/cert.pem"
  tls_key = "/etc/telegraf/telegraf.d/key.pem"
  tls_min_version = "TLS13"
  data_format = "wavefront"
## Additional HTTP headers
[outputs.http.headers]
  Content-Type = "text/plain; charset=utf-8"
  vmId = "<VM_MOR>"
  vcid = "<VC_ID>"
  hostname = "<VM_NAME/HOSTNAME>"
  uuid = ""
[[inputs.exec]]
  commands = ["/bin/bash <Path to mandatory_tags.sh/bat> <path to telegraf executable>"]
  timeout = "5s"
  data_format = "influx"
```

Unmanaged vCenter VMs

```
[agent]
  interval = "300s"
  round_interval = true
  metric_batch_size = 1000
  metric_buffer_limit = 2000
  collection_jitter = "0s"
  flush_interval = "60s"
  flush_jitter = "0s"
  precision = ""
  debug = false
  quiet = false
```

```
  logfile = ""
  hostname = "<VM_NAME/HOSTNAME>"
  omit_hostname = false

# Configuration for HTTP server to send metrics to
[[outputs.http]]
   url = "https://<CP_IP_or_FQDN>/opensource/default/metric"
   timeout = "5s"
   method = "POST"
   insecure_skip_verify = false
   tls_ca = "/etc/telegraf/telegraf.d/ca.pem"
   tls_cert = "/etc/telegraf/telegraf.d/cert.pem"
   tls_key = "/etc/telegraf/telegraf.d/key.pem"
   tls_min_version = "TLS13"
   data_format = "wavefront"
   ## Additional HTTP headers
   [outputs.http.headers]
     Content-Type = "text/plain; charset=utf-8"
     uuid = "<UUID>"
     ip = "<IP_ADDRESS>"
     hostname = "<VM_NAME/HOSTNAME>"
[[inputs.exec]]
  commands = ["/bin/bash <Path to mandatory_tags.sh/bat> <path to telegraf executable>"]
  timeout = "5s"
  data_format = "influx"
```

Telegraf Configuration Details for Supported Application Services

For application services and custom monitoring plugins supported by VCF Operations, there are configuration details that you must follow.

Table 1725: Configuration Details for Supported Application Services

Configuration Details for Supported Application Services
Active Directory Here are the configuration details: `[[inputs.win_perf_counters]]` `  [[inputs.win_perf_counters.object]]` `  ObjectName = "DirectoryServices"` ` Counters = ["LDAP Active Threads","LDAP Client Sessions","LDAP Writes/sec","LDAP Searches/sec","LDAP Successful Binds/sec","LDAP New Connections/sec","LDAP Closed Connections/sec","LDAP UDP operations/sec","DS Threads in Use","DS Directory Writes/sec","DS Directory Reads/sec","DS Directory Searches/sec","DS Client Binds/sec","DS Server Binds/sec","DRA Pending Replication Synchronizations","DRA Sync Requests Made","DRA Sync Requests Successful","DRA Pending Replication Operations","DRA Inbound Objects/sec","DRA Inbound Bytes Total/sec","DRA Outbound Objects/sec","DRA Outbound Bytes Total/sec","Base Searches/sec","Database adds/sec","Database deletes/sec","Database modifys/sec","Database recycles/sec"]` ` Instances = ["*"]` ` Measurement = "ad.active.directory"`  ` [[inputs.win_perf_counters.object]]` ` ObjectName = "Security System-Wide Statistics"` ` Counters = ["NTLM Authentications","Kerberos Authentications","KDC AS Requests","KDC TGS Requests","Digest Authentications"]` ` Instances = ["------"]` ` Measurement = "ad.security.statistics"`  ` [[inputs.win_perf_counters.object]]` ` ObjectName = "Database"` ` Instances = ["*"]` ` Counters = ["Database Cache % Hit","Database Cache Page Faults/sec","Database Page Faults/sec","Database Cache Page Fault Stalls/sec","Database Cache Size"]` ` Measurement = "ad.database"` ` #IncludeTotal=false #Set to true to include _Total instance when querying for all (*).`  ` [[inputs.win_perf_counters.object]]` ` ObjectName = "DFS Namespace Service Referrals"` ` Instances = ["*"]` ` Counters = ["Requests Processed","Requests Failed","Avg. Response Time"]` ` Measurement = "ad.dfsn"`  ` [[inputs.win_perf_counters.object]]` ` ObjectName = "DFS Replication Service Volumes"` ` Instances = ["*"]` ` Counters = ["Data Lookups","Database Lookups","Database Commits"]` ` Measurement = "ad.dfs.replications"`  ` [[inputs.win_perf_counters.object]]` ` ObjectName = "DNS"` ` Counters = ["Dynamic Update Received","Dynamic Update Rejected","Recursive Queries","Recursive Queries Failure","Recursive Query Failure","Secure Update Failure","Total Query Received","Total Response Sent"]` ` Instances = ["*"]` ` Measurement = "ad.dns"`

ActiveMQ
Here are the configuration details:
`# Read Apache ActiveMQ information.`
`[[inputs.jolokia2_agent]]`
`  urls = ["http://localhost:8161/api/jolokia"]`
`  name_prefix = "activemq."`
`  plugin_name_override="activemq"`
`  username = "user"`
`  password = "user"`

Configuration Details for Supported Application Services

Apache HTTPD

Here are the configuration details:

```
# Read Apache status information (mod_status)
[[inputs.apache]]

  ## An array of URLs to gather from, must be directed at the machine
  ## readable version of the mod_status page including the auto query string.
  ## Default is "http://localhost/server-status?auto".
  ##
    ##urls = ["http://localhost/server-status?auto"]
  ##
  urls = ["http://127.0.0.1:80/server-status?auto"]
  ## Credentials for basic HTTP authentication.
    # username = "myuser"
    #password = "mypassword"

  ## Maximum time to receive response.
  # response_timeout = "5s"

  ## Optional SSL Config
      # ssl_ca = "/etc/telegraf/ca.pem"
    # ssl_cert = "/etc/telegraf/cert.pem"
    # ssl_key = "/etc/telegraf/key.pem"

  ## Use SSL but skip chain & host verification
    insecure_skip_verify = true
```

Cassandra

Here are the configuration details:

```
# Read Cassandra metrics through Jolokia
[[inputs.cassandra]]
  context = "/jolokia/read"
  ## List of cassandra servers exposing jolokia read service
  servers = ["myuser:mypassword@10.10.10.1:8778","10.10.10.2:8778",":8778"]
  ## List of metrics collected on above servers
  ## Each metric consists of a jmx path.
  ## This will collect all heap memory usage metrics from the jvm and
  ## ReadLatency metrics for all keyspaces and tables.
  ## "type=Table" in the query works with Cassandra3.0. Older versions might
  ## need to use "type=ColumnFamily"
  metrics  = [
    "/java.lang:type=Memory/HeapMemoryUsage",
    "/org.apache.cassandra.metrics:type=Table,keyspace=*,scope=*,name=ReadLatency"
  ]
```

Configuration Details for Supported Application Services

Hyper-V

Here are the configuration details:

```
 [[inputs.win_perf_counters]]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Virtual Machine Health Summary"
        Instances = ["------"]
        Measurement = "hyperv.vm.health"
        Counters = ["Health Ok", "Health Critical"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Dynamic Memory VM"
        Instances = ["*"]
        Measurement = "hyperv.vm.memory"
        Counters = ["Physical Memory", "Added Memory", "Guest Visible Physical Memory"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Hypervisor Virtual Processor"
        Instances = ["*"]
        Measurement = "hyperv.hypervisor.virtual.processor"
        Counters = ["% Guest Run Time", "% Hypervisor Run Time", "% Total Run Time"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Hypervisor Root Virtual Processor"
        Instances = ["*"]
        Measurement = "hyperv.hypervisor.root.virtual.processor"
        Counters = ["% Guest Run Time", "% Hypervisor Run Time", "% Total Run Time"]
        IncludeTotal = true

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Virtual IDE Controller (Emulated)"
        Instances = ["*"]
        Measurement = "hyperv.virtual.ide.controller"
        Counters = ["Write Bytes/sec", "Read Bytes/sec", "Written Sectors/sec", "Read Sectors/sec"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Virtual Network Adapter"
        Instances = ["*"]
        Measurement = "hyperv.virtual.net.adapter"
        Counters = ["Bytes/sec", "Bytes Received/sec", "Bytes Sent/Sec", "Packets Sent/sec", "Packets
  Received/sec", "Packets/sec"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Legacy Network Adapter"
        Instances = ["*"]
        Measurement = "hyperv.legacy.net.adapter"
        Counters = ["Bytes Dropped", "Bytes Received/sec", "Bytes Sent/Sec"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Hypervisor Partition"
        Instances = ["*"]
        Measurement = "hyperv.hypervisor.partition"
        Counters = ["Virtual Processors"]

      [[inputs.win_perf_counters.object]]
        ObjectName = "Hyper-V Virtual Storage Device"
        Instances = ["*"]
        Measurement = "hyperv.virtual.storage.device"
        Counters = [
                  "Maximum Bandwidth", "Read Bytes/sec", "Write Bytes/sec", "Queue Length",
                  "Lower Latency", "Minimum IO Rate", "Maximum IO Rate", "Latency", "Throughput",
```

Configuration Details for Supported Application Services

Java Plugin

Here are the configuration details:

```
 [[inputs.jolokia2_agent]]
   # Prefix to attach to the measurement name
   name_prefix = "java."
   # Add agents URLs to query
   urls = ["http://localhost:8080/jolokia"]
   #username and password are mandatory for Jolokia 1.6 or later
   #username = <jolokia role username>
   #password = <jolokia role password>
   # response_timeout = "5s"
   ## Optional TLS config
   # tls_ca   = "/var/private/ca.pem"
   # tls_cert = "/var/private/client.pem"
   # tls_key  = "/var/private/client-key.pem"
   # insecure_skip_verify = false
   ### JVM Generic
   [[inputs.jolokia2_agent.metric]]
     name  = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["ProcessCpuLoad","SystemLoadAverage","SystemCpuLoad"]
   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_runtime"
     mbean = "java.lang:type=Runtime"
     paths = ["Uptime"]
   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_memory"
     mbean = "java.lang:type=Memory"
     paths = ["HeapMemoryUsage", "NonHeapMemoryUsage", "ObjectPendingFinalizationCount"]
   [[inputs.jolokia2_agent.metric]]
     name     = "jvm_garbage_collector"
     mbean    = "java.lang:name=*,type=GarbageCollector"
     paths    = ["CollectionTime", "CollectionCount"]
     tag_keys = ["name"]
   [[inputs.jolokia2_agent.metric]]
     name       = "jvm_memory_pool"
     mbean      = "java.lang:name=*,type=MemoryPool"
     paths      = ["Usage", "PeakUsage", "CollectionUsage"]
     tag_keys   = ["name"]
     tag_prefix = "pool_"
   ### TOMCAT
   [[inputs.jolokia2_agent.metric]]
     name     = "GlobalRequestProcessor"
     mbean    = "Catalina:name=*,type=GlobalRequestProcessor"
     paths    = ["requestCount","bytesReceived","bytesSent","processingTime","errorCount"]
     tag_keys = ["name"]
   [[inputs.jolokia2_agent.metric]]
     name     = "JspMonitor"
     mbean    = "Catalina:J2EEApplication=*,J2EEServer=*,WebModule=*,name=jsp,type=JspMonitor"
     paths    = ["jspReloadCount","jspCount","jspUnloadCount"]
     tag_keys = ["J2EEApplication","J2EEServer","WebModule"]
   [[inputs.jolokia2_agent.metric]]
     name     = "ThreadPool"
     mbean    = "Catalina:name=*,type=ThreadPool"
     paths    = ["maxThreads","currentThreadCount","currentThreadsBusy"]
     tag_keys = ["name"]
   [[inputs.jolokia2_agent.metric]]
     name     = "Servlet"
     mbean    = "Catalina:J2EEApplication=*,J2EEServer=*,WebModule=*,j2eeType=Servlet,name=*"
     paths    = ["processingTime","errorCount","requestCount"]
     tag_keys = ["name","J2EEApplication","J2EEServer","WebModule"]
```

Configuration Details for Supported Application Services

JBoss Server

Here are the configuration details:

```
 # Read JBoss status information (mod_status)
 [[inputs.jolokia2_agent]]
   ##urls = ["http://localhost:8380/jolokia"]
   urls = ["http://127.0.0.1:8082/jolokia"]
   name_prefix = "jboss."

   ## Optional SSL Config
     # ssl_ca = "/etc/telegraf/ca.pem"
     # ssl_cert = "/etc/telegraf/cert.pem"
     # ssl_key = "/etc/telegraf/key.pem"
   ## Use SSL but skip chain & host verification
      insecure_skip_verify = true

   ### JVM Generic

   [[inputs.jolokia2_agent.metric]]
     name = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["ProcessCpuLoad","SystemLoadAverage","SystemCpuLoad"]

   [[inputs.jolokia2_agent.metric]]
     name = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["MaxFileDescriptorCount","OpenFileDescriptorCount"]

   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_runtime"
     mbean = "java.lang:type=Runtime"
     paths = ["Uptime"]

   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_memory"
     mbean = "java.lang:type=Memory"
     paths = ["HeapMemoryUsage", "NonHeapMemoryUsage", "ObjectPendingFinalizationCount"]

   [[inputs.jolokia2_agent.metric]]
     name     = "jvm_garbage_collector"
     mbean    = "java.lang:name=*,type=GarbageCollector"
     paths    = ["CollectionTime", "CollectionCount"]
     tag_keys = ["name"]

   [[inputs.jolokia2_agent.metric]]
     name       = "jvm_memory_pool"
     mbean      = "java.lang:name=*,type=MemoryPool"
     paths      = ["Usage", "PeakUsage", "CollectionUsage"]
     tag_keys   = ["name"]
     tag_prefix = "pool_"

   ### JBOSS

   [[inputs.jolokia2_agent.metric]]
     name     = "connectors.http"
     mbean    = "jboss.as:https-listener=*,server=*,subsystem=undertow"
     paths    = ["bytesReceived","bytesSent","errorCount","requestCount"]
     tag_keys = ["server","https-listener"]
```



```
   [[inputs.jolokia2_agent.metric]]
     name     = "connectors.http"
     mbean    = "jboss.as:http-listener=*,server=*,subsystem=undertow"
```

Configuration Details for Supported Application Services

Microsoft IIS

Here are the configuration details:

```
 # Read MSIIS status information (mod_status)
 [[inputs.win_perf_counters]]

      [[inputs.win_perf_counters.object]]
        # IIS, Web Service
        ObjectName = "Web Service"
        Counters = [
                    "Service Uptime", "Current Connections", "Bytes Sent/sec", "Total Bytes Sent",
  "Connection Attempts/sec",
                    "Bytes Received/sec", "Total Bytes Received", "Bytes Total/sec", "Total Bytes
  Transferred",
                    "Get Requests/sec","Total Get Requests", "Post Requests/sec","Total Post Requests",
                    "Put Requests/sec","Total Put Requests", "Delete Requests/sec","Total Delete Requests",
                    "Anonymous Users/sec", "NonAnonymous Users/sec", "Files Sent/sec", "Total Files Sent",
                    "Files Received/sec", "Total Files Received", "Files/sec", "Total Files Transferred",
                    "Not Found Errors/sec", "Locked Errors/sec", "Total Method Requests/sec"
                  ]
        Instances = ["*"]
        Measurement = "iis.websvc"
        IncludeTotal=true #Set to false to not include _Total instance.

      [[inputs.win_perf_counters.object]]
        # Web Service Cache / IIS
        ObjectName = "Web Service Cache"
        Counters = [
                    "Current Files Cached", "Active Flushed Entries", "Total Files Cached", "Total Flushed
  Files",
                    "File Cache Hits", "File Cache Misses", "File Cache Hits %", "File Cache Flushes",
                    "Current File Cache Memory Usage", "Maximum File Cache Memory Usage", "Current URIs
  Cached",
                    "Total URIs Cached", "Total Flushed URIs", "URI Cache Hits", "URI Cache Misses", "URI
  Cache Hits %",
                    "URI Cache Flushes", "Current Metadata Cached", "Total Metadata Cached", "Total Flushed
  Metadata",
                    "Metadata Cache Hits", "Metadata Cache Misses", "Metadata Cache Hits %", "Metadata Cache
  Flushes",
                    "Kernel: Current URIs Cached", "Kernel: Total URIs Cached", "Kernel: Total Flushed
  URIs",
                    "Kernel: URI Cache Hits", "Kernel: Uri Cache Hits/sec", "Kernel: URI Cache Misses",
  "Kernel: URI Cache Hits %",
                    "Kernel: URI Cache Flushes", "Cache Memory Usage"
                  ]
        Instances = ["*"]
        Measurement = "iis.websvc.cache"
        IncludeTotal=true #Set to false to not include _Total instance.

      [[inputs.win_perf_counters.object]]
        # IIS, ASP.NET
        ObjectName = "ASP.NET"
        Counters = ["Application Restarts","Request Wait Time","Requests Current","Requests Queued","Requests
  Rejected"]
        Instances = ["*"]
        Measurement = "iis.aspnet"
        #IncludeTotal=false #Set to true to include _Total instance when querying for all (*).

      [[inputs.win_perf_counters.object]]
        # IIS, ASP.NET Applications
        ObjectName = "ASP.NET Applications"
        Counters = ["Cache Total Entries","Cache Total Hit Ratio","Output Cache Entries","Errors Total/
```

Configuration Details for Supported Application Services

Microsoft SQL Server

Here are the configuration details:

```
[[inputs.sqlserver]]
   name_prefix = "MSSQL."

       #servers = [
       #"Server=<servername>;Port=1433;User Id=telegraf;Password=<mystrongpassword from step 2>;app
 name=telegraf;log=1;",
       #"Server=<servername>;Port=1433;User Id=telegraf;Password=<mystrongpassword from step 2>;app
 name=telegraf;log=1;"
      #]
       servers = [
        "Server=localhost;Port=1433;User Id=sa;Password=Password;app name=telegraf;log=1;"
         ]

   namepass = ["Rows*writes*bytes*sec*", "Rows*reads*bytes*sec*", "Rows*writes*sec*",
 "Rows*reads*sec*", "Query*User*counter*", "Buffer*cache*hit*ratio*", "Page*life*expectancy*",
 "Page*lookups*sec*", "Page*reads*sec*", "Page*writes*sec*", "Lazy*writes*sec*", "Checkpoint*pages*sec*",
 "Log*Apply*Ready*Queue*", "Data*File*s*Size*KB*", "Log*File*s*", "XTP*Memory*Used*KB*",
 "Log*Flushes*sec*", "Write*Transactions*sec*", "Transactions*sec*", "Log*Flush*Wait*Time*",
 "Active*Transactions*", "Log*Bytes*Flushed*sec*", "Processes*blocked*", "User*Connections*",
 "Logins*sec*", "Logouts*sec*", "Active*Temp*Tables*", "Temp*Tables*Creation*Rate*", "Batch*Requests*sec*",
 "SQL*Compilations*sec*", "SQL*Re*Compilations*sec", "Stored*Procedures*Invoked*sec*",
 "Target*Server*Memory*KB*", "Total*Server*Memory*KB*", "SQL*Cache*Memory*KB*", "Log*Pool*Memory*KB*",
 "Connection*Memory*KB*", "Lock*Memory*KB*", "Memory*Grants*Pending*", "Active*memory*grant*amount*KB*",
 "Disk*Read*Bytes*sec*", "Disk*Read*IO*Throttled*sec*", "Disk*Read*IO*sec*", "Disk*Write*Bytes*sec*",
 "Disk*Write*IO*Throttled*sec*", "Used*memory*KB*", "CPU*usage*", "Free*Space*in*tempdb*KB*",
 "Version*Store*Size*KB*", "Transactions*", "Blocked*tasks*", "Active*requests*", "Queued*requests*",
 "Requests*completed*sec*", "Number*of*Deadlocks*sec*", "Lock*Wait*Time*ms*", "Lock*Waits*sec*",
 "Lock*Requests*sec*", "Average*Wait*Time*ms*", "Index*Searches*sec*", "Page*Splits*sec*",
 "Full*Scans*sec*", "CPU*", "Wait*time*ms*", "Wait*tasks*", "State*", "Recovery*Model*"]
```

MongoDB

Here are the configuration details:

```
# Read metrics from one or many MongoDB servers
[[inputs.mongodb]]
  ## An array of URI to gather stats about. Specify an ip or hostname
  ## with optional port add password. ie,
  ##   mongodb://user:auth_key@10.10.3.30:27017,
  ##   mongodb://10.10.3.33:18832,
  ##   10.0.0.1:10000, etc.
  ##servers = ["127.0.0.1:27017"]

  servers = ["localhost:27017"]

  gather_perdb_stats = true

  ## Optional SSL Config
    # ssl_ca = "/etc/telegraf/ca.pem"
    # ssl_cert = "/etc/telegraf/cert.pem"
    # ssl_key = "/etc/telegraf/key.pem"

  ## Use SSL but skip chain & host verification
    # insecure_skip_verify = false
```

Configuration Details for Supported Application Services

MS Exchange Server

Here are the configuration details:

```
 # Read MS Exchange status information (mod_status)
     [[inputs.win_services]]
       name_prefix = "msexchange."
       service_names = [
           "MSExchangeTransport",
           "MSExchangeTransportLogSearch",
           "MSExchangeUM",
           "MSExchangeUMCR",
           "MSExchangeThrottling",
           "MSExchangeServiceHost",
           "MSExchangeFastSearch",
           "MSExchangeRepl",
           "MSExchangeRPC",
           "MSExchangePop3",
           "MSExchangeSubmission",
           "MSExchangeDelivery",
           "MSExchangeMailboxReplication",
           "MSExchangeMailboxAssistants",
           "MSExchangeIS",
           "MSExchangeImap4",
           "MSExchangeHM",
           "MSExchangeFrontendTransport",
           "MSExchangeEdgeSync",
           "MSExchangeDiagnostics",
           "MSExchangeAntispamUpdate",
           "MSExchangeADTopology"
       ]


     [[inputs.win_perf_counters]]
       plugin_name_override="msexchange"
       [[inputs.win_perf_counters.object]]
         ObjectName = "Process"
         Counters = ["% Processor Time","Handle Count","Private Bytes","Thread Count","Virtual
  Bytes","Working Set"]
         Instances = [
           "MSExchangeTransport",
           "MSExchangeTransportLogSearch",
           "umservice",
           "Microsoft.Exchange.UM.CallRouter",
           "MSExchangeThrottling",
           "Microsoft.Exchange.ServiceHost",
           "Microsoft.Exchange.Search.Service",
           "msexchangerepl",
           "Microsoft.Exchange.RpcClientAccess.Service",
           "Microsoft.Exchange.Pop3Service",
           "MSExchangeSubmission",
           "MSExchangeDelivery",
           "MSExchangeMailboxReplication",
           "MSExchangeMailboxAssistants",
           "Microsoft.Exchange.Store.Service",
           "Microsoft.Exchange.Imap4Service",
           "MSExchangeHMHost",
           "MSExchangeFrontendTransport",
           "Microsoft.Exchange.EdgeSyncSvc",
           "Microsoft.Exchange.Diagnostics.Service",
           "Microsoft.Exchange.AntispamUpdateSvc",
           "Microsoft.Exchange.Directory.TopologyService"
         ]
```

Configuration Details for Supported Application Services

MySQL

Here are the configuration details:

```
 [[inputs.mysql]]

  ## specify servers via a url matching:
  ##  [username[:password]@][protocol[(address)]]/[?tls=[true|false|skip-verify]]
  ##  see https://github.com/go-sql-driver/mysql#dsn-data-source-name
  ##  e.g.
  ##    servers = ["user:passwd@tcp(127.0.0.1:3306)/?tls=false"]
  ##    servers = ["user@tcp(127.0.0.1:3306)/?tls=false"]
  #
  ## If no servers are specified, then localhost is used as the host.
  ##servers = ["MS_USER:MS_PASSWORD@tcp(localhost:MS_PORT)/?tls=false"]
  servers = ["root:VMware1!@tcp(localhost:3306)/?tls=false"]
  ## the limits for metrics form perf_events_statements
  perf_events_statements_digest_text_limit  = 120
  perf_events_statements_limit              = 250
  perf_events_statements_time_limit         = 86400
  #
  ## if the list is empty, then metrics are gathered from all databasee tables

  #
  ## gather metrics from INFORMATION_SCHEMA.TABLES for databases provided above list
  gather_table_schema                       = true
  #
  ## gather thread state counts from INFORMATION_SCHEMA.PROCESSLIST
  gather_process_list                       = true
  #
  ## gather auto_increment columns and max values from information schema
  gather_info_schema_auto_inc               = true
  #
  ## gather metrics from SHOW SLAVE STATUS command output
  gather_slave_status                       = true
  #

  ## gather metrics from SHOW BINARY LOGS command output

    gather_binary_logs                        = false

  ## gather thread state counts from INFORMATION_SCHEMA.USER_STATISTICS
  gather_user_statistics                    = false
  #
  ## gather metrics from PERFORMANCE_SCHEMA.TABLE_IO_WAITS_SUMMARY_BY_TABLE
  gather_table_io_waits                     = true
  #
  ## gather metrics from PERFORMANCE_SCHEMA.TABLE_LOCK_WAITS
  gather_table_lock_waits                   = true
  #
  ## gather metrics from PERFORMANCE_SCHEMA.TABLE_IO_WAITS_SUMMARY_BY_INDEX_USAGE
  gather_index_io_waits                     = true
  #
  ## gather metrics from PERFORMANCE_SCHEMA.EVENT_WAITS
  gather_event_waits                        = true
  #
  ## gather metrics from PERFORMANCE_SCHEMA.FILE_SUMMARY_BY_EVENT_NAME
  gather_file_events_stats                  = true
  #
  ## gather metrics from PERFORMANCE_SCHEMA.EVENTS_STATEMENTS_SUMMARY_BY_DIGEST
  gather_perf_events_statements             = true
  #
  ## Some queries we may want to run less often (such as SHOW GLOBAL VARIABLES)
```

Configuration Details for Supported Application Services

NGINX

Here are the configuration details:

```
# Read Nginx's basic status information (ngx_http_stub_status_module)
[[inputs.nginx]]

  ## An array of Nginx stub_status URI to gather stats.
  urls = ["http://127.0.0.1:8081/nginx_status"]

  ## Optional SSL Config
    # ssl_ca = "/etc/telegraf/ca.pem"
    # ssl_cert = "/etc/telegraf/cert.pem"
    # ssl_key = "/etc/telegraf/key.pem"

  ## Use SSL but skip chain & host verification
    # insecure_skip_verify = false

  ## HTTP response timeout (default: 5s)
  response_timeout = "5s"
```

NTPD

Here are the configuration details:

```
[[inputs.ntpq]]
     name_prefix = "ntpd."
     ## If false, set the -n ntpq flag. Can reduce metric gather times.
     dns_lookup = tr
```

OracleDB

- Complete the prerequisites mentioned in the Pre-Requirements for Application Services.
- Download `wavefront_oracle_metrics.py` from cloud proxy to the Telegraf configuration directory. For example, `C:\Telegraf\telegraf-1.20.4\telegraf.d`.
  - Invoke-WebRequest `https://$CP_IP/downloads/salt/content-accessories/wavefront_oracle_metrics.py -OutFile C:\Telegraf\telegraf-1.20.4\telegraf.d\wavefront_oracle_metrics.py`

Here are the configuration details:

```
# Read metrics exposed by chef
[[inputs.exec]]

  # commands = ['python "wavefront_oracle_metrics.py script path" -u "OracleDB Username" -p "OracleDB
 Password" -s "OracleDB SID/DSN"']
  commands = ['python "C:\Telegraf\telegraf-1.20.4\telegraf.d\wavefront_oracle_metrics.py" -u
"orcl.vrops.es" -p "Qw123456" -s "https://VAP-test.vrops.es:1158/em"']

  timeout = "180s"
  data_format = "influx"
  name_prefix = "oracledb."
```

Configuration Details for Supported Application Services

PostgreSQL

Here are the configuration details:

```
 # Read metrics from one or many postgresql servers
 [[inputs.postgresql]]

   ## specify address via a url matching:
   ##   postgres://[pqgotest[:password]]@localhost[/dbname]\
   ##       ?sslmode=[disable|verify-ca|verify-full]
   ## or a simple string:
   ##   host=localhost user=pqotest password=... sslmode=... dbname=app_production
   ##
   ## All connection parameters are optional.
   ##
   ## Without the dbname parameter, the driver will default to a database
   ## with the same name as the user. This dbname is just for instantiating a
   ## connection with the server and doesn't restrict the databases we are trying
   ## to grab metrics for.
   ##
   address = "host=localhost user=postgres password=\t Password1! port=5432"
   ##address = "host=localhost user=postgres  password=\t Password1! port=5432"
   ## A  list of databases to explicitly ignore.  If not specified, metrics for all
   ## databases are gathered.  Do NOT use with the 'databases' option.
     # ignored_databases = ["postgres", "template0", "template1"]


   ## A list of databases to pull metrics about. If not specified, metrics for all
   ## databases are gathered.  Do NOT use with the 'ignored_databases' option.

     # databases = ["app_production", "testing"]

   ## Optional SSL Config
     # ssl_ca = "/etc/telegraf/ca.pem"
     # ssl_cert = "/etc/telegraf/cert.pem"
     # ssl_key = "/etc/telegraf/key.pem"

   ## Use SSL but skip chain & host verification

     # insecure_skip_verify = false
```

Configuration Details for Supported Application Services

Pivotal

Here are the configuration details:

```
 # Read Tomcat status information (mod_status)
 [[inputs.jolokia2_agent]]
   ## urls = ["http://localhost:8080/jolokia"]
   urls = ["http://localhost:8080/jolokia"]
   name_prefix = "pivotalserver."

   ## Optional SSL Config
     # ssl_ca = "/etc/telegraf/ca.pem"
     # ssl_cert = "/etc/telegraf/cert.pem"
     # ssl_key = "/etc/telegraf/key.pem"

   ## Use SSL but skip chain & host verification
     # insecure_skip_verify = false

   ### JVM Generic

   [[inputs.jolokia2_agent.metric]]
     name = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["ProcessCpuLoad","SystemLoadAverage","SystemCpuLoad"]

   [[inputs.jolokia2_agent.metric]]
     name = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["MaxFileDescriptorCount","OpenFileDescriptorCount"]

   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_runtime"
     mbean = "java.lang:type=Runtime"
     paths = ["Uptime"]

   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_memory"
     mbean = "java.lang:type=Memory"
     paths = ["HeapMemoryUsage", "NonHeapMemoryUsage", "ObjectPendingFinalizationCount"]

   [[inputs.jolokia2_agent.metric]]
     name     = "jvm_garbage_collector"
     mbean    = "java.lang:name=*,type=GarbageCollector"
     paths    = ["CollectionTime", "CollectionCount"]
     tag_keys = ["name"]

   [[inputs.jolokia2_agent.metric]]
     name       = "jvm_memory_pool"
     mbean      = "java.lang:name=*,type=MemoryPool"
     paths      = ["Usage", "PeakUsage", "CollectionUsage"]
     tag_keys   = ["name"]
     tag_prefix = "pool_"

   ### TOMCAT
   [[inputs.jolokia2_agent.metric]]
     name     = "GlobalRequestProcessor"
     mbean    = "Catalina:name=*,type=GlobalRequestProcessor"
     paths    = ["requestCount","bytesReceived","bytesSent","processingTime","errorCount"]
     tag_keys = ["name"]
   [[inputs.jolokia2_agent.metric]]
     name     = "JspMonitor"
     mbean    = "Catalina:J2EEApplication=*,J2EEServer=*,WebModule=*,name=jsp,type=JspMonitor"
```

Configuration Details for Supported Application Services

RabbitMQ

Here are the configuration details:

```
# Read metrics from one or many RabbitMQ servers via the management API
[[inputs.rabbitmq]]
arc_service_id = "145d46c3-3f5d-449b-af93-f32a150c4965"

  ##url = "http://localhost:15672"
  url = "http://10.218.135.219:15672"

    metric_exclude = ["federation"]

  # name = "rmq-server-1" # optional tag
    username = "admin"
    password = "admin"

  ## Optional SSL Config
     # ssl_ca = "/etc/telegraf/ca.pem"
    # ssl_cert = "/etc/telegraf/cert.pem"
    # ssl_key = "/etc/telegraf/key.pem"


  ## Use SSL but skip chain & host verification
    # insecure_skip_verify = false

  ## Optional request timeouts
  ##
  ## ResponseHeaderTimeout, if non-zero, specifies the amount of time to wait
  ## for a server's response headers after fully writing the request.
  # header_timeout = "3s"
  ##
  ## client_timeout specifies a time limit for requests made by this client.
  ## Includes connection time, any redirects, and reading the response body.
  # client_timeout = "4s"

  ## A list of nodes to pull metrics about. If not specified, metrics for
  ## all nodes are gathered.
  # nodes = ["rabbit@node1", "rabbit@node2"]
```

Riak

Here are the configuration details:

```
# Read Riak status information (mod_status)
[[inputs.riak]]

  # Specify a list of one or more riak http servers
  ##servers = ["http://localhost:8098"]

  servers = ["http://127.0.0.1:8098"]
```

Configuration Details for Supported Application Services

SharePoint Server

Here are the configuration details:

```
 [[inputs.win_services]]
   name_prefix = "sharepoint."

    service_names = [
    "OSearch15",
    "SPAdminV4",
    "SPSearchHostController",
    "SPTimerV4",
    "SPTraceV4",
    "SPUserCodeV4",
    "SPWriterV4"
 ]

 [[inputs.win_perf_counters]]
   plugin_name_override="sharepoint"

   [[inputs.win_perf_counters.object]]
     ObjectName = "Process"
     Instances = ["mssearch", "WSSADMIN", "hostcontrollerservice", "OWSTIMER", "wsstracing",
  "SPUCHostService", "SPWRITER"]
     Counters = ["% Processor Time", "Thread Count"]
     Measurement = "sharepoint.process"

   [[inputs.win_perf_counters.object]]
     ObjectName = "Web Service"
     Instances = ["*"]
     Counters = ["Bytes Sent/sec", "Bytes Received/sec"]
     Measurement = "sharepoint.webservice"

   [[inputs.win_perf_counters.object]]
     ObjectName = "SharePoint Foundation"
     Counters = ["Object Cache Hit %", "Executing Sql Queries", "Executing Time/Page Request", "Reject Page
  Requests Rate", "Incoming Page Requests Rate", "Active Threads", "Object Cache Hit Count", "Current Page
  Requests", "Sql Query Executing  time", "Responded Page Requests Rate"]
     Instances = ["_total"]
     Measurement = "sharepoint.Foundation"

   [[inputs.win_perf_counters.object]]
     ObjectName = "SharePoint Records Management Counters"
     Counters = ["Search results processed / sec"]
     Instances = ["------"]
     Measurement = "sharepoint.RecordsManagementCounters"

   [[inputs.win_perf_counters.object]]
     ObjectName = "SharePoint Foundation Search Gatherer Projects"
     Counters = ["Crawls in progress", "Filtered Text Rate", "Filtered Office Rate", "Filtered HTML Rate",
  "Accessed File Rate", "Accessed HTTP Rate", "File Errors Rate", "HTTP Errors Rate"]
     Instances = ["_total"]
     Measurement = "sharepoint.FoundationSearchGathererProjects"

   [[inputs.win_perf_counters.object]]
     ObjectName = "SharePoint Foundation Search Schema Plugin"
     Counters = ["Total Documents"]
     Instances = ["_total"]
     Measurement = "sharepoint.FoundationSearchSchemaPlugin"
```



```
   [[inputs.win_perf_counters.object]]
     ObjectName = "SharePoint Foundation BDC Online"
     Counters = ["Total calls failed", "Total calls per second"]
```

Configuration Details for Supported Application Services

Tomcat Server

Here are the configuration details:

```
 # Read Tomcat status information (mod_status)
 [[inputs.jolokia2_agent]]
   ##urls = ["http://localhost:8080/jolokia"]
   urls = ["http://localhost:8083/jolokia"]
   name_prefix = "tomcat."

   ## Optional SSL Config
     # ssl_ca = "/etc/telegraf/ca.pem"
     # ssl_cert = "/etc/telegraf/cert.pem"
     # ssl_key = "/etc/telegraf/key.pem"

   ## Use SSL but skip chain & host verification
     # insecure_skip_verify = false

   ### JVM Generic

   [[inputs.jolokia2_agent.metric]]
     name = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["ProcessCpuLoad","SystemLoadAverage","SystemCpuLoad"]

   [[inputs.jolokia2_agent.metric]]
     name = "OperatingSystem"
     mbean = "java.lang:type=OperatingSystem"
     paths = ["MaxFileDescriptorCount","OpenFileDescriptorCount"]

   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_runtime"
     mbean = "java.lang:type=Runtime"
     paths = ["Uptime"]

   [[inputs.jolokia2_agent.metric]]
     name  = "jvm_memory"
     mbean = "java.lang:type=Memory"
     paths = ["HeapMemoryUsage", "NonHeapMemoryUsage", "ObjectPendingFinalizationCount"]

   [[inputs.jolokia2_agent.metric]]
     name     = "jvm_garbage_collector"
     mbean    = "java.lang:name=*,type=GarbageCollector"
     paths    = ["CollectionTime", "CollectionCount"]
     tag_keys = ["name"]

   [[inputs.jolokia2_agent.metric]]
     name       = "jvm_memory_pool"
     mbean      = "java.lang:name=*,type=MemoryPool"
     paths      = ["Usage", "PeakUsage", "CollectionUsage"]
     tag_keys   = ["name"]
     tag_prefix = "pool_"

   ### TOMCAT
   [[inputs.jolokia2_agent.metric]]
     name     = "GlobalRequestProcessor"
     mbean    = "Catalina:name=*,type=GlobalRequestProcessor"
     paths    = ["requestCount","bytesReceived","bytesSent","processingTime","errorCount"]
     tag_keys = ["name"]
   [[inputs.jolokia2_agent.metric]]
     name     = "JspMonitor"
     mbean    = "Catalina:J2EEApplication=*,J2EEServer=*,WebModule=*,name=jsp,type=JspMonitor"
```

Configuration Details for Supported Application Services

Oracle WebLogic

WebSphere

Here are the configuration details:

```
[[inputs.httpjson]]
  servers = ["http://localhost:9080/wasmonitor/stats"]
  method = "POST"
  name_override="websphere"

  response_timeout = "60s"
  tag_keys = ["name", "process", "node", "cell", "mbeanIdentifier"]
  [inputs.httpjson.headers]
    Authorization = "Basic d2F2ZWZyb250OndhdmVmcm9udA=="

[[inputs.exec]]
  commands = ['C:\\VMware\\UCP\\ucp-minion\\bin\\run-py-script.bat "C:\VMware\UCP\ucp-minion\lib\site-
packages\ucp\application_availability.py" --config="C:\VMware\UCP\salt\conf\grains" "java" "http://
localhost:9080"']
  timeout = "180s"
  data_format = "influx"
  name_prefix = "websphere."
  [inputs.exec.tags]
    server = "http://localhost:9080/wasmonitor/stats"
```

Table 1726: Configuration Details for Custom Monitoring Plugins

Ping check
Here are the configuration details:
`[[inputs.ping]]`
`## Hosts to send ping packets to.`
`urls = [www.vmware.com]`
 	
`## Number of ping packets to send per interval.  Corresponds to the "-c"`
`## option of the ping command.`
`# count = 1`
 	
`## Time to wait between sending ping packets in seconds.  Operates like the`
`## "-i" option of the ping command.`
`# ping_interval = 1.0`
 	
`## If set, the time to wait for a ping response in seconds.  Operates like`
`## the "-W" option of the ping command.`
`# timeout = 1.0`
 	
`## If set, the total ping deadline, in seconds.  Operates like the -w option`
`## of the ping command.`
`# deadline = 10`
 	
`## Interface or source address to send ping from.  Operates like the -I or -S`
`## option of the ping command.`
`# interface = ""`

TCP check
Here are the configuration details:
`[[inputs.net_response]]`
`name_override="netresponse"`
`## Server address (default localhost)`
`address = "10.10.10.10:443"`
`protocol = "tcp"`
 	
`## Set timeout`
`# timeout = "1s"`
 	
`## Set read timeout (only used if expecting a response)`
`# read_timeout = "1s"`

HTTP check

Here are the configuration details:

```
# HTTP/HTTPS request given an address a method and a timeout
[[inputs.http_response]]
name_override="httpresponse"
## List of url to query.
address = https://10.10.10.10/suite-api/api/adapters
method = "GET"

## Set http_proxy.
## Telegraf uses the system wide proxy settings if it's is not set.
# http_proxy = http://localhost:8888

## Set response_timeout (default 5 seconds)
# response_timeout = "5s"

## Whether to follow redirects from the server (defaults to false)
# follow_redirects = false

## Optional name of the field that will contain the body of the response.
## By default it is set to an empty String indicating that the body's content won't be added
# response_body_field = ''

## Optional substring or regex match in body of the response (case sensitive)
# response_string_match = "\"service_status\": \"up\""
# response_string_match = "ok"
# response_string_match = "\".*_status\".?:.?\"up\""


## Optional SSL Config
# ssl_ca = "/etc/telegraf/ca.pem"
# ssl_cert = "/etc/telegraf/cert.pem"
# ssl_key = "/etc/telegraf/key.pem"


## Use SSL but skip chain & host verification
insecure_skip_verify = true

## HTTP Request Headers (all values must be strings)
[inputs.http_response.headers]
accept = "application/json" Authorization = "OpsToken a16f7a2b-
b033-48bc-9827-2daf8e205537::ec11ee5f-8623-4558-a904-8b4ea3f6f47d"
```

Custom scripts

Here are the configuration details:

For Linux Platforms:

```
# Read metrics from command that can output to stdout
[[inputs.exec]]
name_prefix = "executescript."
name_override = "output"
# commands = ["<prefix> <file path> <argument>"]
commands = ["python /opt/scripts/argument.py 10"]

## Data format to consume.
data_format = "value"
data_type = "integer"

## Timeout for each command to complete.
timeout = "300s"

[inputs.exec.tags]

file_path = "/opt/scripts/argument.py"
script_name = "my-py-script on Centos7-VM"
```

For Windows Platforms:

```
# Read metrics from command that can output to stdout
[[inputs.exec]]

name_prefix = "executescript."
name_override = "output"

# commands = ["<prefix> <file path> <argument>"]
commands = ["Powershell -File C:\\\\VMware\\\\Scripts\\\\Arguments.ps1 vmtoolsd"]

## Data format to consume.
data_format = "value"
data_type = "integer"

## Timeout for each command to complete.
timeout = "300s"

[inputs.exec.tags]

file_path = "C:\\VMware\\Scripts\\Arguments.ps1"
script_name = "my-script-for-vmtool on Windows2022-PhysicalServer"
```

Linux Process

Here are the configuration details (via regular expression):

```
# Monitor process cpu and memory usage
[[inputs.procstat]]
  ## pattern as argument for pgrep (ie, pgrep -f <pattern>)
  pattern = ".*ucp-.*"

  fieldpass =  ["running", "cpu_usage", "memory_usage"]
  [inputs.procstat.tags]
    search_pattern = "regex_.#!AsTeRiSk!#ucp-.#!AsTeRiSk!#"
```

Here are the configuration details (via executable name):

```
# Monitor process cpu and memory usage
[[inputs.procstat]]
  ## executable name (ie, pgrep <exe>)
  exe = "top"
  fieldpass =  ["running", "cpu_usage", "memory_usage"]
  [inputs.procstat.tags]

    search_pattern = "exec_top"
```

Here are the configuration details (via process ID):

```
# Monitor process cpu and memory usage
[[inputs.procstat]]
  ## PID file to monitor process
  pid_file = "/var/run/vmtoolsd.pid"

  fieldpass =  ["running", "cpu_usage", "memory_usage"]
  [inputs.procstat.tags]

    search_pattern = "pidfile_/var/run/vmtoolsd.pid"
```

Windows Service

Here are the configuration details (via Windows service name):

```
# Monitor process cpu and memory usage
[[inputs.procstat]]

## Windows service name
win_service = "Dhcp"
pid_finder = "native"

fieldpass =  ["running", "cpu_usage", "memory_usage"]
[inputs.procstat.tags]
search_pattern = "exec_Dhcp"

[[inputs.win_services]]
service_names = ["Dhcp"]
name_override="procstat"

[inputs.win_services.tags]
search_pattern = "exec_Dhcp"
```

Troubleshooting (Open Source Telegraf)

Windows Services

If you configure multiple services and the service names include underscores, then the associated objects have the same display name.

While monitoring a Windows Services plugin using open source Telegraf, if there are multiple services with names such as, `svc1_name1` and `svc1_name2` with an underscore in the service names, then both the objects have the same display name.

If you want the display names to have different names, follow these steps.

1. Manually edit the service name in VCF Operations after the Windows Services object is created so that the correct service name is displayed as the display name of the Service object.
2. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Inventory Management** tile. From the **Objects** tab, select the Service object with the incorrect name and edit it.
3. Add the value as entered in the **Service Name** field to the **Display Name** field.

   For example, while monitoring a Windows service with the service name `SMS_SITE_COMPONENT_MANAGER`, if the display name of the current object is `Services exec SMS on Windows_OS_on_<windows hostname>`, where `SMS` is the display name, change the display name to `Services exec SMS_SITE_COMPONENT_MANAGER on Windows_OS_on_<windows hostname>`.
4. Click **OK**.

Discovering Services and Applications using the Service Discovery Adapter

You can discover services and applications using the Service Discovery adapter.

Service Discovery

Service discovery helps you discover services running in each VM and then builds a relationship or dependency between the services from different VMs. You can view basic metrics based on the services you want to monitor. You can also use the service discovery dashboards to monitor the services.

Service discovery helps you determine the kind of services running on each VM in your environment. You can find out which VM is a part of a service, the impact of shutting down or moving a VM, the impact of an incident, and the right escalation path for a problem. You can also determine which VMs are used to migrate a service and which services are impacted by a planned outage on a VM or an infrastructure component.

Application Discovery

Applications can also be discovered on a VM with services connected to each other and those that talk to each other. You can discover predefined and custom applications. Log management and VCF Operations are the predefined applications. You can view the VMs connected to their services, how the services talk to each other, and those that are connected to different VMs.

Licensing

You can discover and monitor services using VCF Operations Advanced and Enterprise editions.

To discover and monitor services and add and view applications, follow these steps in VCF Operations:

- Configure Service and Application Discovery. For more information, see Configure Service and Application Discovery.
- Manage Services and View Applications. For more information, see Manage Services and View Applications.
- Monitor services using dashboards. For more information, see Service Discovery Dashboards.
- View the services discovered. For more information, see Discovered Services.

Supported Platforms and Products for Service Discovery

Service discovery supports specific platforms and product versions.

You can either provide guest operating system credentials with appropriate privileges or use the credential-less approach to discover services.

Supported Product Versions for Credential-Based Service Discovery

- For ESX, vCenter, and VMware Cloud on AWS versions, see the Compatibility Guide.
- VMware Tools

Supported Product Versions and Other Pre-Requisites for Credential-Less Service Discovery

For information, see KB 78216.

Operating System Versions

Operating Systems	Version
Windows	Windows 7, Windows Server 2008/R2, and above.
Linux	Photon, RHEL, CentOS, SUSE Linux Enterprise Server, OEL, and Ubuntu (all Linux operating systems must be based on kernel version 2.6.25 or above).

Supported Services

Service and application discovery supports several services that are supported in VCF Operations. The supported services are listed here.

Supported Services:

- Active Directory
- Apache HTTP
- Apache Tomcat
- Cassandra
- DB2
- Exchange Client Access Server
- Exchange Edge Transport Server
- Exchange Hub Transport Server
- Exchange Mailbox Server
- Exchange Server
- Exchange Unified Messaging Server
- GemFire
- IIS
- JBoss
- MS-SQL DB
- MySQL DB
- Nginx
- Oracle DB
- RabbitMQ
- SharePoint
- SharePoint Application Server
- SharePoint Server
- SharePoint Web Server
- SRM vCenter Replication Management Server
- SRM vCenter Replication Server
- Sybase DB

- tc Server
- vCenter Site Recovery Manager Server
- vCloud Director
- VMware vCenter
- VMware vCenter (Appliance)
- VMware View Server
- vROps Analytics
- vROps Collector
- vROps GemFire
- vROps Postgres Data
- vROps Postgres Repl
- vROps UI
- vRLI Daemon
- vRLI vInternalization
- vRLI UI
- WebLogic
- WebSphere

Configure Service and Application Discovery

To discover applications and services and their relationships and to access basic monitoring, you can either provide guest operating system credentials with appropriate privileges or use the credential-less approach to discover services.

- You must have a vCenter Adapter instance configured and monitoring the same vCenter that is used to discover services.

  For credential-based service discovery, the configured vCenter user must have the following privileges:
  - key: VirtualMachine.GuestOperations.ModifyAliases, Localization: Guest operations -> Guest operation alias modification
  - key: VirtualMachine.GuestOperations.QueryAliases, Localization: Guest operations -> Guest operation alias query
  - key: VirtualMachine.GuestOperations.Modify, Localization: Guest operations -> Guest operation modifications
  - key: VirtualMachine.GuestOperations.Execute, Localization: Guest operations -> Guest operation program execution
  - key: VirtualMachine.GuestOperations.Query, Localization: Guest operations -> Guest operation queries

  For credential-less service discovery, the configured vCenter user must have the following privileges:

  - key: VirtualMachine.Namespace.Management, Localization: Service Configuration -> Manage service configurations
  - key: VirtualMachine.Namespace.ModifyContent, Localization: Service Configuration -> Modify service configuration
  - key: VirtualMachine.Namespace.Query, Localization: Service Configuration -> Query service configurations
  - key: VirtualMachine.Namespace.ReadContent, Localization: Service Configuration -> Read service configuration
- The ESX instance that hosts the VMs where services should be discovered, must have HTTPS access to port 443 from the collector node on which the service discovery adapter instance is configured.
- Verify that the following types of commands and utilities are used:

Type	Commands and Utilities
UNIX Operating Systems	
Service Discovery	`ps`, `ss`, and `top`
Performance Metrics Collection	: `awk`, `csh`, `ps`, `pgrep`, and `procfs` (file system)
Windows Operating Systems	

Type	Commands and Utilities
Service Discovery	`wmic`, `netstat`, `findstr`, `net`, `reg`, and `sort` **Note:** PowerShell Scripts are now used instead of batch scripts in case of credential-less discovery starting with VMware Tools version 12.3.0 or above. See Deprecated Features for Windows Client for more details. **Note:** If you are using credential-less discovery and VMware Tools version 12.3.0 and above on a Windows VM, ensure that the following modules are installed on the OS:: • Microsoft.PowerShell.Core (default) • Microsoft.PowerShell.Utility • Microsoft.PowerShell.Management • SmbShare (for net share) • netstat utility Also, ensure that the version of the OS is Windows Server 2008R2 SP2 and above.
Performance Metrics Collection	`wmic`, `typeperf`, and `tasklist`

- User Access Restrictions
  - For Linux operating systems, ensure that the user is a root or member of the *sudo* users group.

    Note:
    For non-root users, the NOPASSWD option must be activated in `/etc/sudoers` file to avoid the metrics collector scripts from waiting for the interactive password input.

    Steps to activate the NOPASSWD option for a particular sudo user:

    1. Login to the specific VM as a root user.
    2. Run the `sudo visudo` command that opens an editor.
    3. In the command section, add username ALL=(ALL) NOPASSWD:<ss path>, <awk path>, <netstat path>. The username must be replaced with an existing user name for which this option is activated. Example: vmware ALL=(ALL) NOPASSWD: `/usr/sbin/ss`, `/usr/bin/netstat`, `/user/bin/awk`.
    4. Save the file and close it. It is automatically reloaded.
  - To discover services on Windows, the local administrator account must be configured.

    Note:
    Services will not be discovered for administrator group members that are different from the administrator account itself if the policy setting User Account Control: Run all administrators in Admin Approval Mode is turned on. As a workaround, you can turn off this policy setting to discover services. However, if you turn the policy setting off, the security of the operating system is reduced.
  - To discover services on Windows Active Directory, the domain administrator account must be configured.
- The system clock must be synchronized between the VCF Operations nodes, the vCenter, and the VM if service discovery is working in credential-based mode and guest alias mapping is used for authentication.
- The configured user must have read and write privileges to the temp directory (execute privilege is also required on this directory in Linux systems). For Windows systems, the path can be taken from the environment variable *`TEMP`*. For Linux systems, it is `/tmp` and/or `/var/tmp`.
- The SSO Server URL must be reachable from the VCF Operations node on which the service discovery adapter is located.
- For more information about supported platforms and versions, see Supported Platforms and Products for Service Discovery.

Note:
If more than one VCF Operations instance is monitoring the same vCenter and service discovery is activated for those VCF Operations instances, then service discovery might be unstable, which is a known VMware Tools problem. As a result, guest operations might fail to execute.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Service Discovery** tile.
2. From the **Service Discovery** page, click the **Configure Service Discovery** option.
3. From the **Integrations** page, click the vCenter instance from the list and then select the **Service Discovery** tab.
4. To activate service discovery in this vCenter, activate the **Service Discovery** option.
5. To activate application discovery in this vCenter, select the **Enable Application Discovery** check box.
6. You can choose to add credentials by selecting the **Use alternate credentials** check box.
   a) Click the plus sign and enter the details in the **Manage Credentials** dialog box, which include a credential name and a vCenter user name and password. In addition, enter the user name and password for Windows, Linux, and SRM and click **OK**.
7. Alternatively, if you are using the default user name and password, enter a default user name and password for Windows, Linux, and SRM.
8. Enter a password for the guest user mapping.
9. You can also activate grouping of the application, creation of a business application, and activate application discovery.
10. Click **Save**.

    Note:
    If you specify a non-root user for Linux, services are not discovered unless you activate the option Use Sudo (Linux Non-root user) while editing the associated Service Discovery adapter instance after you create the vCenter Cloud Account. This option is deactivated by default, which means the root user is expected by default when you configure the vCenter Cloud Account.
11. Edit the cloud account created for service discovery.
12. In the **Advanced Settings** section, activate the **Application Discovery** field to discover predefined and custom applications.
13. In the **Advanced Settings** section, to configure credential-less service discovery, select **Activated** from the **Credential-less service discovery status** field.
14. Click **Save**.

You can manage services supported by VCF Operations on specific VMs.

Manage Services

You can manage services supported by VCF Operations on the specific VMs.

Where You Manage Services

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications** panel, select **Manage SDMP Services**. You can also navigate to the Manage SDMP Services tab by clicking **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Service Discovery** tile. Select the Manage Services option from the **Service Discovery** page.

You can view specific details from the options in the data grid.

Table 1727: Datagrid Options

Options	Description
VM Name	Name of the VM.
Operating System	Operating system installed on the VM.
Services Discovered	Displays the names of discovered services or `None`, if services are not discovered on the VM.
Service Monitoring	Displays the current value of the VM's service monitoring setting. If set, services are discovered and service performance metrics are calculated every 5 minutes. Otherwise, service discovery is performed every 24 hours.
Authentications Status	VM authentication status for service discovery. The possible values are: • Unknown • Failed • Guest Alias • Common Credentials • Credential-less
Power State	Power status of the VMs. The possible values are: • Powered On • Powered Off • Suspended • Unknown
Collection State	Displays the collection state of an adapter instance of each object. You can see the name of the adapter instance and its state in a tool tip when you point to the collection state icon. To manage an adapter instance to start and stop collection of data, click **Administration** > **Configurations**, and then click the **Inventory Management** tile.
Collection Status	Displays the collection status of the adapter instance of each object. You can see the name of the adapter instance and its status in a tool tip when you point to the collection status icon. To manage an adapter instance to start and stop collection of data, from the left menu, click **Administration** > **Configurations**, and then click the **Inventory Management** tile. You can view a message for VMs with a failed authentication status in a tool tip when you point to the collection status icon.
vCenter Name	Name of the vCenter Adapter instance to which that VM resource belongs.

Table 1728: Toolbar and Row Options

Options	Description
VM Actions	Displays a list of actions. For more information, see Actions in VCF Operations.
Horizontal Ellipsis > **Provide Password**	Select VMs from the list, click the horizontal ellipsis and then click **Provide Password** to provide a user name and password for the selected VMs to discover the services.

Options	Description
Horizontal Ellipsis > **Activate Service Monitoring**	Select VMs from the list, click the horizontal ellipsis and then click **Activate Service Monitoring** to activate frequent service discovery and service performance metrics calculation (every 5 minutes). **Note:** Selecting too many VMs will potentially result in vCenter degradation which is a known issue.
Horizontal Ellipsis > **Deactivate Service Monitoring**	Select VMs from the list, click the horizontal ellipsis and then click **Deactivate Service Monitoring** to deactivate frequent service discovery and service performance metrics calculation. Service discovery defaults to the 24-hour cycle.
Check box to Select/Deselect all	Selects/clears all VM object selections.
Vertical Ellipsis > **Go To Details**	Navigates to the **Summary** tab for the selected VM.
Filter	You can search through the list of VMs according to the following criteria: VM Name, Operating System, Power State, Authentication Status, and Service.

Create an Application Definition

Create application definitions when you want to discover applications based on user-defined rules such as object names, tags, or properties.

1. In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Application Discovery: Rule Based** tile.
2. Click **Add Application Definition**.
3. In the **Create Application Definition** page, enter the required values.

Table 1729: Create Application Definition

Option	Description
Application Name	Enter a name for the application.
Application Prefix	Enter a prefix for the application. Prefix can help you identify the discovered applications by name. It is prepended to the application name.
Select criteria for grouping discovered applications - This section allows you to define the criteria based on which the objects are grouped into an application.	
Group For Object Type	Select an object type to group objects by
Select	Select the criteria to group the objects by. The options available are: • Properties • Object Name • Tag Category
Add	Click add to add another criteria
Reset	Click reset to clear the criteria
Advanced Settings - This section allows you to specify information for additional filters.	
Select the Object Type that matches all of the following criteria	Use the drop-down list to select the object type which matches the defined criteria

Option	Description
Select	Select the object type. The options available are: • Metrics • Relationships • Properties • Object Name • Tag Based on your selection, you must specify the additional attributes.
Add	Click Add to add another criteria
Reset	Click Reset to clear the defined criteria
Add New Criteria	Click Add New Criteria to add another criteria
Remove Criteria	Click Remove Criteria to delete the criteria

4. Click **Create**.

   The created application definition is displayed in the **Application Discovery: Rule Based** page.

Manage Applications

You can discover applications based on user-defined criteria such as virtual machine names, tags, or properties (rule-based applications). For example, if the application definition is assigned to a cluster compute resource called 'xyz', then the applications would be discovered only within that cluster. The object scope assignments can be made to both service based application definitions and rule based application definitions.

Where You Manage Applications

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications**. From the **Applications**, select **Manage Applications**.

You can view direct or inherited assignments by selecting either **All** or **Self** respectively, in the sliding option at the top of the UI. You must select any of the environment objects in order to see application definitions.

You can view specific details from the options in the data grid.

Table 1730: Datagrid Options

Options	Description
Object Name	Name of the Object.
Direct Assignments	Displays the number of application definitions assigned directly to this object.
Inherited Assignments	Displays the number of application definitions that are inherited from the object’s ancestors.
Object Type	Displays the object type.

Table 1731: Assign Application Definition To Scope of Objects

Options	Description
Vertical Ellipsis	Displays Assign Applications Definitions page.
Select rules from the list below to activate application discovery for selected objects	Displays the list of Application Definition Name and Application Definition Type. Select the required definition, the options available are Service-based application definition and Rule-based application definition. **Note:** You can use the search option to look for application definitions name.
>	Expand the arrow next to the object to view the definitions assigned to the selected object and the discovered applications count. You can also view the inherited assignments for the selected object.
Save	Navigates to the Summary tab for the selected Object.
Vertical Ellipsis > **Go To Details**	Navigates to the **Summary** tab for the selected objects.

View Applications

You can view applications created by VCF Operations on specific VMs.

You can view all resources of the Application type, including:

- VMs connected to their services, how the services talk to each other and are connected to different VMs, using the Service Discovery adapter.
- Discovered applications using the VCF Operations Application Management Pack.

Where You View the Applications

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, select **Workloads** > **Applications** and then click **Applications Home** from the centre pane to view the **Applications Home** page.

You can view the list of applications from the **Applications Home** page. Click an application to view the application in the right pane. Select an application row and then click the **Vertical Ellipsis** > **Go To Details** to view the object details. You can use the Filter option to search through the list of applications according to the following criteria:

- Name
- Object Type

Options to View the Applications

After you click an application from the **Applications Home** page, you can view details of the application or service in the right pane.

Options	Description
View Sphere	Displays the VMs connected to their services in a spherical view.
View Graph	Displays the VMs connected to their services as a graph.
View List	Displays the VMs connected to their services in a list view.
View Links	You can view the links between various services.

Discovered Services

You can discover services using the Service Discovery adapter.

Discovered Services

You can view discovered services, the number of VMs on which each discovered service is running, and you can configure service discovery.

Where You View the Discovered Services

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Service Discovery** tile.

Discovered Services

You see a list of services that are discovered and the number of VMs that have the services running. You see this section after you have configured Service Discovery and the services are discovered.

Known Services

You see a list of all the services supported and those that can be discovered.

Custom Services

You can add a service by clicking **Add Service Definition**. You can add either a process name or Regex from the **Add Custom Service** dialog box.

Table 1732: Add Custom Service

Options	Description
Type	Specify the type as either process or regex. **Process**: The process name must exactly match the name that you see in the guest OS when running commands `ps` in Linux and `wmic` in Windows. Specify a single port for each service. The following characters are not supported: `,`, `\`, and `#`. **Regex** Enter a regular expression that corresponds to the command line (or at least name) of the service, that you see in the guest OS when you run the following commands: `ps` in Linux and `wmic` in Windows. For example, to discover Cassandra services, enter `cass.*dra` as the regex. The following characters are not supported: `,` and `\n`. **Note:** If you specify a service using regex and the service reboots at a later time and all the ports are modified, the current service displays as offline and the new service is discovered again. If the new service has any port overlap with the previous one, a new resource will not be discovered and the previous one will continue collecting data as before.
Process Name	Enter a process name.
Port	Enter the port information.
Display Name	Enter the display name.

Note:
Custom service can be discovered via Service Discovery if there is a permanent listening TCP port or if there is an established UDP connection.

Discovered Applications

You can discover applications based on grouping the services discovered, using the Service Discovery adapter.

Discovered Applications

You can discover predefined applications and custom applications. Log management and VCF Operations are the predefined applications.

Where You View the Discovered Applications

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Application Discovery: Rule Based** tile.

Custom Applications

You can define custom applications. Click **Add Application Definition** to add a custom application.

Table 1733: Add Custom Application

Option	Description
Name	Enter a name for the application.
Prefix	Enter a prefix for the application.
Services	Select a service from the list. If the Service Discovery adapter discovers the service and if the services are connected to each other, a new application is discovered. The new application appears in the **Applications Home** page. Navigate to **Workloads** > **Applications**.

Known Applications

You see a list of the predefined applications supported. Select the predefined application, click the vertical ellipsis, and then click **Preview**. From the relevant Application pane, you can view the application services that can be discovered, and if the connections are identified, they will form an application. You can view the applications that are discovered by clicking **Workloads** > **Applications** > **Applications Home**.

> **Note:**
>
> If the same service instance is a member of more than one application (both known and custom), then the service is a part of the application that has the greatest number of services configured in the application definition.
>
> The discovered set of services that communicate with each other, should match at least 70% of the defined application. Only the matching ones are filtered out based on whether one service is defined on more than one application.

Discovered Rule-Based Applications

VCF Operations discovers applications based on user defined criteria such as object names, tags, or properties.

Where You View the Application Discovery Definitions

In the VCF Operations navigation bar at the top, click **Operate** and in the left navigation pane, click **Administration** > **Configurations,** and then click the **Application Discovery: Rule Based** tile.

Application Discovery: Rule Based

The rule based application discovery provides you the following options.

- Specify grouping criteria for each rule, it can be tags, object names, or properties
- Apply additional filters to the rule to limit the scope of objects considered for grouping

You can discover applications based on the user defined criteria such as object names, tags, or properties. After the rules are created you can assign these rules to specific objects. VCF Operations discovers applications based on these rules and displays them in Applications Home page.

Edit or Delete Application Rule

You can modify or delete the application definition. To edit or delete the application definition, click the vertical ellipsis next to the application and select **Edit** or **Delete**. If you select the edit option you are redirected to the **Edit Application Definition** page, here you can modify the application definition and save the changes. If you select the delete option, you are prompted with a confirmation message, click **Yes** to delete the application definition. Please note that if you delete the application definition, the applications that were discovered by that definition are also deleted.

Note:

Updating the name of the application will not update the Object type (resource kind) of the discovered application.

Alert for Service Unavailability

When a service is unavailable, an alert is triggered for the specific VM.

Alert for Service Unavailability

On a VM that is monitored, if one of the services is down, in the next collection cycle an alert is triggered.

Alert Name	Symptom
One or more monitored service(s) are unavailable on the virtual machine.	Service is not available. When the service is available again, the symptom disappears.

The alert is canceled in the following scenarios:

- When all the discovered services are available again in the monitored VM.
- If the service is not available within 7 days.
- If you deactivate service monitoring for the monitored VM.

Where You Find the Alert

From the **Manage Services** page, ensure that the VM is monitored and one or more service(s) are unavailable on the VM. Select the VM, click **Show Details** to go to the summary page. Click **Alerts** from the toolbar, and then click the **Alerts** tab.

Property for Service Unavailability

You can view the property called Status for a service that has been discovered on the VM. For more information, see Services Properties.

Service Discovery Metrics

Service discovery discovers metrics for several objects. It also discovers CPU and memory metrics for discovered services.

Virtual Machine Metrics

Service Discovery discovers metrics for virtual machines.

Table 1734: Virtual Machine Metrics

Metric Name	Description
Guest OS Services|Total Number of Services	Number of out-of-the-box and user-defined services discovered in the VM.
Guest OS Services|Number of User Defined Services	Number of user-defined services discovered in the VM.
Guest OS Services|Number of OOTB Services	Number of out-of-the-box services discovered in the VM.

Metric Name	Description
Guest OS Services|Number of Outgoing Connections	Number of outgoing connection counts from the discovered services.
Guest OS Services|Number of Incoming Connections	Number of incoming connection counts to the discovered services.

Service Summary Metrics

Service discovery discovers summary metrics for the service object. The object is a single service object.

Table 1735: Service Summary Metrics

Metric Name	Description
Summary|Incoming Connections Count	Number of incoming connections.
Summary|Outgoing Connections Count	Number of outgoing connections.
Summary|Connections Count	Number of incoming and outgoing connections.
Summary|Pid	Process ID.

Service Performance Metrics

Service discovery discovers performance metrics for the service object. The object is a single service object.

Table 1736: Service Performance Metrics

Metric Name	Description
Performance metrics group|CPU	CPU usage in percentage.
Performance metrics group|Memory	Memory usage in KB.
Performance metrics group|IO Read Throughput	IO read throughput in KBps.
Performance metrics group|IO Write Throughput	IO write throughput in KBps.

Service Type Metrics

Service discovery discovers metrics for service type objects.

Table 1737: Service Type Metrics

Metric Name	Description
Number of instances	Number of instances of this service type.